SEMANTIC SEGMENTATION OF FOREST CONDITION IN MT. MAKILING USING A CNN–VIT HYBRID WITH NATIVE TREE SPECIES RECOMMENDATION

A Thesis Presented to the Faculty, College of Computer Studies
Laguna State Polytechnic University
Los Baños Campus
Los Baños, Laguna

In partial fulfillment of the requirement for the degree
Bachelor of Science in Computer Science
Specialized in Intelligent Systems

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JUNE 2026

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ACKNOWLEDGEMENT

The researchers extend their heartfelt gratitude to all individuals and institutions whose support was vital to the completion of this study.

First and foremost, to Almighty God for granting the researchers' strength, wisdom, and perseverance to complete this research.

The researchers would want to thank the Laguna State Polytechnic University Los Baños Campus for providing them the necessary resources. To Dr. Mario R. Briones, the University President and Engr. Manuel Luis R. Alvarez, the OIC Campus Director for their leadership and support.

Special thanks to Dr. Loyd S. Echalar, Associate Dean of the College of Computer Studies and to the faculty members of the College of Computer Studies for their encouragement and guidance.

To Prof. Maribel V. Cardona, the research adviser, they are truly grateful for your patience, valuable insights, and support throughout this journey.

To Prof. Gene Marck B. Catedrilla, Prof. Carolina R. Joval, and Prof. Jhoco S. Millanes, the panel members, for their constructive feedback and recommendations that helped improve the work.

The researchers would also like to extend their gratitude to Dr. Tomas Reyes Jr. and Dr. Cristino L. Tiburan, for lending their knowledge regarding validation measures with the researchers' topic and for showing their support.

The Researchers

DEDICATION

This thesis is dedicated to everyone who walked alongside the researchers during this important chapter of their lives, including those who believed in them, offered support, and gave encouragement throughout the journey.

First and foremost, the researchers give thanks to Almighty God for the strength, guidance, and protection that carried them through this process. Without His presence, completing this study would not have been possible.

The researchers sincerely thank their families, Ms. Angelica Lapitan, Mr. Roberto Garson, Mrs. Delia Lapitan, and Mrs. An Irene; Ms. Maria Teresa D.C. Villaroman; and Mrs. Nimfa U. Dorado and Mr. Reynaldo M. Dorado, for their unconditional love and care throughout the years. Their sacrifices, encouragement, and constant support helped shape the researchers into who they are today and gave them the strength to finish this thesis. This achievement belongs to them just as much as it does to the researchers.

Deep appreciation is also extended to their extended family, Mrs. An Irene and Mrs. Marivi Cabrera, whose kindness and continued support made it possible for the researchers to pursue their studies. Their belief in the researchers helped them grow, persevere, and trust in themselves. This study reflects the positive impact they have had on the researchers' lives.

The researchers are grateful to their friends, whose presence, laughter, and support, sometimes simply by being there, made the journey lighter and reminded them that they were never alone.

They would also like to thank their classmates from BSCS 4B, who shared this journey with them. Their ideas, suggestions, and feedback helped improve the study and brought new perspectives that the researchers might have overlooked. The support and camaraderie of the section are truly appreciated.

Lastly, the researchers express their heartfelt gratitude to their partners, Ms. Sarah Claudine V. Noor, Mr. John Adrian Lapitan, and Ms. Aleanna A. Diaz, for the love, patience, and encouragement they consistently gave. During the most challenging moments, they provided strength, motivation, and reassurance when the researchers felt close to giving up. Their presence made this journey more meaningful, and the researchers are thankful to have shared this chapter of their lives with them as they look forward to the future together.

J.A.L

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ABSTRACT

This study developed and evaluated a deep learning framework for forest monitoring and restoration planning within the Mt. Makiling Forest Reserve (MMFR), utilizing Sentinel-2 satellite imagery and SRTM elevation data. The study implemented a multi-task semantic segmentation pipeline to classify Forest Type, Forest Level, and Forest Health using a 6-channel composite input (RGB + NDVI + EVI + NDWI). Three architectures were compared: U-Net (CNN), SegFormer (ViT), and TransU-Net (Hybrid), across two patch resolutions (64×64 and 128×128). The results identified the Hybrid TransU-Net with 128×128 inputs as the superior architecture, achieving a near-perfect Mean Intersection-over-Union (mIoU) of 99.21% for Forest Health on the 2020 test set, successfully resolving critical minority classes such as "Regrowth" and "Degraded" that simpler models failed to detect. However, temporal evaluation on an independent 2021 dataset revealed a lack of temporal robustness, with Forest Level accuracy dropping to approximately 55% and Forest Health mIoU declining to 33.62%. This indicates that while the model excels at characterizing the specific spatial and spectral conditions of the training year, it suffers from overfitting to static topographical features. The study further demonstrated a rule-based native tree species recommendation module, which functioned logically but was constrained by the upstream classification errors in the multi-temporal test. The research concludes that while the hybrid approach provides a powerful tool for precision forest monitoring, operational deployment requires multi-year training data to ensure temporal generalization.

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DEFINITION OF TERMS

This section defines the essential technical and operational terms relevant to the study. Providing these definitions ensures a clear and common understanding of concepts referenced in the succeeding chapters.

Area of Interest (AOI)	The geographic boundary defined for analysis, specifically the Mount Makiling Forest Reserve (MMFR). All satellite imagery and label data were clipped to this AOI for processing.
CNN (Convolutional Neural Network)	A deep learning architecture designed to automatically learn spatial features from image data. In this study, the U-Net model represents the CNN approach, processing local patterns like textures and edges.
Data Augmentation	The process of artificially increasing the diversity of training data by applying transformations. In this study, techniques such as flipping and rotation were applied to balance rare classes like Degraded and Regrowth.
Degraded Area	A location classified with a significant negative change in NDVI values (typically < -0.10). These areas represent vegetation loss or stress and are prioritized targets for the tree recommendation module.
DEM (Digital Elevation Model)	A raster representation of terrain elevation sourced from SRTM data. It is used to generate the ground-truth labels for Forest Level (Lowland, Lower Montane, Upper Montane).
EVI (Enhanced Vegetation Index)	A vegetation index optimized to correct for soil background signals and atmospheric influences. It is included in the 6-channel input to improve vegetation monitoring in high-biomass areas.
Forest Health	A classification representing the condition of vegetation based on temporal NDVI difference. It is categorized into Degraded, Neutral, or Regrowth.
Forest Level	An elevation-based classification that divides the forest into Lowland, Lower Montane, and Upper

	Montane zones. These labels are derived from SRTM elevation thresholds and dictate the specific tree species recommended by the system.
Forest Type	A land cover classification describing the presence of forest. It distinguishes between Forested and Non-Forest areas, derived from the ESA WorldCover dataset.
GEE (Google Earth Engine)	A cloud-based platform for processing geospatial data. GEE is used for image acquisition, NDVI calculation, cloud masking, and raster export in this study.
Haribon Foundation / DENR Manuals	National ecological references used to guide native tree species selection for restoration, based on forest type, elevation, and site condition. These references support the rule-based recommendation system.
Hybrid TransU-Net	The optimal model architecture identified in the study. It integrates a CNN backbone (ResNet) for local feature extraction with a Vision Transformer (ViT) bottleneck for global context to simultaneously predict forest type, level, and health.
Label Map	A raster layer where each pixel value represents a specific class label. Three label maps were used in this study: Forest Type, Forest Level, and Forest Condition.
Land Cover Map	A raster or vector dataset that categorizes surface features such as forest, agriculture, and water. The ESA WorldCover is used here to derive forest type labels.
mIoU (Mean Intersection-over-Union)	The primary metric used to evaluate segmentation performance. It measures the overlap between the predicted mask and the ground truth mask, averaged across all classes to account for class imbalance.
Mt. Makiling Forest Reserve (MMFR)	A protected area located in Laguna, Philippines, designated as an ASEAN Heritage Park. It serves as the study area for the forest classification and tree recommendation model.

Multi-Task Learning	A training approach where a single model learns to perform multiple prediction tasks at once. In this study, the model predicts three forest labels from a single RGB + indices input patch.
NDVI (Normalized Difference Vegetation Index)	A metric calculated from Near-Infrared and Red bands used to assess vegetation vigor. It serves as both an input feature (in the 6-channel composite) and the source for generating Forest Health labels.
NDVI Difference	The result of subtracting NDVI values from two different time periods to detect changes in vegetation cover. In this study, NDVI difference is used to identify areas of forest degradation, stability, or regrowth.
NDWI (Normalized Difference Water Index)	A spectral index used to monitor changes in water content of leaves. It is included in the input composite to help the model distinguish moisture stress in vegetation.
Patch Extraction	The process of dividing large satellite scenes into smaller, fixed-size samples for training. This study utilized a stratified random sampling method to extract patches at two resolutions: 64×64 and 128×128 pixels.
QGIS	A free and open-source Geographic Information System used to define the Area of Interest (AOI), handle shapefiles, and visualize raster data such as DEM and land cover maps.
Remote Sensing	The process of collecting information about the Earth's surface using satellites or aerial imagery without the need of physical contact. In this study, remote sensing technology is used to gather data on forest cover, elevation, and vegetation health.
Rule-Based Recommendation Module	A logic-based component that suggests native tree species (e.g., *Shorea contorta, Lithocarpus spp.*) by cross-referencing the predicted Forest Level and Forest Health against a lookup table derived from ecological literature.
Semantic Segmentation	A computer vision task where every pixel in an image is assigned a class label. Unlike simple object detection, this provides a precise map of

	where forest boundaries and degraded patches are located.
Sentinel-2	A satellite mission from the European Space Agency providing the multispectral optical imagery used in this study. Bands 2, 3, 4, and 8 were used to generate the RGB, NDVI, EVI, and NDWI inputs.
SRTM (Shuttle Radar Topography Mission)	A satellite-derived dataset that provides global elevation information. It is used to extract forest level information through elevation thresholds.
Tree Recommendation Module	A component of the model that recommends native tree species for reforestation in degraded areas. It applies predefined ecological rules sourced from DENR and Haribon references.
True Color Composite	An RGB (Red, Green, Blue) image created using Sentinel-2 bands B4 (red), B3 (green), and B2 (blue), visually similar to natural color photography. It serves as the input data for the classification model.
ViT (Vision Transformer)	A deep learning architecture represented by the SegFormer model in this study. It uses self-attention mechanisms to capture long-range global spatial relationships across the forest landscape.

CHAPTER 1

INTRODUCTION

This section provides an overview of the research background, the issues that prompted the investigation, and the motivations behind conducting the study. This chapter also defines the objectives of the study, emphasizes its significance and contributions to both scientific and practical applications, and clarifies the scope and limitations that define the research.

Background of the Study

The Philippines is one of the 17 megadiverse countries, a homeland to countless endemic plants and species. Moreover, it remains vulnerable to natural disasters due to deforestation and climate change, driven by population growth, urbanization, and land degradation. Therefore, protecting an ecological area is essential to maintaining soil stability, conserving biodiversity, and absorbing carbon. A well-known protected environmental area in the provinces of Laguna and Batangas is the Mt. Makiling Forest Reserve (MMFR), an ASEAN Heritage Park recognized for its rich biodiversity, being home to endemic plant, tree, and animal species. The forest landscapes have changed primarily due to pressure from human activities, land conversion, and natural disturbances (Combalicer et al., 2011; Malabrigo et al., 2016). Thus, these cases highlight the need for innovative monitoring and restoration strategies.

To monitor forest condition, the Normalized Difference Vegetation Index (NDVI) and other vegetation indices are widely used in remote sensing (Tucker, 1979; Pettorelli et al., 2005). Agencies such as NAMRIA and DENR use these

indices for large-scale monitoring to help differentiate between degraded and healthy vegetation. However, these vegetation indices often lack site-specific systems that can provide restoration insights tailored to locations like Mt. Makiling. Past research by Combalicer et al. (2011) mentioned the potential of vegetation indices in detecting land cover changes in a protected area. Nonetheless, in the context of reforestation, integration between forest condition analysis and species-specific tree recommendations was limited.

Multiple studies of Mt. Makiling highlight its ecological richness, including a wide variety of plant and animal species, and its potential for restoration planning. Researchers Bantayan et al. (2019) developed a digital geodatabase platform incorporating land cover, soils, and biodiversity to support the management of natural resources. Subsequently, Malabrigo et al. (2016) conducted research in a permanent plot at Mt. Makiling, where they documented over 150 native tree species; this result provided valuable information for biodiversity-based reforestation. These studies demonstrate that Mt. Makiling can give a strong ecological foundation for an AI-assisted segmentation capable of distinguishing forest health conditions.

This study addressed these gaps by classifying forest type, level, and health; identifying degraded areas; and recommending native tree species for reforestation efforts. Unlike conventional image-level classification, the study uses semantic segmentation to generate pixel-wise maps of forest type, condition, and health, enabling localized detection of degraded areas. The model integrated NDVI and DEM elevation data with a hybrid CNN–ViT architecture, combining spatial detail with contextual analysis (Lehouel et al., 2024; Roy et al., 2020). By applying deep learning to Mt. Makiling, the system

produced localized, evidence-based outputs to guide restoration. In addition to its local scope, the study also supports the Sustainable Development Goals, specifically SDG 13 – Climate Action and SDG 15 – Life on Land. Between science-based monitoring and restoration, this research not only strengthens conservation research but also community planning for Mt. Makiling's forests.

Objectives

This study aimed to develop a proof-of-concept pipeline for forest classification and native tree species recommendation for reforestation based on multispectral satellite imagery and elevation data, specifically within the Mt. Makiling Forest Reserve (MMFR).

Specifically, it sought to:

1. Collect and process Sentinel-2 Surface Reflectance imagery and SRTM Digital Elevation Models (DEM) to generate a 6-channel composite input (RGB + NDVI + EVI + NDWI) for classification.

2. Develop and compare the performance of three Multi-Task Learning (MTL) deep learning architectures: U-Net (CNN), SegFormer (Transformer), and TransU-Net (Hybrid), in performing semantic segmentation of:

    - Forest Type (Forest, Non-Forest)
    - Forest Level (Lowland, Lower Montane, Upper Montane)
    - Forest Health (Degraded, Neutral, Regrowth)

3. Implement a rule-based recommendation module, informed by ecological literature, that suggests native tree species based on the predicted forest level and forest health.

4. Evaluate the performance of the classification models using standard metrics, including Pixel Accuracy, per-class F1-score, Macro F1-score, per-class Intersection-over-Union (IoU), and Macro Intersection-over-Union (mIoU).

Significance of the Study

This study provided a data-driven approach for monitoring forest condition in the Mt. Makiling Forest Reserve (MMFR). It supported restoration planning by linking observed conditions with ecologically appropriate tree species. The study used NDVI and DEM to generate labels and classified forests using RGB+NDVI input imagery. It employed a CNN–ViT hybrid model for classification and a rule-based module for tree species recommendation. Researchers have widely used NDVI for forest remote sensing because it can distinguish dense from degraded vegetation, as demonstrated by Combalicer et al. (2011). At the same time, combining deep learning with elevation-derived labels helped improve ecological relevance in classification, especially in topographically diverse regions (Lehouel et al., 2024). By narrowing the geographic focus, this system delivered location-sensitive, ecologically grounded outputs to support forest rehabilitation.

The study supported UNESCO's mission to advance the Sustainable Development Goals, particularly SDG 13 (Climate Action) and SDG 15 (Life on

Land). UNESCO promotes the use of digital tools and data science to address ecosystem degradation, a principle also echoed in the Man and the Biosphere (MAB) Programme and UNESCO SDG education reports. This study contributed by integrating remote sensing and AI to guide reforestation planning based on local forest conditions. Furthermore, matching tree species to site-specific forest levels reflected the ecological precision recommended by Malabrigo et al. (2016) in biodiversity-based restoration frameworks. The study, therefore, connected local-scale implementation with global sustainability priorities.

The **DENR** could use the process to strengthen existing forest monitoring tools and visualize condition-based degradation using NDVI, as previously demonstrated in Mt. Makiling studies.

Local government, particularly the University of the Philippines Los Baños (UPLB) and the Mt. Makiling Forest Reserve management, may benefit from automated mapping of degraded areas and matching of tree species recommendations for site-level planning.

Future researchers may use this framework to build scalable models that align deep learning with ecological data, expanding on methods such as those proposed by Bantayan et al. (2019) for geospatial-based restoration. The use of NDVI as a long-term monitoring tool continues to be supported in conservation science (Pettorelli et al., 2005).

Unlike prior NDVI-based studies in MMFR, this study extended beyond monitoring by linking condition detection with site-appropriate tree species

recommendations. As a proof-of-concept system, this study demonstrated the potential of integrating remote sensing, deep learning, and rule-based recommendations for forest monitoring and restoration planning in MMFR. This study also contributed methodologically by demonstrating the feasibility of combining semantic segmentation with a rule-based recommendation system, offering a replicable framework that future research can refine with more extensive datasets or field validation.

Scope and Limitation

This study focused on the Mt. Makiling Forest Reserve (MMFR). The model classified forest type into Forest and Non-Forest (derived from ESA WorldCover); forest level into Lowland, Lower Montane, and Upper Montane (derived from SRTM DEM); and forest condition into Degraded, Neutral, and Regrowth (derived from NDVI Change Detection). The researchers used 6-channel multispectral patches (RGB, NDVI, EVI, NDWI) in the classification. The classification task was implemented as semantic segmentation, assigning each pixel in the imagery to a specific category. The system also provided a lookup-based recommendation of native tree species for degraded forest patches based on their elevation zone.

Limitations:

- **Proof-of-Concept:** The study focused on computationally testing and evaluating the classification and recommendation pipeline. The researchers did not conduct an on-the-ground validation and actual tree planting due to resource constraints.

- **Data Constraints:** Some NDVI difference areas contained NaN values due to persistent cloud cover; these areas were masked during preprocessing to avoid training bias, resulting in reduced spatial coverage in cloud-heavy regions.
- **Geospatial Output and Localization:** Due to stratified random sampling, which divided the study area into discrete, non-contiguous image patches (128x128), the current model output consists of classified image chips rather than a seamless, georeferenced raster map. Consequently, specific geographic coordinates (latitude and longitude) for the detected degraded zones are not explicitly provided in the final output, necessitating an additional post-processing step to map these predicted patches back to their actual physical locations for ground-level intervention.
- **Label Reliance:** Secondary sources, such as ESA World Cover and Thresholded Indices, were used to generate ground-truth labels rather than manual field tagging. Therefore, these generated datasets are the basis for the model's accuracy.
- **Recommendation Scope:** The study based its tree species recommendations on existing literature rather than field validation; users should treat these recommendations as advisory tools that support, not replace, expert forestry assessment.

CHAPTER 2

THEORETICAL FRAMEWORK

Review of Related Literature, Studies, and Systems

The effective monitoring and restoration of tropical ecosystems requires convergence among remote sensing, ecological science, and artificial intelligence. This chapter reviews the theoretical and empirical foundations of these domains, beginning with the utility of vegetation indices, such as NDVI, for assessing forest health and establishing thresholds for degradation. It subsequently examines the evolution of deep learning architectures from Convolutional Neural Networks to advanced Hybrid Transformer models, evaluating their capacity to classify complex forest landscapes with high precision. Furthermore, the review contextualizes these technologies within Philippine ecological settings, specifically analyzing elevational zonation and biodiversity studies in Mt. Makiling to ground the development of a scientifically robust, site-specific tree species recommendation system.

Vegetation Indices for Forest Monitoring

Vegetation indices are essential tools in remote sensing because they help researchers and managers assess plant health, canopy cover, and vegetation distribution using satellite imagery. The most commonly used index is the Normalized Difference Vegetation Index (NDVI), which compares how plants reflect near-infrared (NIR) and red light. NDVI values range from -1 to +1, with higher values indicating healthier, denser vegetation. This index has been applied worldwide for agricultural, forest, and drought monitoring, as well

as for land degradation assessments. However, NDVI is sensitive to soil brightness, sensor differences, and atmospheric conditions, making it less reliable in complex ecosystems like tropical forests (Tucker, 1979).

To address NDVI's weaknesses, researchers developed the Enhanced Vegetation Index (EVI), which adds a blue band to reduce the effects of atmosphere and soil background. EVI performs better in dense vegetation areas, especially in tropical rainforests, where NDVI often saturates and loses sensitivity. It corrects distortions caused by clouds, aerosols, and canopy shadows, producing clearer vegetation signals. Studies by Huete et al. (2002) and others confirmed that EVI can capture subtle variations in vegetation health, even in high-biomass environments. As a result, researchers widely use EVI for forest monitoring and carbon stock estimation projects.

Another index is the Soil-Adjusted Vegetation Index (SAVI), which was designed to reduce the effects of soil brightness in sparsely vegetated or semi-arid regions. SAVI modifies the NDVI formula by introducing a soil-adjustment factor, thereby improving detection in areas with exposed soil. Huete (1988), who developed SAVI, demonstrated its usefulness in areas with low vegetation cover, such as grasslands and degraded forests. Since then, it has been applied in land degradation studies and ecological monitoring in semi-humid regions. In the Philippines, SAVI can provide clearer vegetation insights in upland and transitional forest areas than NDVI alone.

The Green Normalized Difference Vegetation Index (GNDVI) is another variation, which replaces the red band in NDVI with the green band to highlight chlorophyll content. This change makes it particularly effective in monitoring

plant health, crop stress, and early disease detection. Recent studies show that GNDVI, along with NDRE (Normalized Difference Red Edge), can detect nitrogen deficiency and early stress in both temperate and tropical forests (Delegido et al., 2011; Zarco-Tejada et al., 2013). Although not as widely used as NDVI in forests, it shows strong potential for distinguishing healthy from stressed forest patches. In fact, Wu et al. (2021) showed that GNDVI can outperform NDVI in detecting canopy stress in complex, mixed-species forests.

The Modified Soil-Adjusted Vegetation Index (MSAVI) improves on SAVI by automatically adjusting the soil correction factor based on vegetation density in each pixel. This change allows it to better classify areas with mixed vegetation cover, ranging from sparse to moderately dense. Qi et al. (1994) demonstrated that MSAVI reduces noise and improves vegetation mapping accuracy in dryland and transitional forest ecosystems. It is invaluable in environments with seasonal variation, where vegetation cover changes over time. For the Philippines, MSAVI can be valuable in degraded upland forests or recently deforested zones that experience fluctuating vegetation density. In the context of forest condition monitoring, MSAVI may provide better accuracy in mapping areas requiring restoration.

Several comparative studies have examined the strengths and weaknesses of these indices. For instance, Bhandari et al. (2012) explored the effectiveness of NDVI and SAVI for urban vegetation mapping in India, concluding that hybrid index use improved classification accuracy. Gandhi et al. (2015) conducted NDVI differencing in India’s Vellore District and reported improvements in land change detection when additional indices were included.

Similarly, Lu and Weng (2007) emphasized the need to select indices based on environmental context and suggested combining indices with machine learning classifiers to optimize results. In tropical forest environments such as the Philippines, these findings indicate that a multi-index approach offers better vegetation detection and classification, particularly when used as input features in deep learning models.

Vidad et al. (2019) analyzed closed- and open-canopy forests in the Northern Sierra Madre Natural Park using hyperspectral imagery. They found that different canopy structures produce distinct spectral signatures, indicating that vegetation indices such as NDVI can effectively distinguish dense from sparse forests. This proves that vegetation indices are reliable tools for assessing forest conditions in real-world applications. The Sierra Madre study also mirrors the kinds of forest variation expected in Quezon province, strengthening the ecological relevance of NDVI classification. Therefore, NDVI is appropriate as a primary input in this model to assess forest condition in the Quezon part of the Sierra Madre.

NDVI Thresholds for Forest Condition & Degradation

NDVI thresholding is a simple but effective way to evaluate vegetation condition and track land cover changes. Pixel values are commonly grouped into classes such as non-vegetated (<0.2), degraded or sparse (0.2–0.4), moderate (0.4–0.6), and dense, healthy vegetation (>0.6). These thresholds are widely applied in forest assessments and are often adjusted to match regional vegetation dynamics. Sahebjalal and Dashtekian (2013) used this method to detect vegetation loss in Iran, identifying an 18.83% reduction in

forest cover due to urban expansion. Complementing traditional thresholds, recent works have proposed adaptive NDVI thresholding using machine learning to tailor vegetation classifications to regional climate and phenological patterns (Zhang et al., 2020). Adaptive NDVI methods have shown better performance than static thresholds in tropical ecosystems where vegetation cycles change rapidly. In a related study, Gandhi et al. (2015) demonstrated that NDVI thresholds effectively detected a 6% decline in forest area over 5 years, demonstrating its effectiveness in monitoring forest loss. This highlights NDVI's strong role in identifying forest degradation over time.

In mapping forest health, remote sensing studies have successfully distinguished disturbed or degraded forest areas from healthy vegetation and early regrowth. Magallon and Tsuyuki (2024) utilized Landsat and PlanetScope imagery in the Philippines to map typhoon-induced forest damage, using NDVI and EVI to measure vegetation stress and canopy disturbance. Their classification of forest condition, ranging from damaged to regenerating, was based on temporal changes in vegetation indices, which served as indicators of photosynthetic activity and canopy density. By comparing NDVI patterns before and after typhoon events, the authors were able to differentiate areas of degradation from zones showing signs of recovery. This scientific basis aligns with the "forest health" variable used in this study, which uses NDVI-derived thresholds to categorize forest condition as degraded, neutral, or regrowth, representing varying stages of ecosystem vitality and disturbance.

Temporal NDVI analysis, which tracks vegetation values across multiple years, helps detect long-term patterns of forest decline or recovery.

Pettorelli et al. (2005) stressed its importance in studies of seasonal and yearly vegetation shifts. In the Philippines, agencies like NAMRIA and DENR have applied NDVI thresholding with Landsat imagery to classify land into forested, non-forested, and degraded zones, supporting reforestation work and policy-making. By adjusting NDVI thresholds, researchers can automatically highlight degraded areas that need replanting or protection from threats such as logging and agriculture.

Another technique, NDVI differencing, compares NDVI values from two time periods to detect vegetation gains or losses. This approach is practical in revealing subtle land cover changes, especially when combined with other vegetation indices or machine learning models. Studies by Vina et al. (2011) and Yazbeck et al. (2024) showed how NDVI differencing can be used in land surface models to improve predictions of carbon storage and methane emissions. In forests, this method is valuable for tracking reforestation progress, detecting illegal logging, and providing cost-efficient monitoring when paired with high-resolution imagery and field data.

The Normalized Difference Vegetation Index (NDVI) remains one of the most reliable tools for assessing vegetation health through satellite imagery. It has been widely applied in long-term studies to observe forest dynamics, detect ecosystem changes, and monitor deforestation. Ollinger et al. (2008) confirmed that NDVI data from multi-decade archives accurately capture both seasonal and long-term vegetation changes. Their research also showed NDVI’s reliability in tracking trends in biomass, productivity, and land cover condition.

This validates the current study's use of NDVI as the core indicator for forest classification in the Sierra Madre.

Israel et al. (2024) conducted a forest restoration assessment in selected critical watersheds in the Philippines using NDVI and Google Earth Engine (GEE) to evaluate vegetation condition and restoration progress. Their study established NDVI thresholds to classify forest conditions into degraded, stable, and improving states, serving as a basis for prioritizing restoration efforts. By analyzing temporal NDVI changes, the researchers demonstrated that NDVI effectively reflects variations in forest canopy density and health across different regions. This local study reinforces the suitability of NDVI-based analysis for forest condition monitoring in Philippine landscapes, particularly in identifying degraded zones requiring intervention. The findings of Israel et al. (2024) provide empirical support for the NDVI thresholds applied in this study to detect degradation and regrowth in Mt. Makiling.

Deep Learning in Vegetation Classification

The use of deep learning in remote sensing has dramatically improved vegetation classification, offering both higher accuracy and greater automation. Convolutional Neural Networks (CNNs) are among the most popular models because they can extract layered spatial features from satellite images. By applying filters across multiple convolutional layers, CNNs learn to recognize simple elements like edges and textures and gradually build up to more complex vegetation patterns. In vegetation studies, research by Kussul et al. (2017) and Wu et al. (2021) showed that CNNs, when combined with NDVI and other vegetation indices, achieve higher classification accuracy compared to

traditional machine learning models such as random forests or support vector machines. This highlights CNNs as a reliable choice for remote sensing tasks that require strong performance on vegetation detection.

Despite their success, CNNs are limited by their focus on local receptive fields, which means they mainly analyze pixel neighborhoods rather than global patterns. This can be restrictive when mapping vegetation across broad areas or distinguishing visually similar vegetation types spread over different terrains. To address these issues, researchers have tested multi-stream and attention-based CNNs that integrate more diverse spatial-spectral information to achieve stronger performance in mixed landscapes (Roy et al., 2020). Even so, CNNs still fall short in modeling long-range dependencies across images. Vision Transformers (ViTs) emerged as a response, breaking images into non-overlapping patches and using self-attention to evaluate relationships among all patches. This global view allows ViTs to capture broader spatial patterns and achieve improved results in complex vegetation classification tasks (Bazi et al., 2021; Dosovitskiy et al., 2020).

Recent studies have shown that ViTs can outperform CNNs in many remote sensing applications. Amjoud and Amrouch (2023), for instance, reviewed ViT models and reported their effectiveness in tasks such as object detection and land cover classification using high-resolution satellite data. ViTs excel in heterogeneous landscapes, including forest-agriculture boundaries and urban-forest zones, where CNNs often struggle. However, ViTs require large datasets and strong computing power, making them less practical in smaller-scale projects. To make them more accessible, researchers are now

applying transfer learning and self-supervised techniques such as DINO and MAE, which enable ViTs to be pretrained on massive datasets and fine-tuned for vegetation classification with fewer resources (Caron et al., 2021; Chen et al., 2021).

Hybrid architectures that combine CNNs and ViTs have emerged as a promising solution, balancing local and global feature learning. In these models, CNNs extract lower- and mid-level features, which are then fed into ViT modules for global context analysis. Lehouel et al. (2024) demonstrated that CNN-ViT hybrids achieved higher accuracy than standalone CNNs or ViTs, especially in vegetation datasets with high spatial variability. These models also proved more robust to noise and transferable across different environments, which strengthens their appeal for practical use. Given these advantages, the CNN-ViT hybrid model fits this study's objectives of mapping forest types and conditions across diverse landscapes while remaining efficient and scalable.

To further address the limitations of standalone CNNs and standard Vision Transformers, recent research has shifted toward hybrid and hierarchical architectures that merge the feature extraction power of CNNs with the long-range dependency modeling of Transformers. Fusion models, such as TransUNet, were among the first to successfully combine a CNN encoder for local spatial details with a Transformer bottleneck for global context, proving highly effective in segmenting complex boundaries in medical and environmental imagery (Chen et al., 2021). Following this evolution, Swin-UNet introduced a hierarchical Swin Transformer design that processes images in shifting windows. This approach allows the model to capture both local

vegetation textures and broader forest patterns with linear computational complexity, making it superior to traditional ViTs for high-resolution ecological mapping (Cao et al., 2023).

In parallel, SegFormer was developed as a lightweight, robust alternative that eliminates positional encoding and leverages a hierarchical architecture to produce multi-scale features. Xie et al. (2021) demonstrated that SegFormer achieves state-of-the-art accuracy in semantic segmentation while remaining computationally efficient, a critical property for processing large-scale satellite datasets. In the context of remote sensing, these advanced architectures, whether hybrid (CNN-ViT) or hierarchical Transformers, have shown exceptional resilience in distinguishing spectrally similar vegetation classes, such as distinguishing between dense forest and early regrowth (Wang et al., 2022). By integrating these modern "fusion" concepts, this study aims to leverage the Hybrid architecture's specific strengths to resolve the fine-grained transition zones in the Sierra Madre.

In summary, deep learning has transformed remote sensing-based vegetation analysis. CNNs remain effective at capturing local detail, ViTs excel at modeling global relationships, and hybrid CNN-ViT models integrate the best of both. This combined approach supports the development of advanced forest monitoring systems that can classify vegetation more accurately and at larger scales. Future directions, such as self-supervised learning, cloud-based platforms like Google Earth Engine, and high-resolution imagery, are expected to push these systems even further. These advances make AI-powered

monitoring tools increasingly vital for environmental management and forest conservation.

Forest Mapping and Classification Applications

Forest mapping is essential for monitoring land cover, biodiversity, and the overall health of forests. It provides critical baseline information for conservation and restoration planning, especially in countries like the Philippines, where ecosystems are both highly diverse and vulnerable. Remote sensing has become a key tool in this field, offering scalable ways to track forest changes through satellite imagery. National programs such as NAMRIA's land cover maps and DENR's monitoring systems provide essential data, but localized studies often yield more detailed, site-specific insights.

Sentinel-2 is a dual-satellite constellation (Sentinel-2A and Sentinel-2B) with a global revisit frequency of approximately 5 days, enabling dense temporal coverage of terrestrial ecosystems. According to the Copernicus Sentinel-2 Mission Overview, this frequent revisit cycle enhances the utility of multi-date or annual composites because more observations increase the likelihood of obtaining cloud-free pixels (Copernicus ESA, n.d.). The mission also maintains high radiometric stability and geometric accuracy through a rigorous calibration and validation framework, as documented by Szantoi and Strobl (2019). This stability ensures that reflectance values acquired at different dates remain comparable, making temporal aggregation scientifically valid. Consequently, the combination of frequent revisits and consistent radiometry supports the reliable creation of multi-date composites for vegetation monitoring.

In tropical regions with persistent cloud cover, these short revisit intervals are significant because they reduce temporal data gaps and improve the chances of capturing usable surface reflectance. The Sentinel-2 mission explicitly emphasizes its high temporal resolution as a key feature that enables consistent surface monitoring across different seasons (Copernicus ESA, n.d.). When paired with its robust radiometric calibration, this design allows multi-date or annual composites to mitigate cloud contamination, minimize noise, and stabilize spectral signatures. Szantoi and Strobl (2019) further highlight that Sentinel-2's radiometric uncertainty remains below approximately 5%, supporting reliable multi-temporal analyses. Therefore, the mission's technical characteristics provide strong justification for the use of annual or multi-date composites in forest condition mapping.

In the Mt. Makiling Forest Reserve (MMFR), forest mapping has been widely applied due to its status as a protected area and research site. As a natural laboratory for forestry and ecology, Mt. Makiling provides a valuable setting to study forest dynamics. Combalicer et al. (2011) used NDVI and unsupervised classification to analyze long-term landscape changes, revealing both deforestation and forest recovery. Their study highlighted the usefulness of vegetation indices as sensitive indicators of forest condition and temporal change.

Studies on Philippine mountain ecosystems have shown that vegetation composition strongly follows elevation gradients, validating the use of forest level as an ecological variable. Villanueva and Buot (2018) conducted a vegetation analysis along the altitudinal gradient of Mount Ilong in the Halcon

Range of Mindoro Island. They identified distinct forest zones ranging from lowland to mossy forests. Their findings demonstrate that elevation significantly influences species distribution, forest structure, and canopy composition. This supports the forest level classification used in this study, Lowland, Lower Montane, Upper Montane, and Mossy, as a valid ecological framework for representing the natural stratification of Mt. Makiling.

Several Philippine ecological studies substantiate the use of elevation-based forest level categories by demonstrating distinct changes in vegetation composition, structure, and species richness at different altitudes. For example, Alcala (2019) sampled moss diversity in Mt. Makiling along a gradient (plots every 100 m asl) and found increasing species richness and abundance with elevation, especially in less disturbed zones. Gevaña and Pampolina (2017) conducted broadleaf vegetation and carbon stock assessments on Mt. Makiling's slopes, delineating three elevation zones (50 - 300 m, 301 - 700 m, and 701 - 1114 m asl) and found notable differences in species composition and structural forest attributes among these zones. Similarly, Aureo et al. (2021) studied floristic composition along the elevation gradient of Balinsasayao Twin Lakes Natural Park. They reported that species diversity increases from the lowland to submontane zones, then declines toward higher montane elevations. These findings provide local, empirical support for defining forest-level classes in this study and justify the use of distinct elevational strata to capture ecological variation in forest structure and health.

Many ecological studies in the Philippine mountains demonstrate that vegetation structure varies significantly with elevation, supporting the use of

forest level classifications such as Lowland, Lower Montane, Upper Montane, and Mossy. For example, Buot and Okitsu (1998) examined tree height, species composition, and canopy density across altitudinal zones in Mt. Pulog, where they identified clear vertical forest stratification. Their classification was based on temperature gradients, elevation ranges, and distinct ecological transitions observed at specific altitudes, principles consistent with environmental zonation in other Philippine mountain systems, such as Mt. Makiling. The upper montane and mossy forests in their study were characterized by shorter, stunted trees and dense cloud cover. In contrast, lower montane forests contained taller, richer canopies similar to those observed in Makiling's mid-elevation zones. These findings provide the ecological foundation for the "forest level" variable in this study, validating the use of elevation-based categories to represent natural variations in vegetation structure and composition.

Beyond NDVI analysis, researchers have also created integrated geospatial systems for Mt. Makiling. Bantayan et al. (2019) developed a geodatabase that combines land cover, elevation, soil, and biodiversity data to support restoration and conservation initiatives. This work demonstrates how merging ecological and spatial datasets can guide reforestation strategies and biodiversity protection within a unified framework.

Machine learning has further advanced these approaches by boosting classification accuracy and automating complex forest assessments. Models such as Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) can detect subtle differences in forest type and condition, particularly in tropical landscapes. Applying these methods in Mt. Makiling enables researchers to

refine forest-type classification, zonation, and health monitoring, thereby complementing existing geospatial and ecological studies.

Overall, Mt. Makiling illustrates the potential of advanced classification methods to support both academic research and practical forest management. Its rich biodiversity, strong history of ecological research, and conservation status make it an ideal site to showcase the integration of remote sensing, GIS, and deep learning. These combined approaches not only improve the precision of forest mapping but also provide a stronger foundation for targeted restoration and long-term sustainability.

Reforestation Detection and Tree Recommendation Systems

Reforestation is widely regarded as one of the most effective ways to address forest degradation and biodiversity loss in the Philippines. Modern approaches, however, go beyond simply replanting trees and focus on restoring ecological integrity by prioritizing native species adapted to specific site conditions. To guide this process, many systems integrate monitoring tools that can detect deforestation and recommend appropriate restoration strategies. In the Philippine setting, the Department of Environment and Natural Resources (DENR) has issued planting guides for different forest zones. At the same time, non-governmental groups like the Haribon Foundation created the Rainforestation Guide, which promotes the use of ecologically appropriate native trees. These resources provide broad national frameworks that can serve as foundations for more localized recommendation systems.

The need for localized references is evident in the Mt. Makiling Forest Reserve (MMFR), which serves as both a protected area and a natural laboratory for ecological research. Studies in Makiling reveal the richness of native tree diversity, showing that restoration should not rely solely on national lists but must incorporate site-specific biodiversity information. Malabrigo et al. (2016), for instance, documented over 150 tree species—including endemic and threatened species—within a permanent biodiversity monitoring plot in Mt. Makiling. Their findings not only highlight the site's ecological richness but also describe its forest structure, which is crucial for identifying suitable species for restoration. Using such data ensures that reforestation efforts prioritize species already thriving under Makiling's ecological conditions.

Building on this, studies of Mt. Makiling's natural vegetation highlight its potential as a direct source of planting materials. They showed that canopy dominants such as Swietenia macrophylla and Diplodiscus paniculatus, along with key understory species, can be used for restoration. The study also emphasized that species composition changes with elevation, underscoring the need to match species to site-specific gradients. This perspective strengthens the logic of a rule-based recommendation system, ensuring that degraded lowland forests are planted with lowland-adapted species. At the same time, montane areas receive trees suited to higher altitudes.

Combining national and local references creates a stronger scientific basis for species recommendation systems. National guides like those from DENR and Haribon provide broad ecological guidance, while Makiling-specific studies provide fine-scale details on diversity, structure, and vegetation

patterns. This dual approach ensures that recommendations are both scientifically sound and contextually relevant. For example, Narra (Pterocarpus indicus) and White Lauan (Shorea contorta) are consistently recommended in both national and local sources. Still, Makiling-specific studies also highlight less-cited native species that are critical for ecological balance and biodiversity.

The study's simplified land cover classes: Forested and Non-Forest, are consistent with the class structure used in ESA WorldCover, a 10-meter global land cover product derived from Sentinel-1 and Sentinel-2 data. WorldCover categorizes land into Tree Cover, Built-up, and several non-forest vegetation types, all of which can be aggregated into the high-level groups adopted in this research (Zanaga et al., 2021). This alignment ensures that the study's class definitions follow internationally recognized land cover standards explicitly designed for high-resolution environmental monitoring. Furthermore, label aggregation follows the same principles as ESA CCI's Land Cover Classification System, in which classes are merged to increase spectral separability and mapping accuracy. Using WorldCover as a conceptual reference strengthens the validity of the study's simplified labeling scheme.

Determining the appropriate number of samples or patches is a critical component of training deep learning models for land cover mapping. A meta-analysis of remote-sensing deep learning applications reported that most studies use hundreds to several thousand samples per class, with a median of approximately 242 annotated samples per class, highlighting the importance of sufficient class representation for model stability (Ma et al., 2019). Patch-based semantic segmentation experiments further show that accuracy improves as

the number of representative patches increases, especially when patches are sampled across spatial and spectral variability within the study area (Maggiori et al., 2017). These findings support the use of stratified random sampling, which ensures that samples are distributed proportionally and capture the heterogeneity of different land-cover categories (Olofsson et al., 2014). Additionally, empirical sampling heuristics suggest that selecting approximately 4% of the total available patches can be an effective training fraction for large-area mapping tasks when stratified by class and geographic region (Reese et al., 2018). Together, these studies establish that deep-learning land-cover models require well-distributed sample sets, typically in the hundreds per class or proportional fractions of the study area, and that stratified sampling yields more reliable and generalizable model performance.

A rule-based system built on this integrated knowledge addresses the shortcomings of purely generic recommendations. Instead of relying solely on broad forest-type categories, it fine-tunes species suggestions to match those historically and ecologically suited to Makiling. This increases tree survival, accelerates forest recovery, and ensures restored areas align with the reserve's natural biodiversity. Moreover, drawing on Makiling's research adds credibility, showing that recommendations are backed by both authoritative national frameworks and localized ecological studies.

Ultimately, this approach positions Mt. Makiling as a model for reforestation detection and recommendation systems in the Philippines. By combining national guidelines with site-specific biodiversity studies, it moves beyond generic models and demonstrates how ecological knowledge can be

transformed into practical restoration tools. This not only allows the detection of degraded areas but also ensures that species recommended for replanting are ecologically appropriate, socially relevant, and contextually aligned. As a result, reforestation can contribute more effectively to forest resilience and long-term biodiversity conservation.

Synopsis of the Review of Related Literature

The review of related literature points to the growing reliance on remote sensing, vegetation indices, and machine learning for forest monitoring and restoration. Tools like NDVI have been widely used to evaluate canopy health, detect degraded areas, and track forest regrowth. Researchers have also developed threshold values that separate healthy from degraded vegetation, giving a quantitative basis for analysis. These indices continue to serve as reliable measures of ecological conditions across forest landscapes. For this reason, they are considered vital for developing automated systems to detect and monitor forest changes.

Recent advances in deep learning have improved vegetation classification by enhancing the automated detection of forest types and health conditions. Convolutional Neural Networks (CNNs) are effective at identifying local details such as canopy structure and texture, while Vision Transformers (ViTs) capture larger spatial relationships. When applied together, they show strong potential for interpreting tropical forest imagery, where spatial complexity often limits the effectiveness of traditional classification methods. These approaches demonstrate how artificial intelligence can provide both accuracy

and flexibility in ecological monitoring. With these capabilities, they have become key tools for addressing large-scale environmental challenges.

Studies in the Mt. Makiling Forest Reserve (MMFR) illustrate how these methods can be applied in localized settings. Research on forest mapping in Makiling shows that combining satellite data with ecological information produces a more complete understanding of forest dynamics. Biodiversity studies also highlight the wide variety of native tree species, stressing the need for site-specific references. These findings strengthen the rationale for using Makiling as a model site for classification and recommendation systems. This ensures that restoration practices remain consistent with the area's ecological conditions.

Reforestation studies further underline the importance of using ecologically appropriate tree recommendation systems. National frameworks from the DENR and Haribon Foundation provide general planting guidelines that support large-scale reforestation programs. However, Makiling-specific studies contribute additional detail by identifying species that naturally thrive in its environment. Combining both national references and local biodiversity data ensures that tree recommendations are scientifically sound and ecologically relevant. This dual foundation enhances the potential success of reforestation and restoration initiatives.

Overall, the reviewed literature establishes a clear foundation for the present study. Remote sensing and vegetation indices provide practical tools for detecting forest conditions. Deep learning enhances classification accuracy and addresses the complexity of tropical ecosystems. Ecological references,

particularly from Makiling, guide the recommendation of native tree species suited to the area. Together, these elements justify the development of an integrated forest classification and recommendation system.

Literature Matrix

Table 1. Literature Matrix

Author	Title	Methodology	Key Findings
Bhandari et. al. (2012)	Feature extraction using Normalized Difference Vegetation Index (NDVI): A case study of Jabalpul City	NDVI threshold analysis	Combining NDVI with other indices improves classification in mixed land use areas.
Dosoyitskiy et al. (2020)	An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale	Transformer-based image classification	ViTs are effective for capturing long-range spatial relationships in imagery.
Hansen et al. (2013)	High-resolution global maps of 21st-century forest cover change	High-resolution Landsat maps	Mapped annual forest cover loss and gain globally with high resolution.
Huete (1988, 2002)	A soil-adjusted vegetation index (SAVI) and overview of MODIS vegetation indices	Spectral index development	SAVI reduces soil interference; EVI improves vegetation analysis in dense canopies.
Israel et al. (2024)	Deforestation and forest degradation analysis of Southern Sierra Madre using GEE and community mapping	NDVI differencing using Google Earth Engine	GEE and NDVI are effective for detecting forest loss in Sierra Madre
Kussul et al. (2017)	Deep Learning Classification of Land Cover and Crop Types Using Remote Sensing Data	Multi-source deep learning (Landsat-8, Sentinel-1A)	CNNs outperformed RF and MLP, achieving >85%

			accuracy in crop classification
Lehouel et al. (2024)	Hybrid deep learning models for NDVI-based vegetation classification	CNN-ViT fusion model	CNN-ViT hybrid improves accuracy to 90.4%, surpassing standalone ViTs
Ollinger et al. (2008)	Long-Term NDVI Datasets for Detecting Ecosystem Functional Changes	Long-term NDVI time series	NDVI reliably tracks long-term ecosystem change
Palma et al. (2020)	GIS-based species recommendation tool for Philippine reforestation	GIS-based tool	GIS-based tool recommends native trees based on site characteristics.
Pettorelli et al. (2005)	Using the satellite-derived NDVI to assess ecological responses to environmental change differencing and classification methods	Time-series NDVI analysis	NDVI can detect seasonal and ecological vegetation changes.
Roy et al. (2020)	Landsat-based science vision for global land cover monitoring with AI	Multitemporal Landsat analysis	AI and multitemporal satellite data improve land cover classification.
Saatchi et al. (2011)	Benchmark map of forest carbon stocks in tropical regions across three continents	Global biomass estimation	Used satellite data to map carbon stocks in tropical forests.
Sahebjalal & Dashtekian (2013)	Analysis of NDVI to assess land cover change in a semi-arid region	NDVI thresholding	NDVI is effective for distinguishing vegetated and degraded land
Shao et al. (2022)	Dual-branch CNN–Transformer networks for remote sensing scene classification	Dual-branch deep learning	Dual-branch models combining CNN and ViT improve accuracy and robustness.

Strassburg et al. (2020)	Global priority areas for ecosystem restoration	Spatial ecosystem modeling	Spatial modeling helps prioritize cost-effective reforestation zones.
Tucker (1979)	Red and photographic infrared linear combinations for monitoring vegetation	Remote sensing	NDVI is a simple but effective way to detect vegetation health using satellite imagery.
Uso et al. (2020)	GIS-Based Land Suitability Analysis for Solar-Powered Irrigation in Davao	Used NAMRIA land cover maps for GIS-based land analysis	NAMRIA data is accurate for supervised land classification
Vina et al. (2011)	Comparison of vegetation indices for estimating green leaf area index of crops	Index comparison	GNDVI offers improved performance over NDVI in detecting canopy stress.
Wu et al. (2021)	Deep learning and vegetation index fusion for forest degradation detection in tropical areas	Deep learning + index fusion	Combining NDVI with deep learning improves forest degradation detection.

Conceptual Framework

This study integrates multi-temporal satellite imagery, vegetation indices, elevation data, and a hybrid deep learning model to classify forest conditions and support reforestation planning in Mt. Makiling. The framework begins with the preparation of core geospatial datasets: Sentinel-2 imagery (2019–2021), NDVI and NDVI-change layers for assessing forest health, SRTM-derived elevation for classifying forest level, and ESA WorldCover maps for identifying forest type. These datasets are preprocessed through cloud masking, clipping, and band calculations to generate three label categories:

Forest Type (Forested, Others), Forest Level (Lowland, Lower Montane, Upper Montane), and Forest Health (Degraded, Neutral, Regrowth).

A six-band input composite (RGB, NDVI, EVI, NDWI) is created from the 2020 and 2021 imagery to enrich spectral and vegetation information. The 2020 composite will be the focus for model training. Using stratified sampling, 64×64 patches are extracted with a ≥5% rare-class threshold to maintain representation of minority classes. The dataset is split into training, validation, and an internal (2020) test set. An external 2021 test set is generated from the exact patch coordinates, enabling temporal generalization testing.

Minority classes are augmented, while dominant classes remain unaltered to preserve the natural spectral structure. The study will train three models: a standalone CNN, a standalone ViT, and a Hybrid CNN–ViT model. Outputs are produced through three parallel segmentation heads corresponding to forest type, level, and health. Training uses class-weighted loss, AdamW optimization, checkpointing, and early stopping.

Performance is evaluated on both internal (2020) and external (2021) test sets using accuracy, precision, recall, macro F1-score, macro IoU, and confusion matrices. External evaluation provides insight into model stability across different years.

Finally, degraded areas identified by the Health head are grouped into spatial clusters. Their dominant predicted forest type and elevation level are matched with a rule-based tree species recommendation system based on DENR, Haribon, and reviewed studies. This results in site-appropriate native species suggestions that support ecological restoration planning.

The framework produces forest condition maps, temporal comparison results, and recommended species for each degraded region, forming a decision-support system for forest monitoring and reforestation in the Mt. Makiling landscape.

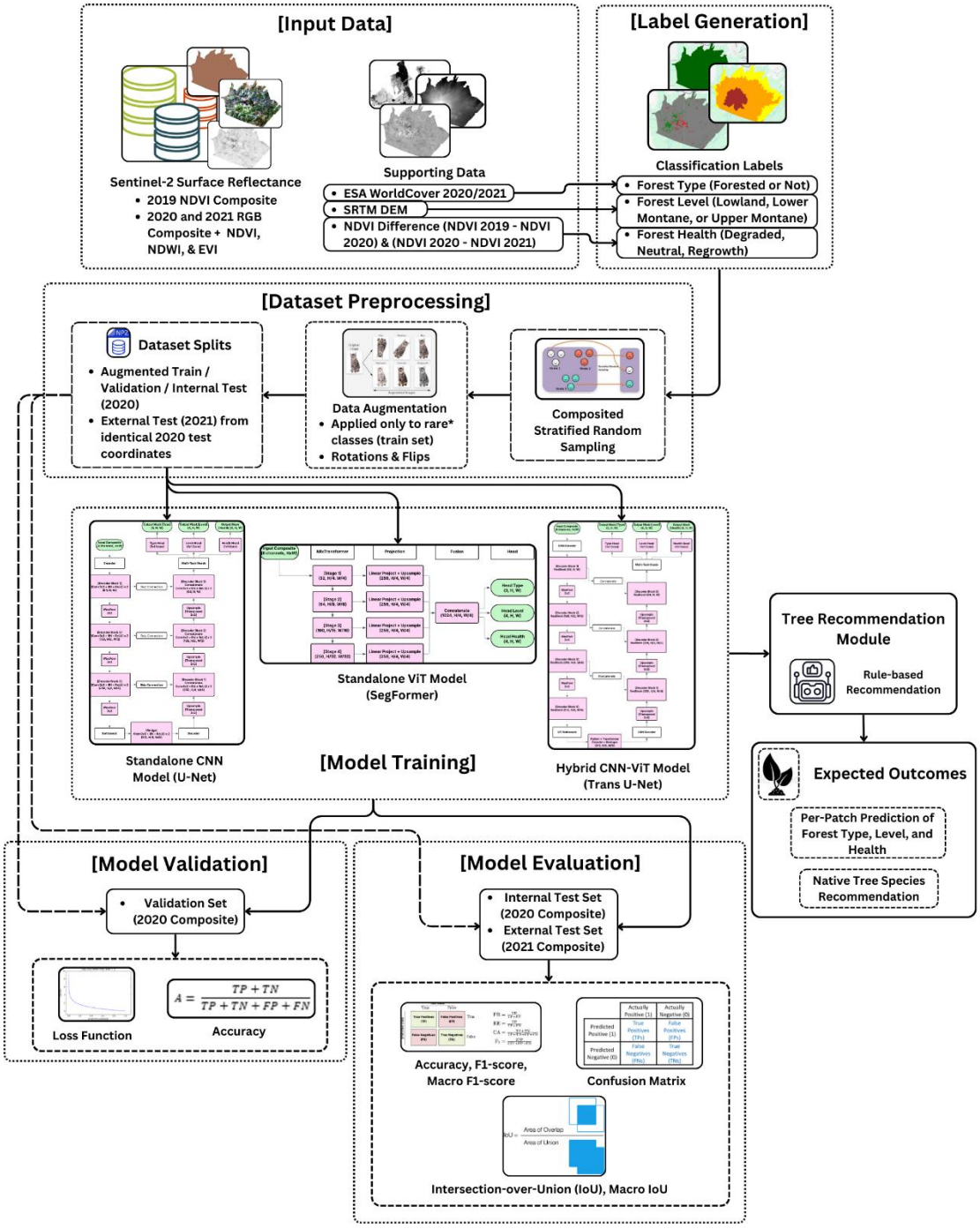


Figure 1. Conceptual Framework

CHAPTER 3

RESEARCH METHODOLOGY

The study integrated multi-temporal remote sensing analysis for ground-truth label generation, a comparative analysis of three Multi-Task Learning (MTL) architectures: a baseline CNN (U-Net), a Transformer (SegFormer), and a Hybrid CNN–ViT (TransU-Net), and a rule-based module for native tree species recommendation. The methodology emphasized computational optimization through binary data serialization and systematic evaluation using pixel-wise diagnosis.

Study Area

This study covered the Mt. Makiling Forest Reserve (MMFR), which is a protected landscape in Southern Luzon, Philippines. The reserve spans approximately 4,244 hectares, with terrain that transitions from lowland dipterocarp forest at the base to montane forest near the summit. For this study, the official boundary of Mt. Makiling was obtained from the World Database on Protected Areas (WDPA) and adopted as the Area of Interest (AOI), which was used for all other data processing.



Figure 2. Visualization of the Study Area

Research Design

The study utilized a quantitative experimental design focused on semantic segmentation. The workflow was divided into four primary phases: Data Collection and Preprocessing, Dataset Generation and Stratification, Model Architecture Development and Training, and Performance Evaluation and Application.

A key feature of the design was the Dual-Scale Experiment, where all models were trained and tested on two distinct patch sizes to evaluate the trade-off between local texture and global context:

- Experiment A (64×64 pixels): Covers approximately 40.96 hectares per patch (at 10m resolution). Focuses on high-frequency details, such as canopy texture and edges.
- Experiment B (128×128 pixels): Covers approximately 163.84 hectares per patch. Focuses on broader spatial context and elevation gradients.

Theoretical Foundation

This study was based on two main ideas: spectral vegetation analysis, which explained how NDVI values can indicate vegetation health and abundance by analyzing how plants reflect light, and computer vision models. Convolutional Neural Networks (CNNs) were effective at finding local patterns in images, while Vision Transformers (ViTs) were effective at analyzing long-range relationships. By combining them, the CNN–ViT model utilized the strengths of both approaches. For the recommendation module, the study was

guided by principles of ecological restoration, especially the idea of matching the right native tree species to the appropriate forest type and elevation.

Materials and Data Acquisition

This study used publicly available remote sensing datasets for forest monitoring, ecological zoning, and vegetation condition assessment. Forest health labeling required multi-year NDVI inputs; forest level labeling required reliable elevation data; and forest type labeling required authoritative land-cover references. The integration of these products supported the study's multi-label semantic segmentation framework. To ensure the model learned vegetative health properties beyond the visible spectrum, a 6-Channel Composite input tensor was constructed.

Table 2. Data Types and Sources

Data Type	Source/Tool	Notes
Satellite Imagery	Sentinel-2 Surface Reflectance (2019-2021) via GEE	2019 year-round NDVI Composite was used to compute Forest Health; 2020 Composite was used for Model Training; 2021 Composite were used for independent Model Testing
Forest Level Labels (Elevation Data)	SRTM DEM (30m) via GEE	Used to generate forest level labels for all dataset
Forest Type Labels (Land Cover Classification)	ESA WorldCover 2020 / 2021	Used to generate forest type labels for testing dataset
Shapefile Boundary	World Database on Protected Areas (WPDA) – Mount Makiling Forest Reserve (MMFR)	Defines the study area (AOI)
NDVI Composites	Google Earth Engine	Computed median NDVI per year
NDVI Difference	GEE (NDVI of Prior Year – NDVI of Current Year)	Label for forest health, whether training or for testing

To ensure high-quality spectral inputs, the study utilized Google Earth Engine (GEE) to process the full archive of Sentinel-2 Level-2A (Surface Reflectance) imagery intersecting the Mt. Makiling AOI. A total of 245 individual satellite scenes were processed to generate the annual maps used in the study.

The preprocessing pipeline implemented a rigorous pixel-level masking algorithm using the Scene Classification Layer (SCL) band. Pixels classified as Dark Area (Class 2), Cloud Shadows (Class 3), Clouds (Classes 8, 9), and Cirrus (Class 10) were removed from every image in the stack. To maximize data availability across the tropical study area, the initial scene-level cloud filter was set to <60%.

Two distinct compositing strategies were applied based on the dataset's role:

- 2019 Baseline (NDVI Only): A Median Composite was generated to provide a stable, noise-reduced representation of the vegetation state prior to the monitoring period.
- 2020 and 2021 (Training, Validation, and Testing): A Quality Mosaic strategy was implemented. The image collection was sorted by cloud cover percentage in ascending order, and the mosaic reducer was applied. This prioritized pixels from the most explicit available scenes, preserving the sharp spatial details and texture required for the Deep Learning model to classify forest structure accurately.

Table 3. Summary of Sentinel-2 Satellite Imagery Processing

Year	Composite Method	Bands Generated	Cloud Filter	Total Scenes Processed
2019	Median Reducer (Stable Baseline)	NDVI Only		93
2020	Quality Mosaic (Clearest Pixel)	RGB + NDVI + EVI + NDWI	< 60%	86
2021	Quality Mosaic (Clearest Pixel)	RGB + NDVI + EVI + NDWI		66
			Total:	245

To understand the data distribution before normalization, a statistical analysis of the full Mount Makiling Forest Reserve (MMFR) scene was conducted using the 2020 Sentinel-2 composite. The raw satellite imagery covers a spatial extent of 990 × 762 pixels. This results in a total available pixel count of 754,380 pixels across the study area. With the inclusion of the six (6)

spectral channels (RGB, NIR, EVI, NDWI), the model processes approximately 4.5 million data points per full-scene inference.

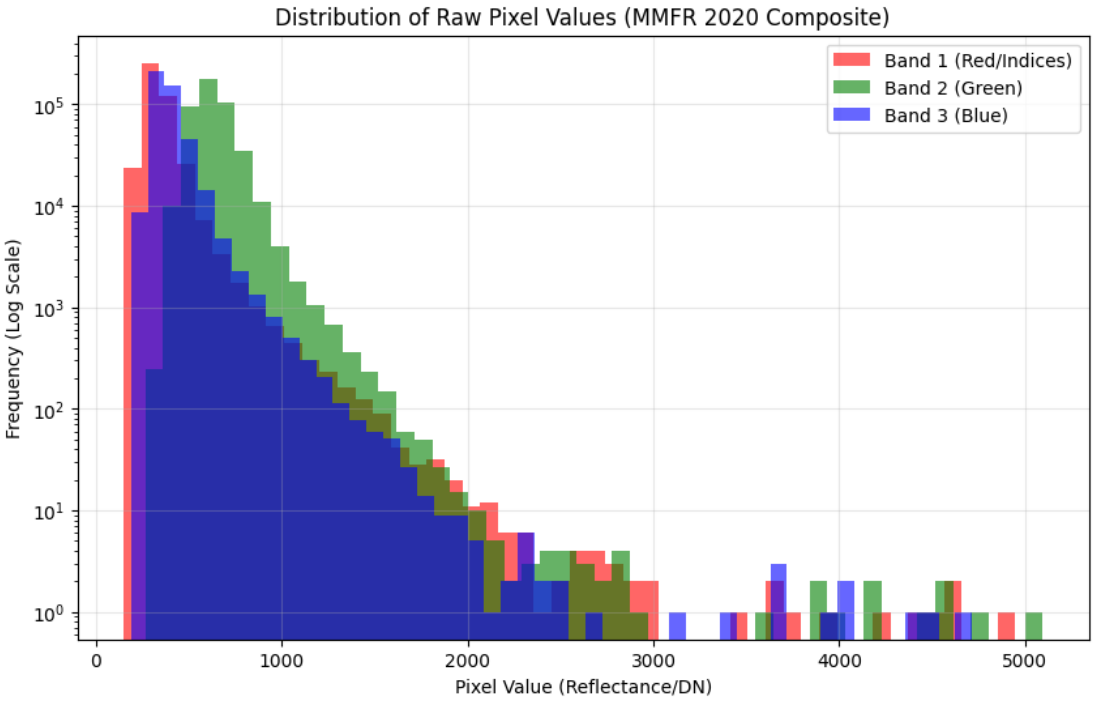


Figure 3. Histogram of Raw Pixel Distributions for 2020 Composite

Software and Tools

To effectively manage spatial data, train deep learning models, and implement post-processing tasks, a combination of geographic information systems and machine learning libraries was used. These tools enabled the extraction of NDVI features, spatial mapping, model training, and classification evaluation.

Table 4. Software and Tools Used

Tool	Purpose
QGIS	Used to manually and visually check the GeoTIFF files used in the study
Google Earth Engine (GEE)	Imagery download, NDVI calculation, DEM extraction, cloud masking, label generation, also for visualizations
Google Colaboratory (using T4 GPU)	Served as the main coding and model training environment, providing GPU acceleration and seamless integration with Google Drive for managing datasets, checkpoints, and output visualizations
Python + OpenCV	Aligned RGB and label maps, performed patch extraction and data augmentation
TensorFlow / PyTorch	Trained the CNN–ViT hybrid segmentation model
Scikit-learn, NumPy, Pandas	Data preprocessing, model evaluation, and statistical analysis

Spectral Indices Computation and Preprocessing

In addition to the standard Red, Green, and Blue (RGB) bands, three spectral indices were computed and stacked into the input tensors to enhance the model's ability to distinguish vegetation condition, structure, and moisture content. Spectral indices derived from multispectral imagery are widely used because they amplify biophysical properties of vegetation that raw reflectance values alone cannot fully capture. They improve model interpretability and classification performance by highlighting canopy vigor, water status, and biochemical responses to stress (Xue & Su, 2017).

Normalized Difference Vegetation Index (NDVI). NDVI was calculated using the formula:

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

NDVI is one of the most widely used remote sensing indices for vegetation monitoring. It exploits the strong absorption of red light by chlorophyll and the high reflectance of near-infrared light by plant cell structure (Rouse et al., 1974). High NDVI values correspond to healthy, dense vegetation, while lower values indicate degraded or sparse cover. NDVI has been proven effective for forest condition assessment, land cover mapping, and vegetation change detection, making it essential for modeling forest type, condition, and health in Mt. Makiling.

Enhanced Vegetation Index (EVI). EVI was calculated using the formula:

$$EVI = 2.5\ x\ \frac{NIR - Red}{NIR + 6\ x\ Red - 7.5\ X\ Blue + 1}$$

EVI was developed to address NDVI's limitations, particularly its tendency to saturate in high-biomass tropical forests (Huete et al., 2002). It improves sensitivity in dense vegetation by incorporating the Blue band to correct for atmospheric scattering and soil background effects. Because Mt. Makiling includes dense forest zones, EVI helps the model differentiate subtle canopy variations and is more robust in areas with high biomass, making it valuable for assessing forest health and vegetation structure.

Normalized Difference Water Index (NDWI). NDWI was calculated using the formula:

$$NDWI = \frac{Green - NIR}{Green + NIR}$$

NDWI enhances the detection of vegetation water content by exploiting differences in reflectance between the Green and NIR wavelengths (Gao, 1996). It is beneficial for detecting water stress, which can be linked to forest degradation, disturbance, or regrowth dynamics. Including NDWI enables the model to detect better moisture-related changes in vegetation health, an essential factor for distinguishing degraded from healthy forest scenes.

All indices were derived from Sentinel-2 Level-2A surface reflectance data, specifically Bands 2 (Blue), 3 (Green), 4 (Red), and 8 (NIR). Sentinel-2's 10-meter spatial resolution and high spectral richness make it well-suited for vegetation analysis (Drusch et al., 2012). Level-2A data processed using atmospheric correction ensures radiometric consistency across scenes, which is critical for year-to-year NDVI–EVI–NDWI comparisons and reliable forest health mapping. Input tensors were dynamically normalized. Raw pixel values (0-255) were scaled to the [0, 1] floating-point range during the data loading stage to ensure compatibility with the pre-trained weights of the SegFormer and ResNet backbones.

Label Generation from Remote Sensing Indices

To enable supervised learning for forest classification, three distinct label maps were prepared: forest type, forest level, and forest health. Each represents a critical ecological attribute for monitoring forest dynamics and guiding reforestation planning. These labels were generated from established geospatial datasets and standardized through thresholding and class remapping to ensure consistency.

For forest type (Forested or Non-Forest) classification was based on the ESA WorldCover Map with the following remapping:

- Forested: Class 10 (Tree Cover)
- Non-Forest: All other land cover classes, including Built-Up, Shrubland, Cropland, etc.

This simplified scheme eliminates agricultural tree-cover confusion and focuses on structural land cover distinctions relevant to ecological analysis.

The ESA WorldCover dataset is a high-resolution global land-cover product produced by the European Space Agency using Sentinel-1 and Sentinel-2 satellite imagery at 10-meter spatial resolution. It provides an updated and standardized classification of Earth's surface into eleven land-cover classes, including tree cover, shrubland, grassland, water bodies, cropland, and built-up areas. WorldCover is widely used in environmental research because of its accuracy, consistent methodology, and global coverage, making it suitable for analyses requiring reliable land-cover labels. For this study, the WorldCover classification was adopted to generate forest type labels, ensuring that the model was trained on a robust and internationally recognized land-cover standard. This dataset serves as the foundational reference for mapping forest conditions within the Mt. Makiling Forest Reserve.

Table 5. Land Cover Classes from ESA WorldCover

ID	Description
10	Tree Cover
20	Shrubland
30	Grassland
40	Cropland
50	Built-up
60	Bare / Sparse Vegetation
70	Snow and Ice
80	Permanent Water Bodies
90	Herbaceous Wetland
95	Mangroves
100	Moss & Lichen

The forest level labels follow the elevation zones identified by Gevaña & Pampolina (2017) for Mt. Makiling, with three main classes:

- Lowland (Low elevation): < 300 meters above sea level (m asl)
- Lower Montane (Middle elevation): 301-700 m asl
- Upper Montane (High elevation): 701-1114 m asl

These thresholds were applied to the Shuttle Radar Topography Mission (SRTM) - Digital Elevation Model (DEM) and masked to include only forested pixels. An uppermost ‘mossy’ forest belt (≥1000 m asl) was computed as an ancillary layer; however, because mossy forest occupies a very small fraction of the study area (<1.1% of total forest area), the primary analysis used the three-level scheme. This approach preserves local ecological relevance while avoiding severely under-represented classes that could destabilize model training.

```
Mossy forest area (sq meters):
410741.60288238525

Total forest area (sq meters):
37989231.862225845

Mossy forest as % of total forest:
1.0812053383232565
```

Figure 4. Distribution of Mossy Forest Area Using GEE

For forest health labels, it was derived from temporal NDVI differencing, following precedents from Philippine watershed studies (Israel et al., 2024) and typhoon/disturbance mapping research (Magallon & Tsuyuki, 2024). The NDVI difference ($\Delta NDVI = NDVI_{curr} - NDVI_{prev}$) was computed. To classify vegetation health, thresholding was applied using values supported by NDVI change-detection literature. Studies have shown that NDVI decline greater than –0.065 to –0.10 is indicative of forest disturbance or degradation (Stroppiana et al., 2020), while positive NDVI changes in the +0.10 to +0.20 range often signal greening or canopy recovery (Walz & Weber, 2021) because typical Sentinel-2 NDVI noise is within ±0.03–0.05, a threshold of ±0.10 lies safely above the noise range and captures only meaningful vegetation change.

Table 6. NDVI Difference Threshold

NDVI Difference	Label	Class
< -0.10	Degraded	1
-0.10 to 0.10	Neutral	2
> 0.10	Regrowth	3

In summary, the forest type labels were derived from ESA WorldCover. Classes such as Class 10 (Tree Cover) were remapped into Forested, while other classes were remapped into Non-Forest. Forest level labels were derived

from SRTM-DEM, which were thresholded into Lowland, Lower Montane, and Upper Montane. Lastly, forest health labels: Degraded, Neutral, and Regrowth, were derived from the computed and thresholded NDVI differences.

Table 7. Labels for Model Training

Label Type	Source	Method
Forest Type	ESA WorldCover 2020/2021	Remapped classes: Forested and Non-Forest
Forest Level	DEM (SRTM)	Altitudinal classes: Lowland, Lower Montane, Upper Montane
Forest Health	NDVI Difference	Thresholded into degraded, neutral, and regrowth

Patch Extraction and Label Association

To prepare the inputs for training the segmentation model, the multispectral imagery and label data underwent a Composite Stratified Random Sampling procedure. This sampling framework simultaneously accounts for Forest Type, Forest Level, and Forest Health labels, ensuring that all combinations of classes are systematically represented in the training, validation, and testing sets.

This method operates by generating a unique stratum for every possible combination of the three label maps. A Stratum ID was assigned to each pixel based on the following class groupings:

- Forest Type (2 classes): Forest, Non-Forest
- Forest Level (3 classes): Lowland, Lower Montane, Upper Montane
- Forest Health (3 classes): Degraded, Neutral, Regrowth

This produces a total of 18 unique potential strata.

This stratification ensures that the sampling process recognizes and extracts data from all ecologically meaningful intersections, for example, *“Forest + Upper Montane + Degraded”*, which may be visually subtle but critically important for model learning.

The sampling design is grounded on remote sensing accuracy assessment standards. Congalton and Green (2019) emphasize the need for at least 50–75 samples per class to achieve statistically valid and balanced accuracy reporting. By ensuring adequate representation of each stratum, the model avoids bias toward more common conditions, such as (*Forested + Lowland + Neutral*), and it gains sensitivity to rarer combinations.

For this study, the complete set of existing strata was first identified from the label maps. A constant value of N = 500 was then defined as the maximum number of sampling points to be collected. From within each active stratum, the sampling algorithm attempted to identify a balanced number of pixels until either the stratum quota was met or the global cap of 500 points was reached. Each of these sampled points was subsequently used as the center pixel for patch extraction.

To visualize this coverage, Figure 5 illustrates the representative spatial distribution of the sampling points. As shown, the stratified approach ensures that training data is not clustered in a single region but is distributed across the varied elevation gradients and land cover types of the reserve.

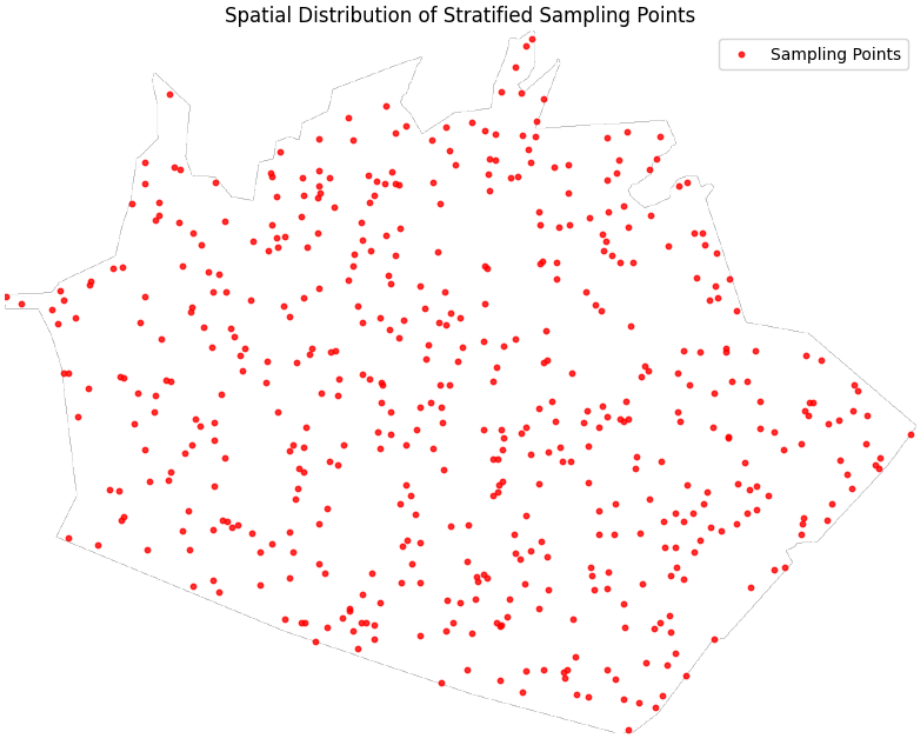


Figure 5. Representative Spatial Distribution of Stratified Sampling Points

A distinct advantage of the Stratified Random Sampling method is its efficiency compared to a brute-force sliding window approach. A calculation was performed to determine the theoretical maximum number of extractable patches if the entire MMFR scene were scanned pixel-by-pixel. This calculation utilizes the formula: $(W - P + 1)\ x\ (H - P + 1)$, which accounts for the boundary limits to ensure the sliding window remains strictly within the image extent.

As detailed in Table 8, a full sliding-window scan would yield over 600,000 potential patches. However, processing this volume would introduce significant spatial redundancy. By using stratified sampling, the study reduced this massive theoretical volume to a curated dataset of 4,438 unique sampling points, ensuring ecological representativeness without computational waste.

Two patch sizes were then extracted: 64×64 and 128×128 pixels. The use of two resolutions supports the differences in spatial learning behavior between the CNN and ViT components of the hybrid model:

- 64×64 patches favor convolutional feature extraction by focusing on small-scale textures, canopy patterns, and local gradients.
- 128×128 patches provide the Vision Transformer with broader spatial context, enabling it to capture structural forest patterns, elevation transitions, and larger ecological gradients.

Using both patch sizes improves generalization by exposing the model to multiple spatial scales during training.

Table 8. Theoretical Maximum vs. Actual Sampled Patches

Patch Dimension	Extraction Method	Calculation	Total Patches
64 x 64	Theoretical Full Scan	(990 − 63) × (762 − 63)	647,973
64 x 64	Stratified Sampling	Curated Dataset	4,438
128 x 128	Theoretical Full Scan	(990 − 127) × (762 − 127)	548,005
128 x 128	Stratified Sampling	Curated Dataset	4,438

To ensure the ecological defensibility of the ground-truth labels, the association criteria were reviewed by a GIS and remote sensing expert from UPLB-CFNR on October 30th and November 21st of 2025. The expert offered critical insights regarding potential label inaccuracies, which became relevant in the model's final evaluation (Chapter 4). Specifically:

1. While the Forest Type label derived from the ESA WorldCover dataset was considered reliable due to its standardized global classification system, the expert noted that its basis on 2020 imagery introduced a potential temporal mismatch when paired with more recent satellite data. To address this, the study used 2020 WorldCover classifications for training and reserved 2021 satellite imagery for testing, ensuring temporal consistency and reducing discrepancies between land-cover labels and input imagery.
2. The Forest Level partition based purely on elevation was considered generally correct but not discretely defined, as actual ecological zonation is highly dependent on other factors like slope, soil, and microclimate.
3. The Forest Health labeling based on a cited NDVI difference threshold was deemed acceptable, provided the source and derivation are clearly documented. These limitations underscore that the model was trained on labels containing inherent temporal and contextual noise, contributing to the poor generalization demonstrated on the unseen test set.

To ensure a sufficient class representation, a minimum of 5% rare class pixel threshold was applied during patch selection. Only the patches where at least 5% of the pixels belonged to a rare class, specifically, Built-Up, Degraded, and Regrowth, were flagged for augmentation in the next phase. Each .npz file contained four synchronized arrays: the 6-channel image tensor and the three corresponding ground-truth label maps (Type, Level, Health), ensuring atomic consistency during data loading.

Table 9. Components of a Single Patch

Component	Description
Input Patch	64×64px or 128x128px patch containing 6 channels: (3) RGB, (1) NDVI, (1) EVI, and (1) NDWI bands from 2020
Forest Type	From remapped ESA WorldCover 2020/2021 (Forested, Non-Forest)
Forest Level	From elevation zones (Lowland, Lower Montane, Upper Montane)
Forest Health	Labeled from NDVI difference (2019-2020, 2020-2021) as Degraded, Neutral, or Regrowth

Data Splitting

Following the patch extraction, the total curated dataset consisted of 4,438 patches per scale. This dataset was then divided into training, validation, and test subsets in a ratio of 70:15:15. This resulted in 3,106 patches for training, 666 patches for validation, and 666 patches for internal testing. The splitting process was conducted prior to any augmentations, ensuring that the model's performance evaluations remain unaffected by having altered samples.

The training and validation patches for the 2020 composite were then used for Model Training and Validation, while the test patches from the 2020 composite were saved and used to derive actual test patches in the composites of 2021 to evaluate temporal generalization, meaning that 666 patches from 2021 were used and allotted for external testing.

Data Augmentation

After the split, augmentation was applied only to the training set to enrich the class balance and prevent overfitting. This was especially important for rare

classes such as Non-Forest in type and Degraded and Regrowth in health, which naturally occur less frequently in the dataset.

The augmentation pipeline significantly increased the training data volume to prevent overfitting and address class imbalance.

- For the 64 x 64 experiment, the training set expanded from 3,106 to 11,771 patches.
- For the 128 x 128 experiment, the training set expanded from 3,106 to 9,986 patches.

Table 10. Augmentation Strategy per Patch Category

Patch Type	Criteria Met	Augmentation Strategy
Rare Class Patches	Contains ≥5% pixels of Non-Forest, Upper Montane, Mossy, or Regrowth	Augmented ×8 with horizontal/vertical flips and 90°/180°/270° rotations.
Normal Patches	No rare classes present	No augmentation applied to preserve class balance and reduce redundancy

Model Architecture and Implementation

This study implemented three distinct deep learning architectures to perform pixel-level classification of forest type, forest level, and forest health: a Convolutional Neural Network (U-Net), a Transformer (SegFormer), and a Hybrid model (TransU-Net). To evaluate the trade-off between capturing fine details and understanding the broader landscape, all models were trained and tested on two different image scales: smaller 64×64 pixel patches and larger 128×128 pixel patches. Each architecture followed a multi-task segmentation

framework, meaning a single model was responsible for simultaneously generating three distinct output maps from one input image.

The Multi-Head U-Net model utilizes a standard encoder-decoder structure with variable input dimensions (6, H, W). The encoder (left) acts like a funnel, progressively downsampling the image to extract essential texture features. The decoder (right) expands these features back to the original image size (H, W) to create the classification maps. Critical "skip connections" bridge the encoder and decoder, allowing the model to retain high-resolution spatial information, such as forest boundaries, that are often lost during downsampling. The final output is split into three parallel branches to predict the Forest Type, Level, and Health masks independently. In this study, the model was trained under two configurations: Experiment A (where H and W are 64) and Experiment B (where H and W are 128).

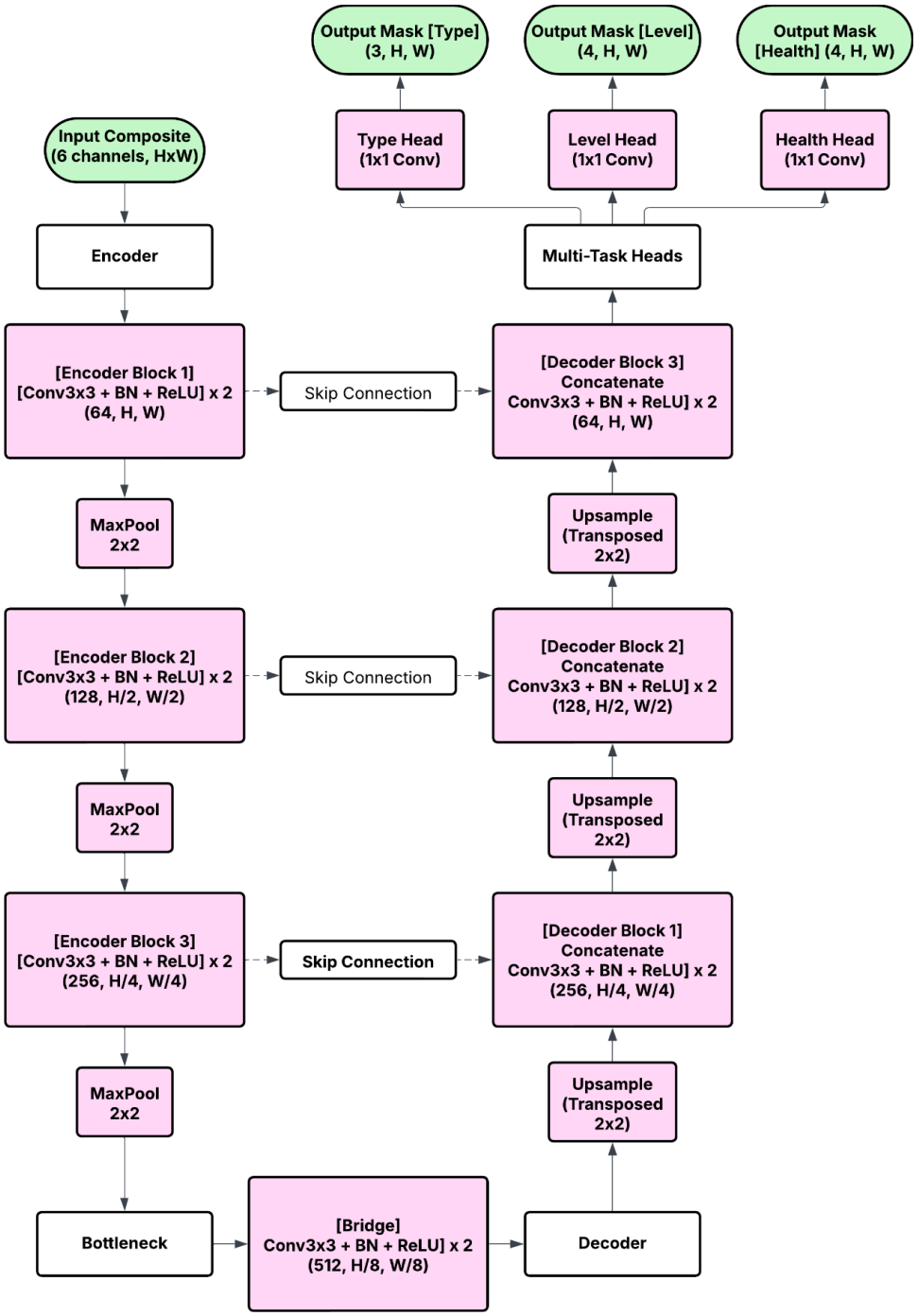


Figure 6. CNN (Multi-Head U-Net) Model Architecture

The Multi-Task SegFormer represents a more advanced approach using Transformer technology. Unlike standard Vision Transformers that split an image into a single sequence of blocks, the SegFormer uses a hierarchical "MixTransformer" structure. This allows the model to analyze features at four

different scales simultaneously, from fine details (1/4 size) to broad context (1/32 size), making it highly efficient. The model's input patch embedding layer was specifically modified to accept the study's 6-channel composite (RGB plus vegetation indices), allowing it to learn from spectral health data directly. A custom MLP (Multi-Layer Perceptron) decoder was implemented to combine the features from all four scales into a unified representation, which is then projected by three separate convolutional heads into the final full-resolution forest maps.

While the original "Vanilla" Vision Transformer (ViT) (Dosovitskiy et al., 2020) pioneered the use of attention in vision, it was deemed unsuitable as a baseline for this study due to its fixed-resolution columnar structure, which often results in coarse segmentation masks and poor boundary adherence. SegFormer was selected as the representative Transformer model because it successfully adapts the Self-Attention mechanism, which is the defining feature of Transformers for dense prediction tasks. Although it incorporates lightweight convolutional operations within its feed-forward networks (Mix-FFN) to eliminate the need for positional embeddings, its primary learning capacity relies on global attention modeling. This distinguishes it fundamentally from the purely convolutional U-Net (Model A) and the serial CNN-ViT Hybrid (Model C), positioning it as the state-of-the-art pure Transformer architecture for efficient semantic segmentation.

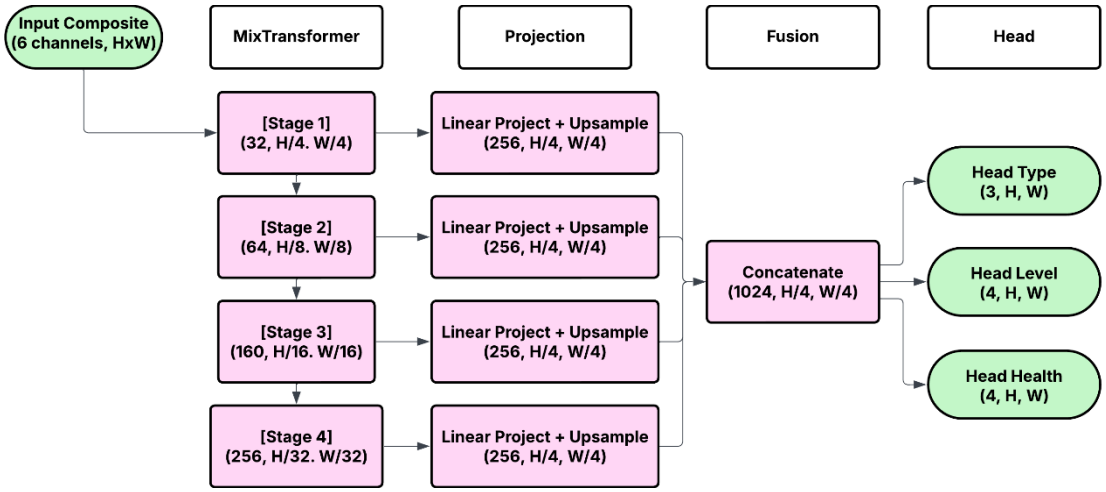


Figure 7. ViT (SegFormer) Model Architecture

Figure 8 illustrates the architecture and data flow of the Hybrid TransU-Net. The model follows a U-shaped trajectory, designed to combine the complementary strengths of Convolutional Neural Networks and Transformers. It operates on the conceptual framework of having a CNN "eye" for detail and a Transformer "brain" for context.

- Stage 1 (Encoder): The model employs a ResNet-style CNN backbone to extract high-resolution local features, such as leaf textures and boundary edges. As the data flows downstream, the image is progressively downsampled; crucial feature maps at each resolution are preserved to serve as spatial references for the reconstruction phase.
- Stage 2 (Bottleneck): The lowest-resolution feature map is passed through a Transformer Encoder. By flattening the features into tokens, this stage allows every pixel to attend to every other pixel via self-attention. This mechanism enables the model to understand the global context. For example, recognizing that a degraded patch is spatially

correlated with a specific elevation zone before reshaping the data back into a spatial grid.

- Stage 3 (Decoder): Finally, the data flows upstream where a decoder reassembles the features into a precise spatial map. It utilizes skip connections to mathematically concatenate the upsampled context features (from the bottleneck) with the preserved local features (from the encoder). This fusion ensures that the final prediction masks retain the sharp boundaries captured by the CNN while benefiting from the semantic understanding of the Transformer.

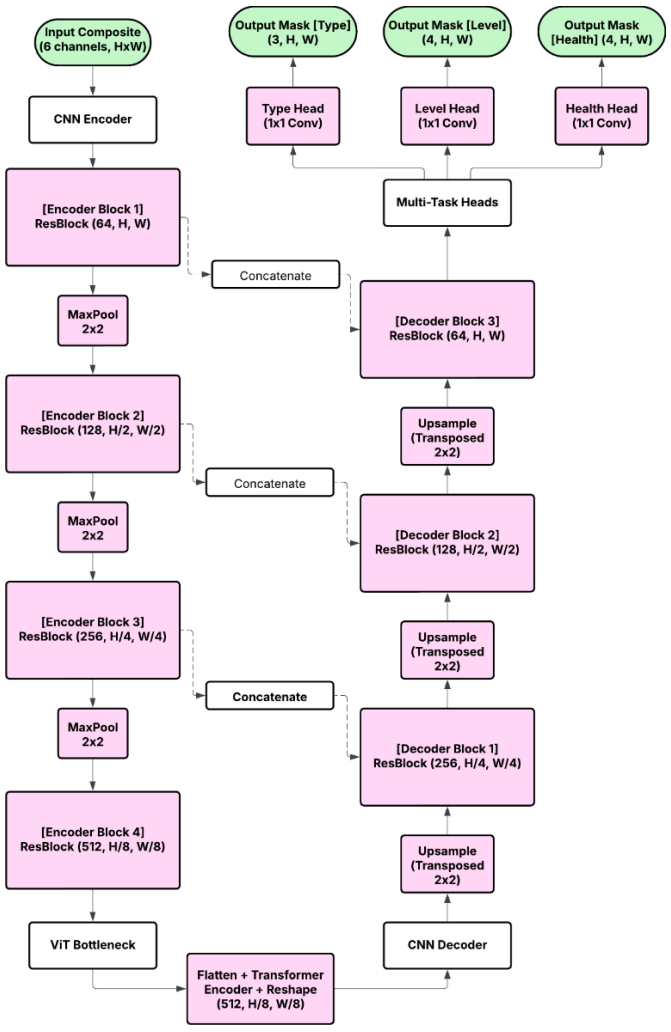


Figure 8. Hybrid CNN-ViT (Trans U-Net) Model Architecture

Hyperparameter Configuration and Training Strategy

To ensure a fair comparison, all models were trained in a standardized environment on Google Colab using a single NVIDIA T4 GPU. Rather than performing an exhaustive automated search for hyperparameters (which is computationally expensive), this study adopted empirical best practices established in recent deep learning literature for semantic segmentation. The configuration prioritized stability and convergence speed.

Key Training Parameters:

- **Batch Size:** Adjusted based on the experiment to maximize GPU memory usage. A batch size of 32 was used for 64×64 patches, while a size of 16 was used for 128×128 patches to prevent memory overflows.
- **Learning Rate:** Set to 1e-4 (0.0001) for the U-Net and 6e-5 for the Transformer-based models, as pre-trained Transformers typically require lower learning rates to preserve their learned features.
- **Epochs:** Models were trained for 25-30 epochs, with an Early Stopping mechanism that halted training if no improvement was observed for five consecutive epochs (Patience = 5).
- **Optimization:** The AdamW optimizer was employed for the SegFormer and Hybrid models to apply weight decay, which helps prevent overfitting in complex architectures. The standard Adam optimizer was used for the baseline U-Net.

- Reproducibility: A fixed random seed of 42 was applied across all libraries (Python, NumPy, PyTorch) and GPU operations. This ensured that the weight initialization and data shuffling sequences were identical across all models, guaranteeing that performance differences were due to architectural capabilities rather than randomness.

Algorithm Details and Loss Function

The training process used Mixed Precision (AMP), a technique that performs calculations using half-precision (16-bit) floating-point numbers instead of the standard 32-bit. This optimization significantly reduced memory consumption and increased training speed without compromising model accuracy.

Learning Rate Scheduler. Unlike standard CNN training, which often uses a fixed learning rate, this study implemented a Cosine Schedule with Warmup. For the first 10% of the training steps, the learning rate gradually increased from 0 to the target value. This "warmup" phase is critical for Transformer-based models (SegFormer and TransU-Net) to stabilize the self-attention mechanism before the learning rate gradually decays following a cosine curve.

Multi-Task Loss Function: Weighted Cross-Entropy. Since the models must predict three different maps (Type, Level, and Health) simultaneously, the system minimizes a combined Multi-Task Weighted Cross-Entropy Loss. This loss function forces the model to learn features that benefit all three tasks. The total loss is calculated as:

$$L_{total} = L_{type} + L_{level} + L_{health}$$

Where each L_{task} is the weighted cross-entropy defined as:

$$L_{task} = -\sum_{C=1}^{C} w_C * y_C * log\ log\ (\hat{y}_C)$$

- C is the number of classes
- y_C is the ground truth for class C.
- $\hat{y}_C$ is the predicted probability for class C.
- w_C is the class weight for class C.

These weights were dynamically calculated from the training data statistics to assign higher importance to rare classes (such as Regrowth or Degraded) and lower importance to dominant classes (such as Neutral), ensuring the model does not ignore minority cases. These class weights helped improve model sensitivity to underrepresented classes.

Training and Validation Strategy

A "Fast-Train, Rich-Validation" strategy was employed to balance the need for detailed monitoring with computational speed. During the training loop, the primary focus was on throughput; therefore, the system only calculated the Loss and simple Pixel Accuracy. This streamlined approach allowed the GPU to dedicate its resources to backpropagation without being slowed by complex metric calculations.

Conversely, the validation loop was designed to assess generalization performance. At the end of every epoch, the model was evaluated on a separate Validation Set containing data the model had never seen. During this phase, a comprehensive suite of metrics, including Macro F1-Score and Macro Intersection-over-Union (mIoU), was calculated to provide an accurate performance measure. The validation loss served as the primary signal for the Early Stopping mechanism. If the validation loss failed to decrease for 10 consecutive epochs (the patience parameter), the training was automatically terminated, and the model weights from the best-performing epoch were saved. This protocol effectively prevented overfitting, ensuring that the final model was robust enough to handle new, unseen satellite imagery.

Model Evaluation

To ensure a rigorous assessment of the model's generalization capacity, evaluation was performed exclusively on two held-out datasets: the Internal Test Set (2020) to assess baseline performance and the External Test Set (2021) to evaluate temporal robustness. The validation metrics recorded during training were used only for early stopping; the final results are reported entirely based on unbiased test-set performance. During inference, a robustness check was implemented to handle corrupted or 'Not-a-Number' (NaN) pixel values, which are often encountered in edge cases in satellite patches. Such artifacts were automatically zeroed out to prevent model instability during visualization.

The quantitative assessment utilized standard semantic segmentation metrics. Because the dataset is imbalanced (e.g., Regrowth is rarer than

Neutral), Macro-Averaged versions of F1 and IoU were calculated to ensure that performance on minority classes contributed equally to the final score.

- **Accuracy:** Measures the overall proportion of correctly classified pixels across the image.

$$A = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision & Recall:** Precision measures the correctness of positive predictions, while recall measures the model's ability to detect all relevant instances (especially for degraded areas).

- **F1-score:** Harmonic mean of Precision and Recall, providing a balance between sensitivity and specificity.

$$F_1 = 2 \cdot \frac{P \cdot R}{P + R}$$

Where P is Precision and R is Recall. They are computed using the following formulas:

$$P = \frac{TP}{TP + FP}$$

$$R = \frac{TP}{TP + FN}$$

- **Macro F1-score**: F1-scores are averaged equally across all classes in a segmentation head, ensuring all classes contribute the same regardless of frequency.

$$F_1 macro = \frac{1}{C} \sum_{i=1}^{C} F_{1,i}$$

Where ***TP*** is True Positive, ***TN*** is True Negative, ***FP*** is False Positive, ***FN*** is False Negative, ***C*** is the number of classes, and $F_{1,I}$ is the F1-score of class ***i***.

- **Intersection-over-Union (IoU):** Evaluates spatial accuracy by comparing predicted forest masks against ground-truth labels.

$$IoU = \frac{TP}{TP + FP + FN}$$

- **Macro Intersection-over-Union (Macro IoU):** Class-wise IoUs averaged across all classes, giving equal weight to rare and dominant classes.
    - $IoU_{macro} = \frac{1}{C} \sum_{i=1}^{C} IoU_{,i}$

For this multi-class segmentation task, both F1-scores and IoU values were calculated as Macro-averages to ensure fair representation and prevent scores from being dominated by the largest class.

Confusion Matrix (C): Visualizes performance per class, helping identify misclassified forest types, levels, or health statuses.

- **Confusion Matrix (C):** Visualizes performance per class, helping identify misclassified forest types, levels, or health statuses.

The Confusion Matrix is an ***N x N*** matrix (where ***N*** is the number of classes, including Background) that visualizes the classification algorithm's

performance. Each cell C_{ij} showed the count of pixels that were actually in class ***i*** but were predicted to be in class ***j***. This visualization was crucial for diagnosing specific misclassification patterns across segmentation tasks (Type, Level, Health), helping quantify the model's tendency to confuse rare classes (such as Mossy or Degraded) with dominant classes (such as Closed or Neutral). The matrix directly provided the necessary components (***TP***, ***TN***, ***FP***, ***FN***) required for calculating the Macro-averaged metrics.

These metrics ensured a robust evaluation of both forest type classification and health condition detection. Additionally, Error Maps were generated by subtracting the Ground Truth mask from the predicted mask. These binary heatmaps visually highlight the exact spatial locations of misclassifications, revealing whether errors are random or clustered around specific features, such as forest edges.

Lastly, to rigorously determine if the performance differences between the models were statistically significant or merely due to random variance, a Two-Way Analysis of Variance (ANOVA) was employed. This statistical test analyzes the influence of two independent variables on the model performance (Mean IoU):

1. **Factor A:** Model Architecture (Levels: U-Net, SegFormer, Hybrid TransU-Net).
2. **Factor B:** Patch Size (Levels: 64×64, 128×128).

The ANOVA output determines whether there is a significant main effect for architecture or patch size, and whether there is an interaction between them.

If significant differences were detected ($p < 0.05$), a Tukey's Honest Significant Difference (HSD) post-hoc test was performed to identify precisely which specific pairs of models (e.g., TransU-Net 64 vs. SegFormer 64) differed significantly in performance.

Tree Species Recommendation Module

Our model uses a rule-based module to recommend native tree species for restoration. It first detects degraded areas within the landscape and then determines whether each area is forested. To enhance interpretability, a 'Locator Visualization' technique was developed. This module identifies connected components (blobs) of degraded pixels, calculates their centroids, and overlays a coordinate grid and unique ID labels directly onto the RGB imagery. This allows for precise spatial localization of reforestation targets. If the area is classified as forested, the forest level will be collected for each degraded patch. Based on this elevation classification (lowland, lower montane, or upper montane), the module applies predefined ecological rules that match each forest level to its corresponding native tree species. However, there will be no tree recommendations for areas identified as Non-Forest, as our study does not cover reforestation efforts for other land cover types.

Since Philippine forests naturally change in composition with increasing altitude, this rule-based approach ensures that the recommended species are aligned with the forest type naturally expected at that elevation. In this way, the system provides ecologically appropriate, site-specific restoration recommendations using a transparent, consistent decision logic.

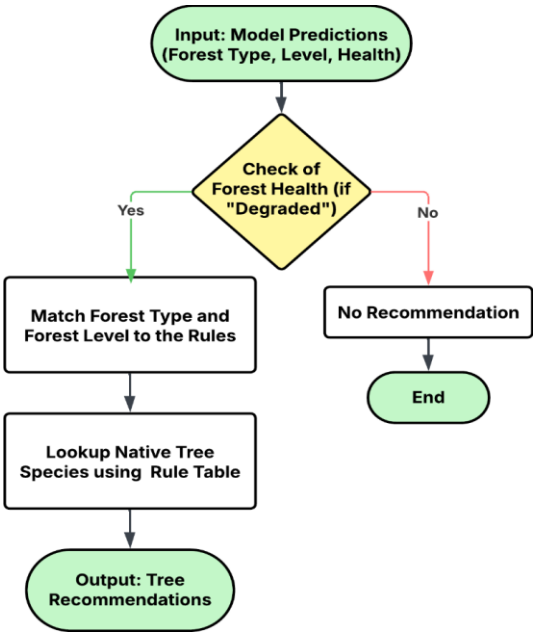


Figure 9. Tree Recommendation Module Logic

The rule-based recommendations in our model are grounded in established ecological studies and authoritative references on Philippine forest vegetation. Key sources include the works of Fernando et al. (2008), Malabrigo et al. (2016), Buot and Okitsu (1998), Villanueva and Buot (2018), and Alcala (2021), which provide detailed descriptions of species distributions across lowland, lower montane, and upper montane forest zones. These publications served as the scientific basis for organizing the native species listed in Table 9, ensuring that each elevation band is matched with ecologically appropriate trees. By relying on these well-documented ecological gradients, the system maintains consistency with accepted forest ecology literature. As a result, the model's rule-based outputs remain evidence-based, transparent, and aligned with current scientific understanding of Philippine forest ecosystems.

Table 11. Tree Recommendation Lookup Table

Forest Type	Elevation Level	Recommended Tree Species	Sources
Forested	Lowland (<300 m)	• *Dipterocarpus grandiflorus* (Apitong) • *Shorea contorta* (White Lauan) • *Shorea negrosensis* (Red Lauan) • *Pterocarpus indicus* (Narra) • *Vitex parviflora* (Molave) • *Artocarpus blancoi* (Antipolo) • *Canarium ovatum* (Pili)	Malabrigo et al. (2016); Haribon Foundation (2010); Fernando et al. (2008)
	Lower Montane (301–700 m)	• *Lithocarpus spp.* (Oak) • *Syzygium spp*. (Stone Oak) • *Elaeocarpus cumingii* (Elaeocarpus tree) • *Podocarpus rumphii* (Rumph's Yellowwood) • *Cinnamomum mercadoi* (Kalingag) • *Ficus balete* (Balete tree) • *Weinmannia luzoniensis* (Luzon Weinmannia)	Fernando et al. (2008); Villanueva & Buot (2018); Alcala (2021)
	Upper Montane (700–1000 m)	• *Dacrycarpus cumingii* (Cuming's Yellowwood) • *Ilex philippinensis* (Philippine holly) • *Vaccinium spp*. (Blueberries) • *Rapanea spp.*(Kalingag-kalingag) • *Clethra tomentella* • *Weinmannia subsessilis*	Buot & Okitsu (1998); Villanueva & Buot (2018); Alcala (2021)

Ethical Considerations

This study follows ethical practices by using only open, government-approved data, including Sentinel-2 satellite images, SRTM elevation data, and ESA WorldCover. These datasets are free to use and do not contain any personal or sensitive information. Since the study only uses satellite data and does not involve people or field surveys, there are no privacy risks.

The data were processed using standard steps like removing clouds, clipping to the study area, and applying NDVI thresholds. No changes were made beyond these normal steps, and areas with missing data were excluded and reported. This makes the results more honest and reliable.

All software used, including Google Earth Engine, QGIS, Python libraries, and machine learning tools, is open-source or under academic licenses. Proper credit was given to all data providers. This supports open science, making it easier for other researchers to repeat or build on the study.

Lastly, the tree species recommendations were based on trusted sources such as the Haribon Foundation's Rainforestation Guide and DENR planting manuals. The species chosen are native and suited to the local environment. These suggestions are not strict rules but serve as guidance to help plan reforestation sustainably.

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the results of implementing the CNN–ViT hybrid semantic segmentation model for classifying forest type, forest level, and forest health in the Mt. Makiling Forest Reserve (MMFR). It also discusses the outputs of the rule-based recommendation module, which suggest native tree species for degraded areas. Results were analyzed using both quantitative evaluation metrics (accuracy, precision, recall, F1-score, Intersection-over-Union, and confusion matrices) and qualitative visualizations of the classification maps and recommendations. Results presented here included both training/validation metrics used to monitor learning and final evaluation metrics on the held-out test set. The discussion situates these results within the broader objectives of forest monitoring and restoration planning in MMFR.

Dataset Overview

Before presenting the model performance, it is essential to analyze the distribution of the training data. The original dataset exhibited significant class imbalances, which can lead to model bias. To mitigate this, data augmentation techniques were applied to increase the representation of minority classes artificially. The following tables detail the pixel counts before and after augmentation for the 64x64 and 128x128 patch sizes training sets.

Table 12 highlights the extreme disparity between the "Forested" and "Non-Forest" classes, characteristic of the dense vegetation cover in Mt. Makiling. The "Non-Forest" class initially comprised only a minute fraction of the dataset (approx. 59,000 pixels). To prevent the model from ignoring this minority class, aggressive augmentation was applied, increasing the number of non-forest samples by 5.24x. While the dataset remains heavily skewed toward forested areas, this targeted increase ensures the model encounters sufficient examples of non-vegetated features (such as roads or bare soil) to define forest boundaries accurately.

Table 12. Augmentation Impact on Forest Type (64x64)

Class	Original Pixels	Augmented Pixels	Augmentation Factor
Forested	12,408,542	45,198,582	3.64x
Non-Forest	59,329	310,679	5.24x

Table 13 details the distribution of samples across the three elevational zones. While the original pixel counts were relatively comparable across classes, the augmentation strategy prioritized the "Upper Montane" zone, increasing its representation by 5.64x. This strategic oversampling was implemented to ensure the model captures the complex, cloud-affected features often found at higher altitudes. Conversely, the "Lowland" class, which typically exhibits distinct and stable spectral signatures, required less augmentation (1.98x) to achieve convergence.

Table 13. Augmentation Impact on Forest Level (64x64)

Class	Original Pixels	Augmented Pixels	Augmentation Factor
Lowland	3,242,372	6,426,097	1.98x
Lower Montane	4,989,068	15,334,123	3.07x
Upper Montane	4,205,663	23,709,293	5.64x

Table 14 reveals the most critical imbalance in the study: the "Neutral" (healthy) class dominates the original dataset, with over 10.8 million pixels. In contrast, the critical restoration classes, "Degraded" and "Regrowth," were significantly underrepresented. To address this, the augmentation pipeline prioritized these minority classes, boosting "Regrowth" by 5.69x and "Degraded" by 5.40x. This rebalancing is vital for the study's objectives, as it forces the deep learning models to learn the subtle spectral patterns of vegetation stress and recovery rather than defaulting to predicting the majority "Neutral" class.

Table 14. Augmentation Impact on Forest Health (64x64)

Class	Original Pixels	Augmented Pixels	Augmentation Factor
Degraded	586,549	3,169,344	5.40×
Neutral	10,806,377	36,224,427	3.35×
Regrowth	1,074,945	6,115,490	5.69×

Table 15 presents the distribution for the larger 128x128 input patches. A notable observation here is the massive scale of the "Forested" class, which grew to nearly 160 million pixels after augmentation, reflecting the increased spatial coverage of the larger patches. In contrast, the "Non-Forest" class shows a minimal increase factor of 1.04x, maintaining a pixel count close to its original value. This disparity is likely due to the scarcity of pure non-forest

patches in the dataset; since the 128x128 patches cover a larger area, they are statistically more likely to contain forest pixels, resulting in an overwhelming dominance of the "Forested" class. Despite this extreme imbalance (approx. 1:3000), the high segmentation accuracy achieved in the test phase suggests that the models still learned the distinct spectral features of non-forest elements effectively.

Table 15. Augmentation Impact on Forest Type (128x128)

Class	Original Pixels	Augmented Pixels	Augmentation Factor
Forested	46,974,326	159,594,016	3.40×
Non-Forest	49,056	51,146	1.04×

Table 16 illustrates a targeted augmentation strategy based on elevation complexity. The "Lowland" class, which typically presents clearer spectral signatures and fewer atmospheric obstructions, received negligible augmentation (1.01x). In contrast, the "Upper Montane" zone, which is often subjected to cloud cover and complex topographic shadows, was aggressively upsampled by a factor of 5.70x, resulting in the highest final pixel count of over 88 million. This strategic boosting ensures that deep learning models receive significantly more training examples for the rugged terrain at high elevations, compensating for the natural environmental noise in those regions.

Table 16. Augmentation Impact on Forest Level (128x128)

Class	Original Pixels	Augmented Pixels	Augmentation Factor
Lowland	10,236,355	10,385,105	1.01×
Lower Montane	21,200,614	60,911,724	2.87×
Upper Montane	15,495,283	88,253,253	5.70×

Table 17 demonstrates the study's continued emphasis on addressing the rarity of health-related features. Similar to the 64x64 dataset, the "Neutral" class remains dominant, accounting for over 132 million augmented pixels. However, the critical minority classes, "Degraded" and "Regrowth," experienced the highest increase factors of 5.31x and 5.44x, respectively. This resulted in a substantial boost in training data, raising the "Degraded" pixel count from roughly 1.9 million to over 10.5 million. This massive injection of synthetic data is a primary factor contributing to the Hybrid 128x128 model's ability to achieve ~99% IoU on these classes in the 2020 test, as the model was provided with ample variations of vegetation stress and recovery patterns during training.

Table 17. Augmentation Impact on Forest Health (128x128)

Class	Original Pixels	Augmented Pixels	Augmentation Factor
Degraded	1,985,933	10,550,753	5.31×
Neutral	41,914,565	132,108,170	3.15×
Regrowth	3,122,884	16,986,239	5.44×

Table 18 summarizes the pixel distribution of the validation set used to monitor model performance during training. Unlike the training set, the validation data was not augmented, preserving the natural class imbalances inherent in the Mt. Makiling ecosystem. The data shows a massive skew in the Forest Type category, with "Forested" pixels outnumbering "Non-Forest" by roughly 243:1. Similarly, the Forest Health category is heavily dominated by the "Neutral" class (approx. 2.37 million pixels). In contrast, the critical "Degraded" and "Regrowth" classes comprise only a small fraction of the total area. This natural imbalance explains the volatility observed in the validation loss graphs,

as the models initially struggled to generalize to these minority classes without synthetic augmentation.

Table 18. Class Distribution of the Validation Set (64x64)

Class	Original Pixels	Augmented Pixels
Forest Type	Forested	11,072
	Non-Forest	677,700
Forest Level	Lowland	1,119,555
	Lower Montane	904,722
	Upper Montane	123,939
Forest Health	Degraded	2,369,354
	Neutral	214,224
	Regrowth	11,072

Table 19 details the pixel composition of the validation set utilized for the 128x128 models. Consistent with the 64x64 configuration, this dataset remains non-augmented to provide a realistic benchmark of model generalization. The larger patch size significantly inflates the "Forested" pixel count to over 10 million, while the "Non-Forest" class remains stagnant at approximately 10,000 pixels. This creates an extreme imbalance ratio of roughly 1000:1, rendering the non-forest class statistically negligible without synthetic oversampling during training. In the Forest Health category, the "Neutral" class dominates with nearly 9 million pixels, overshadowing the "Degraded" (455,408) and "Regrowth" (684,986) classes. This disparity highlights the challenge models face in the 128x128 configuration: they must identify critical health indicators that account for less than 12% of the total validation data, underscoring the importance of class-weighted augmentation strategies.

Table 19. Augmentation Impact on Forest Level (128x128)

Class	Original Pixels	Augmented Pixels
Forest Type	Forested	10,088,854
	Non-Forest	10,581
Forest Level	Lowland	2,035,794
	Lower Montane	4,537,829
	Upper Montane	3,508,120
Forest Health	Degraded	455,408
	Neutral	8,959,041
	Regrowth	684,986

Table 20 quantifies the shift in ground-truth distributions between the training year (2020) and the test year (2021). While the dominant "Forested" class remained remarkably stable with a negligible decrease of 0.12%, the "Non-Forest" class exhibited a significant relative surge of +26.05%. Although the absolute increase in non-forest pixels is small relative to the total area, this substantial percentage jump represents a "distributional drift." This implies that non-vegetated features (such as clearings, roads, or bare soil) became more frequent or prominent in the 2021 imagery. This shift presents a challenge for the deep learning models, as the statistical probability of encountering a "Non-Forest" pixel in 2021 was significantly higher than what the model learned from the 2020 distribution, contributing to the reduced intersection-over-union (IoU) scores observed in the temporal test.

Table 20. Class Distribution Drift: 2020 vs. 2021 (Type 64x64)

Class	2020 Pixel Count	2021 Pixel Count	Shift (%)
Forested	2,673,730	2,670,645	−0.12%
Non-Forest	11,841	14,926	+26.05%

Table 21 confirms the absolute stability of the Forest Level variable, showing a 0.0% shift across all classes between 2020 and 2021. This consistency is expected, as the ground truth for these categories is derived from a Digital Elevation Model (DEM), and the topographical features of Mt. Makiling are static over a single year. This stability serves as a crucial control variable for the study. Since the ground-truth labels for the Lowland and Montane zones remained identical, the significant drop in Forest Level classification accuracy observed in the 2021 test results cannot be attributed to a shift in data distribution. Instead, it definitively shows that the deep learning models failed to recognize the visual features of these elevations when the forest's spectral appearance changed, confirming that the models relied on overfitting to the 2020 imagery rather than learning generalized topographical cues.

Table 21. Class Distribution Drift: 2020 vs. 2021 (Level 64x64)

Class	2020 Pixel Count	2021 Pixel Count	Shift (%)
Lowland	712,101	712,101	0.0%
Lower Montane	1,132,193	1,132,193	0.0%
Upper Montane	835,354	835,354	0.0%

Table 22 reveals a drastic ecological shift between the two years, characterized by massive degradation of forest cover. The "Degraded" class exploded by nearly 390%, rising from roughly 123,000 pixels in 2020 to over 600,000 in 2021. Concurrently, the "Regrowth" class, which represents recovering vegetation, collapsed by 68.76%. This inverse relationship suggests that many areas previously classified as stable or recovering deteriorated significantly in the subsequent year. For the deep learning models, this represents an extreme case of "distribution shift." The models were trained on

2020 data, where degradation was a minority feature; when faced with 2021 data, where degradation was a dominant feature, the models lacked the learned representations to handle the high volume of stressed vegetation, directly contributing to the poor IoU scores observed in the temporal test.

Table 22. Class Distribution Drift: 2020 vs. 2021 (Health 64x64)

Class	2020 Pixel Count	2021 Pixel Count	Shift (%)
Degraded	123,139	602,965	+389.66%
Neutral	2,354,416	2,017,625	−14.30%
Regrowth	208,016	64,981	−68.76%

Table 23 illustrates a sharp divergence in the prevalence of non-vegetated features between the two study years for the 128x128 dataset. While the "Forested" class remained stable, with a negligible decline of 0.15%, the "Non-Forest" class pixel count more than doubled, increasing by 139.32%. This surge from approximately 10,000 to nearly 26,000 pixels indicates that the 2021 test set contained significantly more clearings, roads, or bare soil patches than the 2020 training set. This drastic distribution shift challenges deep learning models, which were optimized on a dataset where non-forest features were extremely rare. The sudden abundance of these features in the 2021 data likely contributed to the lower precision scores observed in the temporal test, as the models struggled to adapt to the new frequency of non-forest occurrences.

Table 23. Class Distribution Drift: 2020 vs. 2021 (Type 128x128)

Class	2020 Pixel Count	2021 Pixel Count	Shift (%)
Forested	10,057,350	10,042,246	−0.15%
Non-Forest	10,841	25,945	+139.32%

Table 24 confirms the absolute stability of the Forest Level variable, demonstrating a 0.0% shift across all elevational classes between the 2020 and 2021 datasets. This consistency is strictly anticipated, as the ground truth for these categories is derived from a Digital Elevation Model (DEM), and the topographical signatures of the study area remain static over the observation period. This stability effectively serves as a crucial control variable for the longitudinal evaluation. Since the ground-truth definitions for the Lowland, Lower Montane, and Upper Montane zones remained unchanged, any decline in Forest Level classification accuracy observed in the 2021 test results cannot be attributed to shifts in the data distribution. Instead, it definitively indicates that the deep learning models failed to decouple elevation from spectral vegetation features, suggesting that the architectures overfitted to the specific visual conditions of 2020 rather than learning generalized topographical cues.

Table 24. Class Distribution Drift: 2020 vs. 2021 (Level 128x128)

Class	2020 Pixel Count	2021 Pixel Count	Shift (%)
Lowland	2,148,856	2,148,856	0.0%
Lower Montane	4,495,397	4,495,397	0.0%
Upper Montane	3,404,268	3,404,268	0.0%

Table 25 reveals a drastic ecological shift between the two years, characterized by massive degradation of forest cover. The "Degraded" class exploded by over 426%, rising from roughly 444,000 pixels in 2020 to over 2.34 million in 2021. Concurrently, the "Regrowth" class collapsed by 65.49%, while the stable "Neutral" class also declined by 16.44%. This inverse relationship suggests that vast areas previously classified as stable or recovering deteriorated significantly in the subsequent year. For the deep learning models,

this represents an extreme case of "distribution shift." The models were trained on 2020 data, where degradation was a minority feature; when faced with 2021 data, where degradation became prevalent, the models lacked the learned representations to handle the high volume of stressed vegetation, directly contributing to the poor IoU scores observed in the temporal test.

Table 25. Class Distribution Drift: 2020 vs. 2021 (Health 128x128)

Class	2020 Pixel Count	2021 Pixel Count	Shift (%)
Degraded	444,455	2,340,079	+426.51%
Neutral	8,984,929	7,507,652	−16.44%
Regrowth	638,807	220,460	−65.49%

Figure 10 visually illustrates the critical difference in spatial context provided by the two input patch dimensions. The 64x64 configuration (Left) functions as a "local texture" analyzer, isolating high-frequency spectral details and specific pixel-level variations; however, its restricted field of view limits the model's ability to perceive the surrounding landscape. In contrast, the 128x128 configuration (Right) expands the receptive field to capture "global context," revealing macro-scale patterns such as canopy density gradients, clearing edges, and topographic transitions. This visual comparison elucidates why the 128x128 models achieved superior performance in the quantitative results: the broader spatial context allows the architecture to differentiate between isolated noise and genuine degradation zones by referencing the health and structure of the neighboring vegetation.

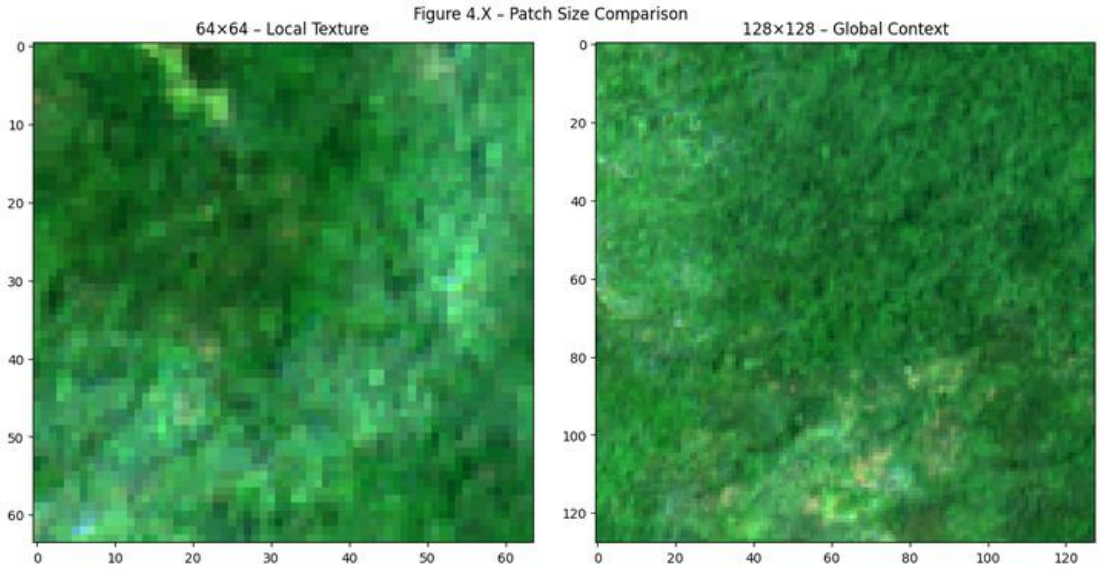


Figure 10. Class Distribution Drift: 2020 vs. 2021 (Health 128x128)

Model Training and Validation Loss

This section evaluates the training efficacy of the developed deep learning models by analyzing the progression of loss values during training and validation. The performance of the Convolutional Neural Network (CNN), Vision Transformer (ViT), and Hybrid architectures was rigorously monitored to assess their ability to minimize error in detecting forest degradation. To optimize classification accuracy, each architecture was trained and validated at two distinct image patch resolutions: 64x64 and 128x128 pixels. The subsequent graphs illustrate the loss function trajectories for each configuration, highlighting the convergence rates and generalization capabilities of the respective models. These comparative results serve as the primary basis for identifying the most robust model configuration to drive the system's tree recommendation logic.

Figure 11 illustrates the training and validation loss performance for the Convolutional Neural Network (CNN) model processed on 64x64 image

patches over 24 steps. The training loss, represented by the blue line, exhibits a smooth, consistent optimization trajectory, gradually decreasing to a final smoothed value of 0.1902. In contrast, the validation loss initially exhibits extreme volatility, with sharp spikes exceeding 2.0, which suggests early challenges in generalizing to unseen data. However, the model stabilizes significantly after the tenth step, with the validation curve dropping steeply to achieve a final smoothed value of 0.1853. This eventual convergence, where the validation loss falls slightly below the training loss, indicates that the 64x64 model effectively overcomes early instability to achieve robust degradation classification.

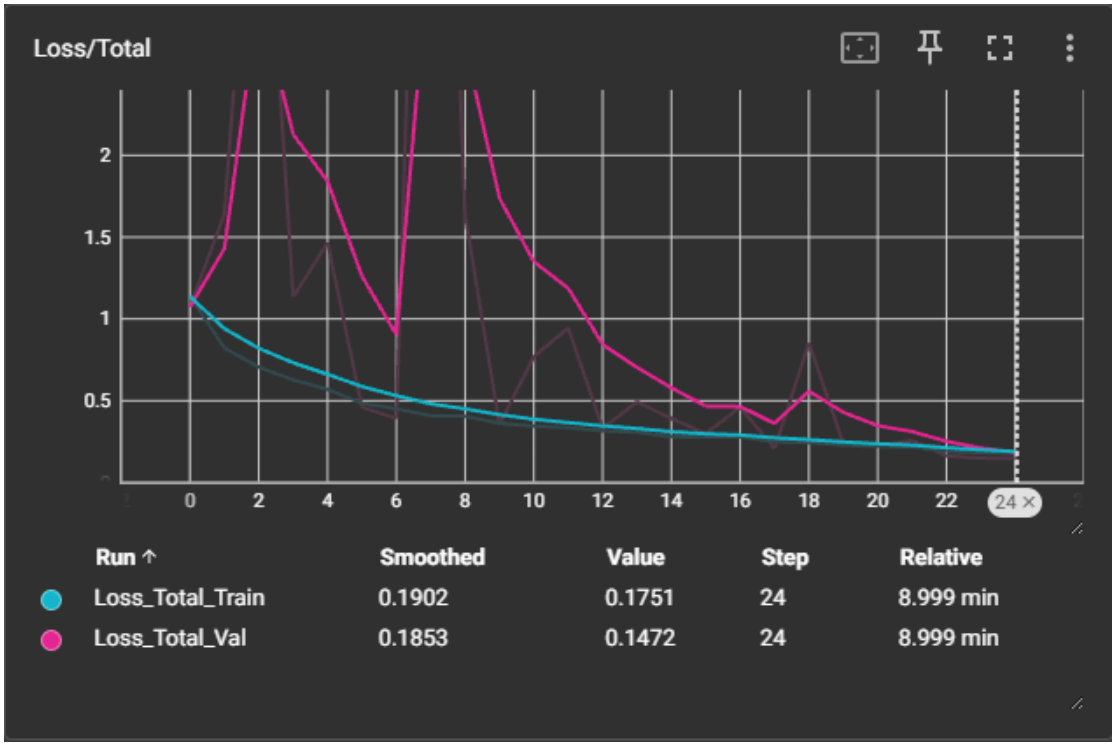


Figure 11. CNN 64x64 Input - Training Loss

Figure 12 depicts the loss metrics for the Convolutional Neural Network (CNN) trained on larger 128x128 image patches, revealing a critical disparity between learning and generalization capabilities. The training loss, indicated by

the blue curve, demonstrates exceptional optimization, descending smoothly from an initial value near 0.8 to a near-zero smoothed value of 0.0208. Conversely, the validation loss exhibits severe instability throughout the training process, characterized by erratic fluctuations and multiple spikes despite temporary periods of improvement around step 14. A major divergence occurs in the final steps, where the validation loss surges dramatically to a smoothed value of 0.5808, significantly detaching from the convergent training metric. This drastic gap between the low training error and the spiking validation error indicates that the 128x128 model is suffering from significant overfitting, where it memorizes the training data but fails to accurately classify new forest samples.

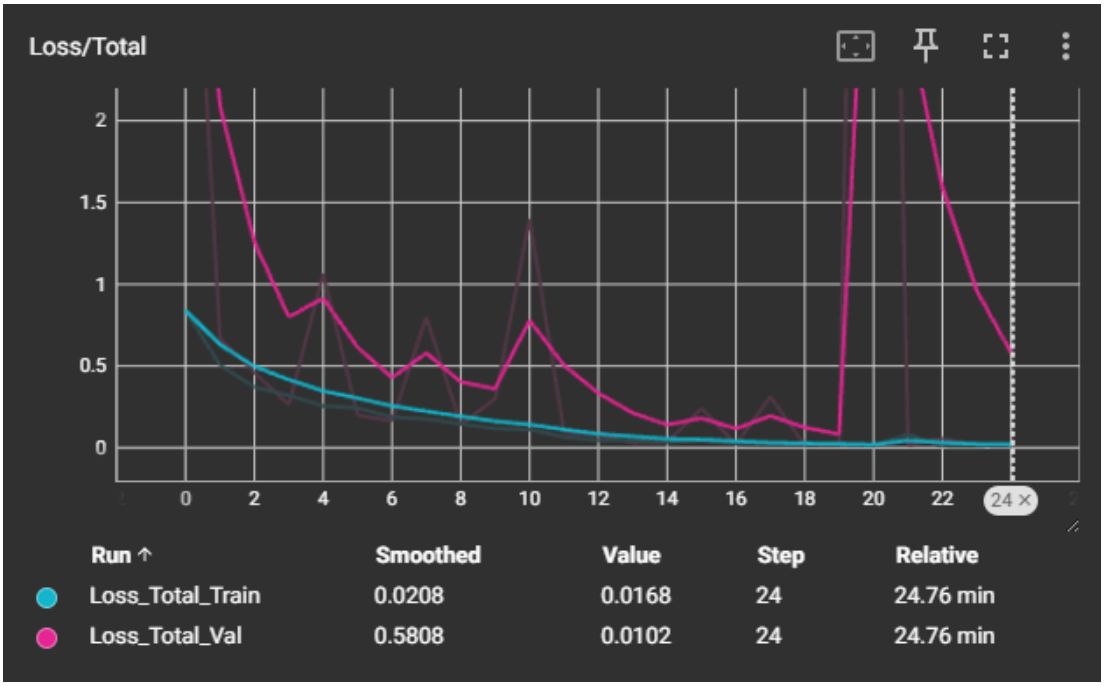


Figure 12. CNN 128x128 Input - Validation Loss

The comparison between CNN input sizes reveals a distinct trade-off between model stability and training optimization. The 64x64 (Figure 11) model demonstrated robust generalization, with validation loss closely tracking the

training metrics, whereas the 128x128 (Figure 12) model suffered from severe instability and overfitting. While the 128x128 configuration achieved a lower training loss, its validation loss spiked drastically, indicating that the model memorized noise in the larger images rather than learning meaningful forest features. In contrast, the 64x64 input provided a more balanced performance, proving that the simpler CNN architecture functions more reliably with lower-resolution data for this specific task. Consequently, the 64x64 resolution is deemed more suitable for the standalone CNN model to ensure consistent degradation detection.

Figure 13 presents the loss metrics for the Vision Transformer (ViT) model trained on 64x64 input patches, displaying a highly stable and efficient learning curve over 29 steps. The training loss, shown in blue, follows a classic exponential decay pattern, steadily decreasing from an initial high to a final smoothed value of 0.3244. In a promising trend, the validation loss, represented by the pink line, consistently outperforms the training metric, dropping rapidly to stabilize at a lower smoothed value of 0.2517. The absence of fluctuations or spikes in the validation curve indicates that the model is generalizing exceptionally well to new data without any signs of overfitting. This smooth convergence confirms that the ViT architecture with 64x64 resolution effectively captures the necessary spectral features for accurate forest degradation classification.

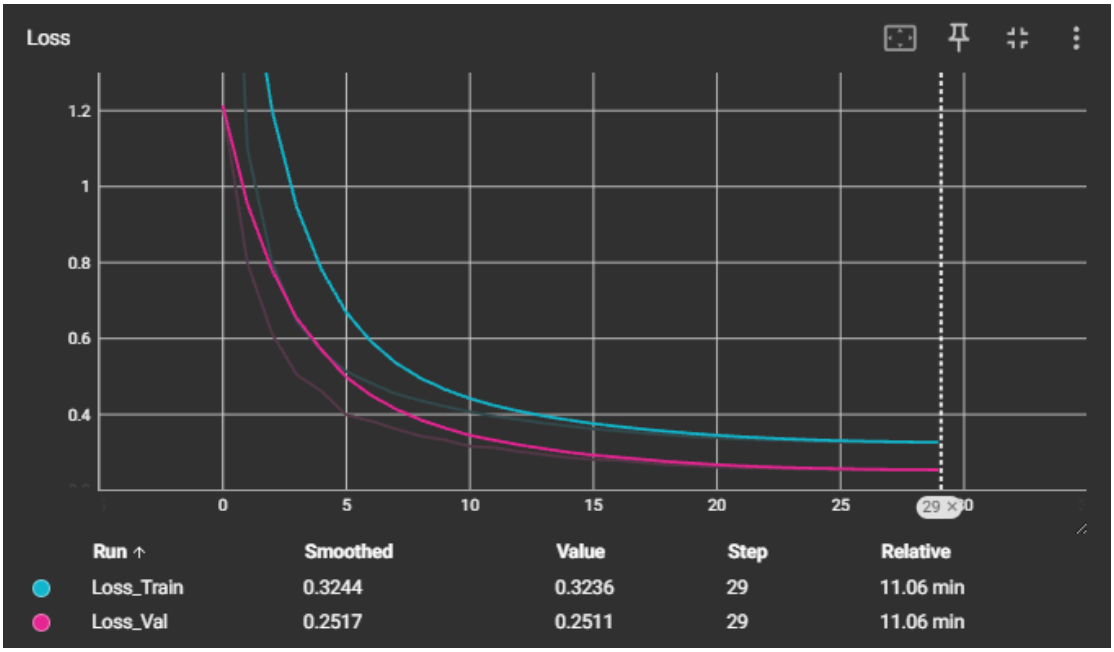


Figure 13. ViT 64x64 Input - Training Loss

Figure 14 illustrates the loss performance of the Vision Transformer (ViT) model utilizing 128x128 image patches over a span of 29 training steps. The training loss, depicted by the blue curve, exhibits a steady and smooth exponential decay, starting above 1.0 and gradually settling at a smoothed value of 0.2372. Concurrently, the validation loss, shown in pink, demonstrates superior performance by dropping rapidly and maintaining a trajectory consistently below the training curve. The model achieves excellent convergence, with the validation metric stabilizing at a final smoothed value of 0.1692, which is significantly lower than the final training loss. This substantial gap, where validation error is minimized effectively, suggests that the 128x128 ViT model possesses robust generalization capabilities and identifies forest degradation patterns without overfitting.

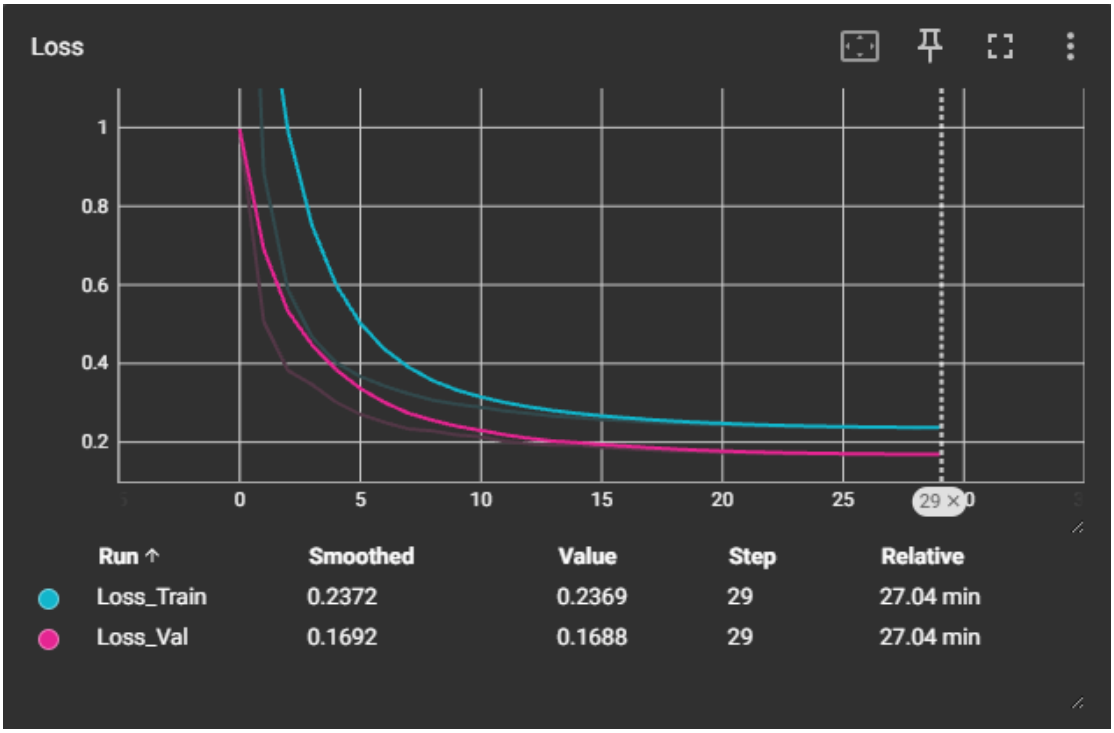


Figure 14. ViT 128x128 Input - Training Loss

The Vision Transformer (ViT) architecture exhibited improved predictive performance when processed with larger 128x128 (Figure 14) input patches compared to the 64x64 (Figure 13) configuration. While both resolutions maintained stable and convergent learning curves, the 128x128 model achieved a significantly lower final validation loss of 0.1692 compared to 0.2517 for the smaller inputs. This improvement suggests that the self-attention mechanisms within the ViT benefit substantially from the increased spatial resolution, allowing the model to capture finer spectral details of the forest cover. Unlike the CNN, the ViT scaled effectively with the larger data dimensions without overfitting, validating its capacity to handle complex spatial relationships. Therefore, the 128x128 input size is the superior configuration for maximizing the accuracy of the Vision Transformer.

Figure 15 illustrates the total loss progression for the Hybrid model utilizing 64x64 input dimensions, tracking performance over twenty-nine training steps. The training loss, depicted by the blue curve, exhibits a rapid and smooth descent, effectively minimizing error to reach a remarkably low smoothed value of 0.0211. Closely mirroring this trajectory, the validation loss in pink demonstrates high stability and converges tightly with the training data, settling at a final smoothed value of 0.0344. The exceptionally narrow gap between the training and validation metrics indicates that the hybrid architecture successfully leverages both local spatial features and global context without overfitting. Overall, these near-zero loss values suggest that the Hybrid 64x64 model offers the most precise and reliable performance among the tested architectures for forest degradation detection.

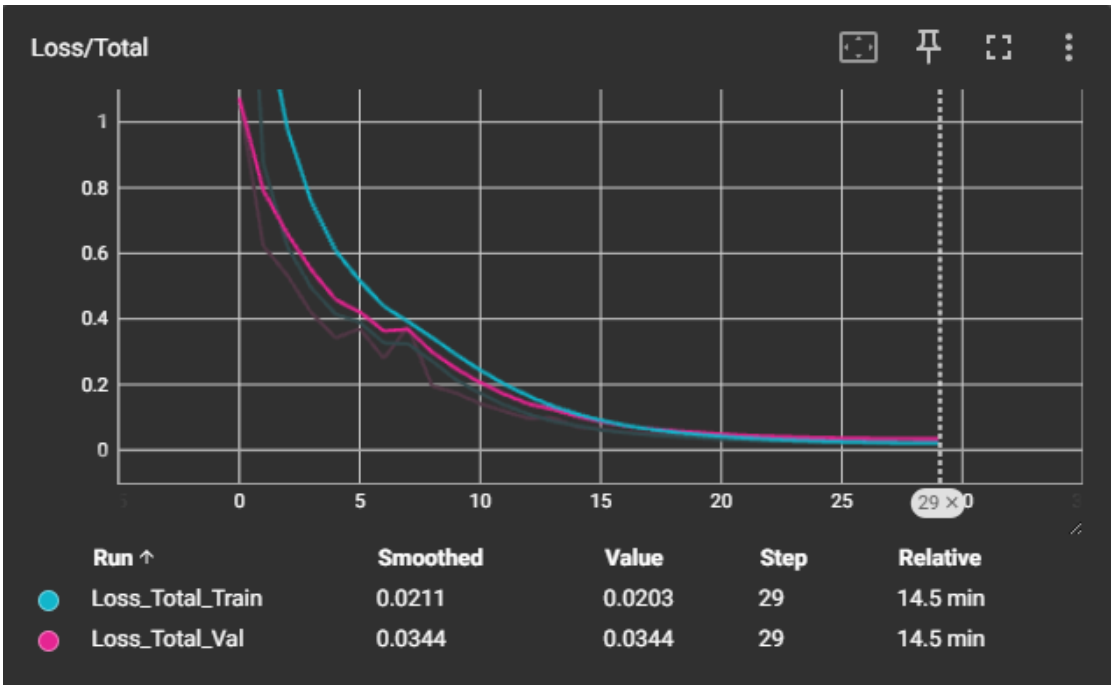


Figure 15. Hybrid 64x64 Input - Training Loss

Figure 16 depicts the loss trajectory for the Hybrid model processed on larger 128x128 inputs, demonstrating the most significant error reduction

among all tested architectures. Although the validation curve, shown in pink, experiences transient spikes around steps 7 and 15, the model exhibits strong resilience and quickly realigns with the optimization path. The training loss achieves an exceptionally low smoothed value of 0.0043, indicating that the model has reached a near-perfect understanding of the training dataset features. Similarly, the validation loss converges to a minute value of 0.0056, confirming that the model generalizes with high precision despite the increased complexity of the larger image patches. These extremely low and tightly aligned final values suggest that the Hybrid 128x128 configuration provides the optimal balance of spatial detail and global context for the study.

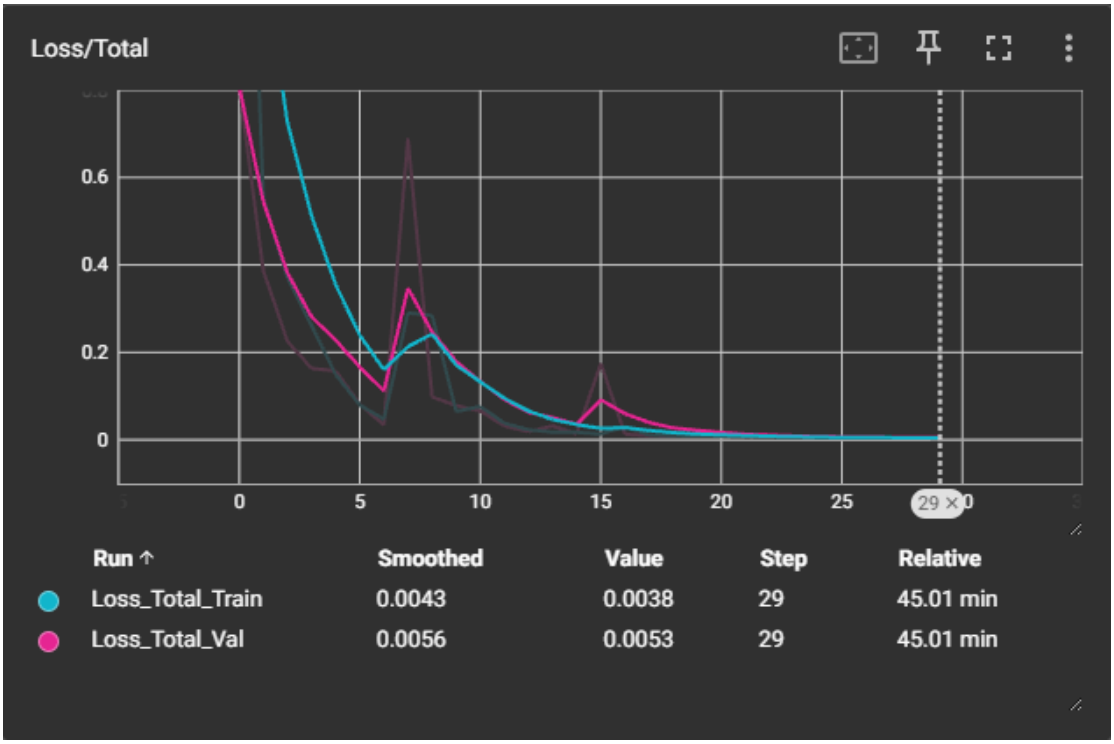


Figure 16. Hybrid 128x128 Input - Training Loss

The Hybrid model displayed exceptional performance across both input sizes, but the 128x128 (Figure 16) configuration yielded the definitive superior result for the study. The 64x64 (Figure 15) model already achieved an

impressive low loss of 0.0344, but increasing the resolution to 128x128 further drove the validation error down to a minute 0.0056. This substantial reduction in loss demonstrates that the Hybrid architecture effectively synthesizes local CNN features and global ViT context to fully leverage the additional pixel data. Although the 128x128 model showed slight initial volatility, it quickly stabilized to produce the lowest error rate among all tested architectures. Ultimately, the 128x128 Hybrid model offers the most precise forest degradation detection, justifying the use of higher-resolution inputs for the final system.

Training and Validation Metrics Per Head

While the loss curves provide insight into the model's overall learning trajectory, the head-specific performance metrics reveal how effectively the CNN–ViT hybrid addressed the distinct classification tasks, such as forest type, forest level, and forest health. It is important to clarify that the metrics discussed in this section, which are Accuracy, F1 Macro, and IoU Macro, were tracked in each epoch using the training and validation sets only. These metrics track optimization but do not represent the final test evaluation. Their purpose is to monitor the learning behavior and stability during the optimization. The final evaluation on the unseen test set, which provides the ultimate measure of model generalization, is presented in a later section.

Figure 17 details the validation accuracy for the CNN model using 64x64 inputs, tracking performance across Forest Type, Level, and Health categories over 24 steps. The Forest Type classification, illustrated by the blue line, exhibits near-perfect stability from the onset, maintaining a consistently high accuracy that concludes at 0.9995. In contrast, the Forest Level accuracy,

shown in grey, begins with significant instability and low values below 0.70 but improves drastically after the eighth step to reach a final smoothed value of 0.9831. The Forest Health metric, represented by the orange curve, displays early fluctuations with distinct drops before steadily ascending to achieve a reliable final accuracy of 0.95. These results indicate that while the model easily distinguishes basic forest types, the more complex variables of level and health require additional training steps to reach optimal precision.

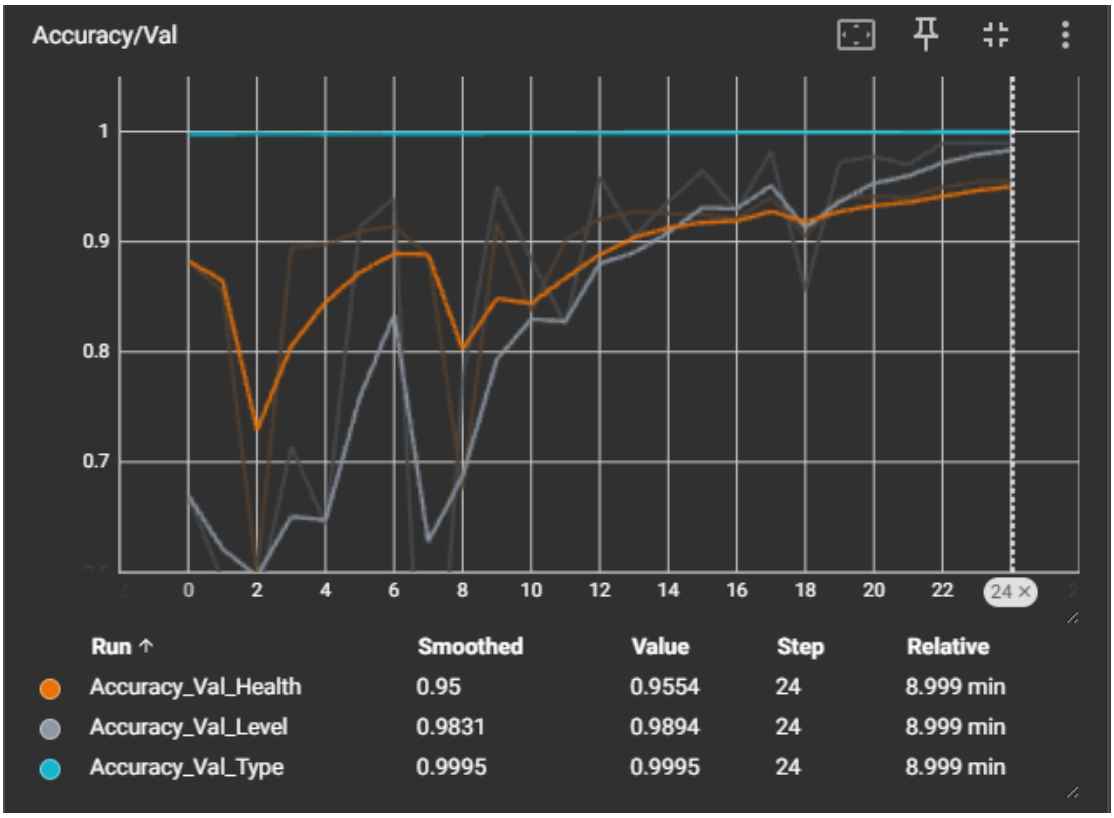


Figure 17. CNN with 64x64 Input Validation Accuracy

Figure 18 depicts the validation accuracy for the CNN model trained on larger 128x128 images, highlighting high peak performance marred by significant volatility across the three target variables. The Forest Type classification, shown in blue, remains exceptionally stable and accurate throughout the process, achieving a near-perfect smoothed value of 0.9999. In

contrast, the Forest Health (orange) and Forest Level (grey) metrics exhibit erratic behavior, characterized by sharp declines and recoveries, most notably a severe drop around the twentieth step. Despite these fluctuations, the model recovers to finish with strong smoothed accuracy scores of 0.9894 for Forest Health and 0.9748 for Forest Level. This instability indicates that while the 128x128 resolution provides sufficient data for high accuracy, it makes the model prone to significant performance swings during the validation phase.

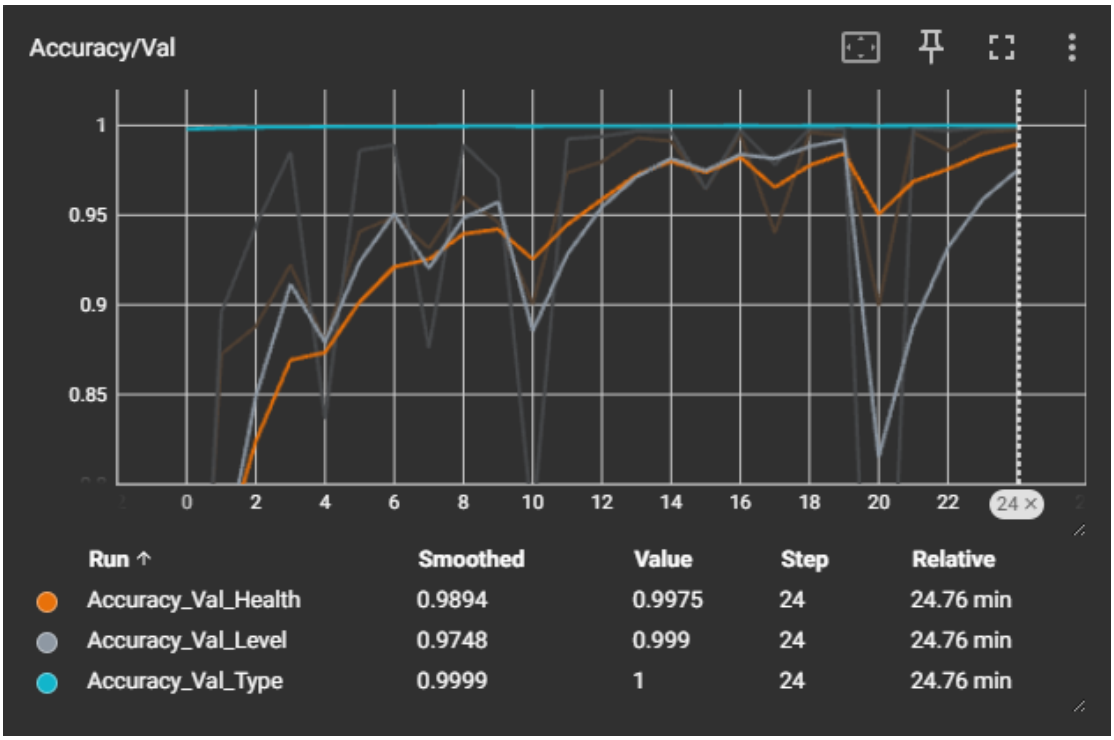


Figure 18. CNN with 128x128 Input Validation Accuracy

Comparing the accuracy metrics between Figure 16 and Figure 17 reveals a distinct trade-off where the larger 128x128 input size enhances peak performance but introduces critical stability issues indicative of overfitting. The 128x128 model in Figure 17 achieved superior numerical precision, reaching a near-perfect smoothed accuracy of 0.9999 for Forest Type and surpassing the 64x64 model in Forest Health classification. However, this high accuracy is

marred by severe volatility, most notably a drastic performance drop at step 20, which contrasts sharply with the smooth and dependable learning curve observed in Figure 11. This erratic behavior suggests that the 128x128 model is memorizing specific training patterns rather than generalizing, rendering it fragile despite its high scores. Consequently, while Figure 10 displays higher potential accuracy, the stable consistency of the 64x64 model makes it a more robust and reliable choice for practical degradation detection.

Figure 19 illustrates the validation accuracy trajectories for the Vision Transformer (ViT) model using 64x64 inputs across the three target classification categories over 29 steps. The Forest Type classification, represented by the green line, demonstrates immediate and sustained superiority, maintaining a near-perfect smoothed accuracy of 0.998 throughout the entire training phase. In contrast, the Forest Level metric (purple) exhibits a rapid initial learning curve, climbing steeply from the starting steps to stabilize at a high accuracy of 0.9808. The Forest Health accuracy, shown in orange, reflects a more gradual but consistent optimization process, steadily rising to a final smoothed value of 0.9244. The remarkable smoothness of all three curves, free from any significant fluctuations or spikes, confirms that the ViT architecture provides a highly stable and reliable learning environment for these variables.

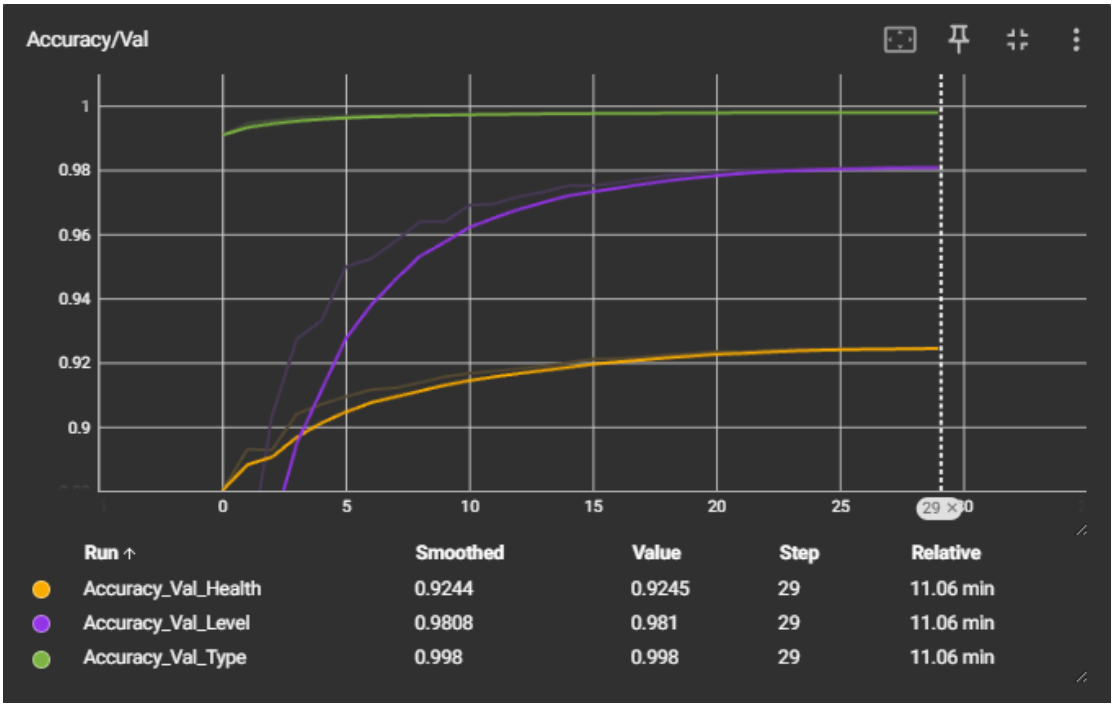


Figure 19. ViT with 64x64 Input Validation Accuracy

Figure 20 displays the validation accuracy for the Vision Transformer (ViT) model trained on 128x128 input images, showcasing exceptional performance across all target variables. Forest Type classification, indicated by the green curve, achieves near-perfection almost immediately, sustaining a smoothed accuracy of 0.9993. The Forest Level metric, shown in purple, exhibits a robust logarithmic ascent, quickly stabilizing at an impressive final value of 0.9931. Meanwhile, the Forest Health accuracy (orange) demonstrates a steady and continuous improvement, reaching a final smoothed value of 0.9402. The remarkably smooth trajectories of all three curves indicate that the increased resolution provides the ViT model with sufficient spatial detail to learn complex features without the volatility seen in other architectures.

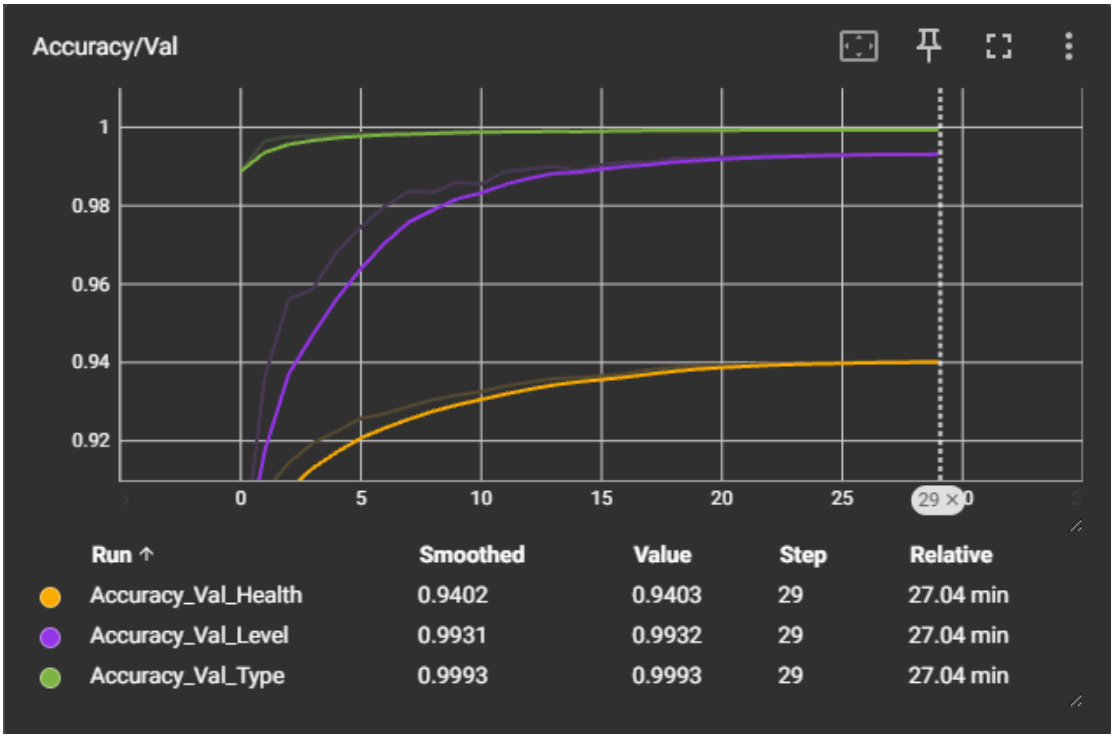


Figure 20. ViT with 128x128 Input Validation Accuracy

Comparing the validation accuracy between Figure 19 and Figure 20 reveals that increasing the resolution to 128x128 consistently enhances the Vision Transformer's performance without compromising stability. The 128x128 configuration demonstrated superior precision across all metrics, boosting the Forest Level accuracy to 0.9931 compared to 0.9808 in the 64x64 model. A similar improvement is observed in the more challenging Forest Health category, where the larger inputs raised the final accuracy from 0.9244 to 0.9402. Critically, unlike the CNN results, both ViT graphs display remarkably smooth learning curves, confirming that the architecture effectively utilizes the additional spatial data to refine its features rather than overfitting. Therefore, the 128x128 resolution proves to be the optimal configuration for the ViT model, maximizing classification power while retaining robust generalization capabilities.

Figure 21 displays the validation accuracy trajectories for the Hybrid model utilizing 64x64 input patches, tracking performance across the three key forest variables over 29 steps. The Forest Type classification, represented by the blue line, exhibits exceptional stability from the outset, maintaining a near-perfect smoothed accuracy of 0.9999 throughout the training duration. The Forest Level metric, shown in grey, overcomes initial volatility in the first five steps to achieve a rapid ascent, eventually stabilizing at a high smoothed value of 0.9961. Similarly, the Forest Health accuracy, depicted by the orange curve, demonstrates a strong and continuous improvement, rising from initial values below 0.90 to a final smoothed score of 0.9939. These high convergence rates across all categories indicate that the Hybrid architecture effectively extracts meaningful spatial and spectral features even at the lower 64x64 resolution.

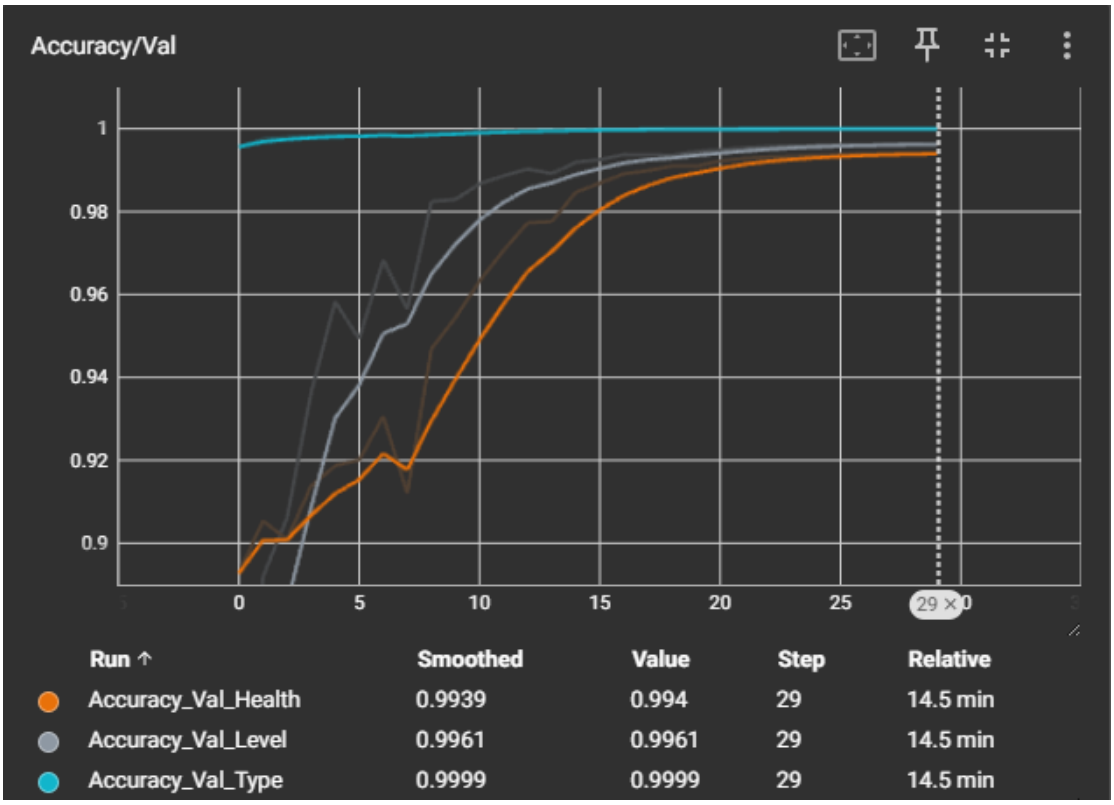


Figure 21. Hybrid with 64x64 Input Validation Accuracy

Figure 22 illustrates the validation accuracy for the Hybrid model trained on 128x128 input images, demonstrating near-perfect final classification capabilities despite temporary mid-training volatility. The Forest Type classification, shown in blue, maintains an immaculate performance record, sustaining a perfect smoothed accuracy of 1.0 throughout the entire validation process. Conversely, the Forest Health (orange) and Forest Level (grey) metrics exhibit transient instability, characterized by sharp distinct drops in accuracy around steps 7 and 15 before immediately rebounding. Despite these fluctuations, the model demonstrates exceptional resilience, rapidly correcting these errors to align all three variables near the maximum accuracy threshold. Ultimately, the training concludes with superior smoothed values of 0.9988 for Health and 0.9995 for Level, confirming that the 128x128 Hybrid configuration delivers the most precise results of the study.

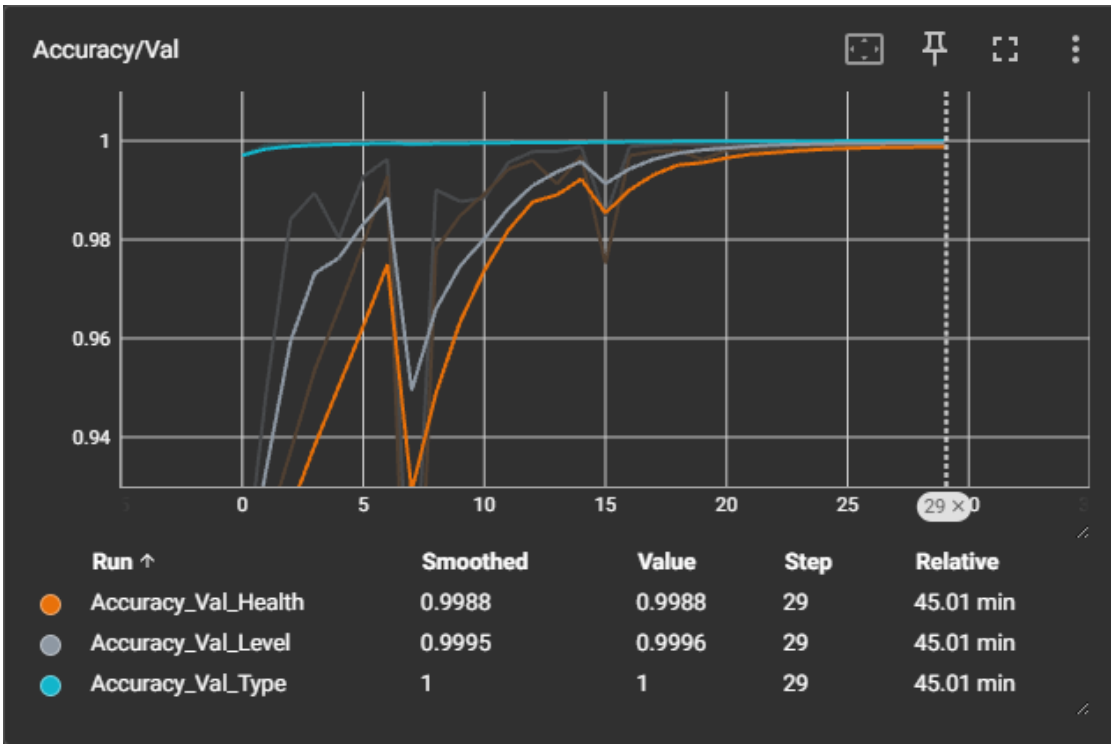


Figure 22. Hybrid with 128x128 Input Validation Accuracy

The comparison between the Hybrid model inputs reveals that while the 64x64 configuration offers superior training stability, the 128x128 resolution ultimately yields higher predictive precision. The 64x64 model demonstrated a remarkably smooth learning curve with minimal fluctuations, achieving strong final accuracy scores of 0.9961 for Forest Level and 0.9939 for Forest Health. Conversely, the 128x128 configuration experienced distinct mid-training volatility, evidenced by sharp drops around steps 7 and 15, yet it recovered swiftly to surpass the smaller model's performance. Unlike the instability seen in the CNN architecture, this volatility did not signal overfitting, as the model achieved a near-perfect final accuracy of 0.9988 for Health supported by extremely low loss values. Thus, despite the transient fluctuations, the 128x128 Hybrid model proves to be the most effective architecture, leveraging the increased spatial resolution to maximize classification accuracy across all variables.

Figure 23 presents the F1 Macro scores for the CNN model trained on 64x64 inputs, evaluating performance across Forest Type, Level, and Health over 24 validation steps. The Forest Type metric, shown in purple, demonstrates consistent stability from the beginning, maintaining high precision throughout the process to conclude with a strong smoothed score of 0.976. Forest Level classification, represented by the yellow curve, initially exhibits significant fluctuation but achieves a rapid recovery to eventually reach the highest final smoothed value of 0.9831. In contrast, the Forest Health metric displayed in pink proves to be the most challenging variable, starting with low values around 0.60 before gradually climbing to a final smoothed score of 0.875. These distinct trajectories indicate that while the model effectively identifies broad categories

like forest type and level, it requires more training effort to resolve the complex nuances of forest health status.



Figure 23. CNN with 64x64 Validation F1-Score

Figure 24 illustrates the validation F1 Macro scores for the CNN model utilizing 128x128 input images, revealing high peak performance tempered by significant instability. The Forest Type classification, depicted in purple, maintains the most consistent trajectory among the variables, culminating in a strong smoothed score of 0.9852. However, the model exhibits severe volatility, most notably characterized by a precipitous decline in performance across all categories around the twentieth training step. Despite this drastic fluctuation, the metrics for Forest Level (yellow) and Forest Health (pink) successfully rebound to achieve final smoothed values of 0.9735 and 0.9627, respectively. While these final scores are numerically superior to the 64x64 configuration,

the erratic behavior and deep performance spikes underscore the model's struggle with overfitting on the larger image dimensions.

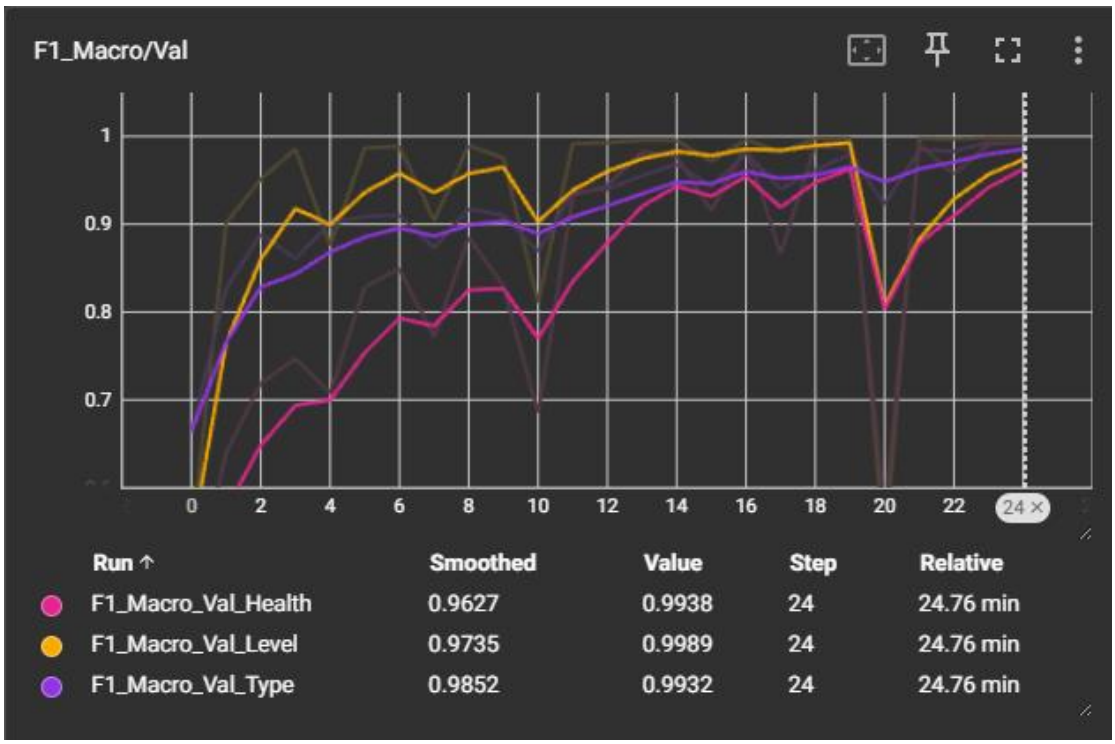


Figure 24. CNN with 128x128 Validation F1-Score

The comparison of F1 scores between Figures 23 and 24 underscores a critical divergence between predictive reliability and peak classification power. The 128x128 model achieved significantly higher final smoothed F1 scores, particularly in the challenging Forest Health category where it reached 0.9627 compared to only 0.875 for the 64x64 configuration. However, this performance advantage is compromised by extreme volatility, most notably a drastic simultaneous drop in all metrics around step 20, which signals that the model is overfitting to the training data. In contrast, the 64x64 model displays a consistent and gradual improvement trajectory without such severe fluctuations, indicating a more stable, albeit slightly less precise, learning process. Therefore, while the 128x128 inputs unlock higher potential for

identifying complex health statuses, the 64x64 inputs provide a safer and more generalizable foundation for the CNN architecture.

Figure 25 displays the validation F1 scores for the Vision Transformer (ViT) model using 64x64 inputs, illustrating a stable but stratified performance across the three classification categories. The Forest Type classification shown in blue demonstrates consistent improvement, settling at a solid value of 0.9343 by the end of the training session. Similarly, the Forest Level metric, represented by the grey curve, exhibits the most robust learning trajectory, rising steeply to achieve a superior final smoothed score of 0.9796. However, the Forest Health metric, depicted in orange, lags significantly behind, starting from a low baseline and ascending slowly to reach a final smoothed value of only 0.7998. This disparity highlights that while the 64x64 ViT model effectively handles broader spatial categories, it struggles to resolve the finer spectral details necessary for high-precision forest health analysis.

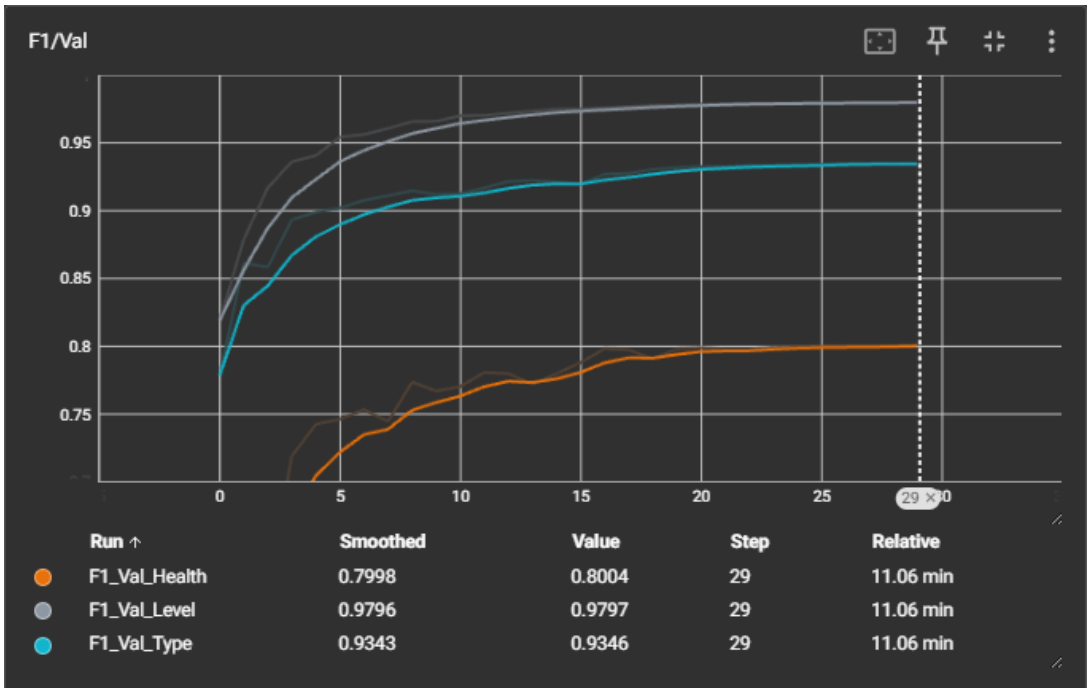


Figure 25. ViT with 64x64 Validation F1-Score

Figure 26 illustrates the validation F1 Macro scores for the Vision Transformer (ViT) model trained on 128x128 inputs, showing distinct performance tiers across the three forest variables. The Forest Type metric, shown in blue, experiences notable volatility around the seventh step but recovers to maintain a stable trajectory, ending with a reliable value of 0.9278. Forest Level classification, depicted by the grey line, achieves exceptional results, rising smoothly to secure a near-perfect final smoothed score of 0.9914. In contrast, Forest Health classification (orange) remains the most difficult task, showing a steady but slower rate of improvement to conclude at a moderate score of 0.8297. While the increased resolution boosted the Forest Level and Health scores compared to the smaller input size, the persistent gap between Health and the other variables indicates that complex degradation patterns remain challenging for the standalone ViT architecture.

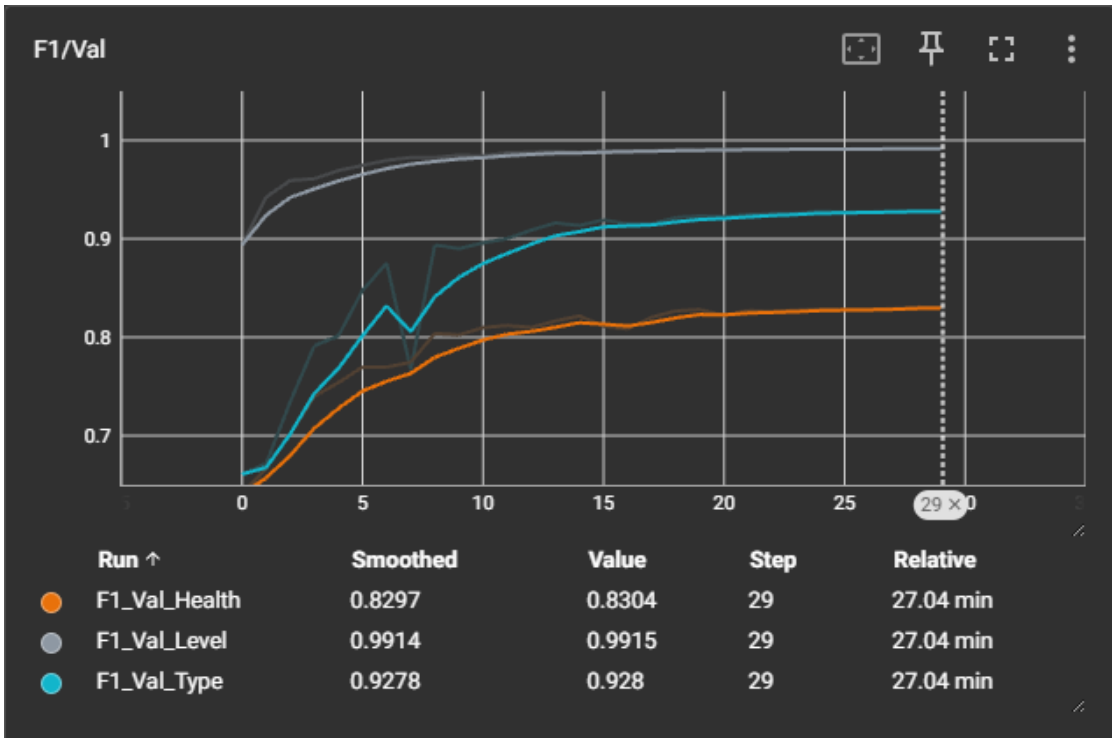


Figure 26. ViT with 128x128 Validation F1-Score

The comparison of F1 scores reveals that increasing the input resolution to 128x128 enhances the Vision Transformer's ability to classify complex forest attributes, particularly in the more difficult categories. Specifically, the 128x128 model achieved a near-perfect smoothed score of 0.9914 for Forest Level, distinctively surpassing the 0.9796 peak achieved by the 64x64 configuration. Crucially, the larger input size also improved performance in the challenging Forest Health category, raising the final F1 score from 0.7998 to 0.8297. Although the 128x128 model exhibited minor early volatility in the Forest Type metric, both resolutions maintained the characteristic stability of the ViT architecture without the drastic crashes observed in the CNN models. Ultimately, the 128x128 configuration proves to be the superior choice, as the added spatial detail allows for more accurate differentiation of critical variables like health and level.

Figure 27 displays the validation F1 Macro scores for the Hybrid model utilizing 64x64 input patches, tracking performance across the three target variables over 29 steps. The metrics for Forest Type, shown in purple, and Forest Level, in yellow, demonstrate immediate and robust reliability, climbing smoothly to achieve near-perfect final smoothed scores of 0.9965 and 0.9952. In contrast, the Forest Health metric (pink) exhibits significant initial difficulty with low values and erratic fluctuations during the first eight steps. However, the model rapidly corrects this volatility, executing a steep upward convergence to reach a highly impressive final score of 0.986. This trajectory confirms that while the 64x64 Hybrid model requires a brief warm-up period for complex health features, it ultimately attains exceptional classification balance across all categories.

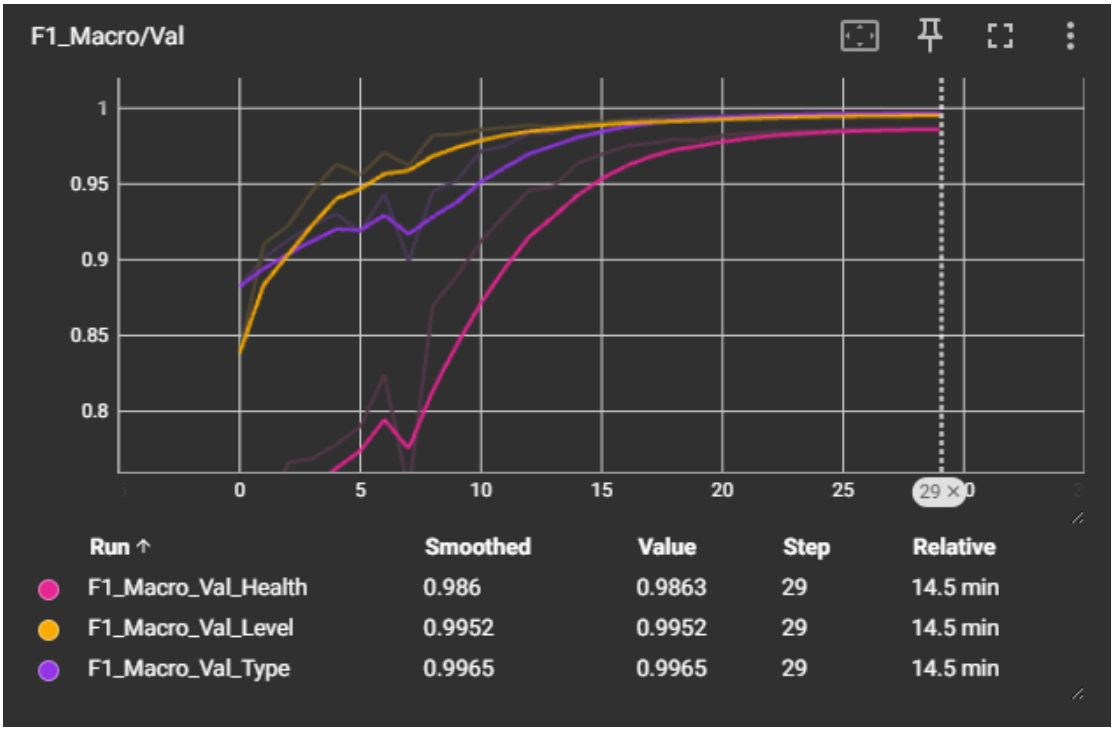


Figure 27. Hybrid with 64x64 Validation F1-Score

Figure 28 illustrates the validation F1 Macro performance for the Hybrid model utilizing larger 128x128 input patches over the course of 29 training steps. The Forest Type in purple follows a steady upward trend to achieve a final smoothed score of 0.9954, while Forest Level metric, depicted in yellow, maintains superior stability throughout the training process. Conversely, the Forest Health variable (pink) demonstrates significant early instability, characterized by a sharp, deep decline near step seven before initiating a rapid recovery. Following this volatility, the model effectively stabilizes all parameters, with the Health metric rebounding aggressively to reach an impressive final value of 0.997. This performance indicates that while the larger 128x128 input size introduces transient noise during early epochs, it ultimately facilitates superior feature convergence and near-perfect classification accuracy.

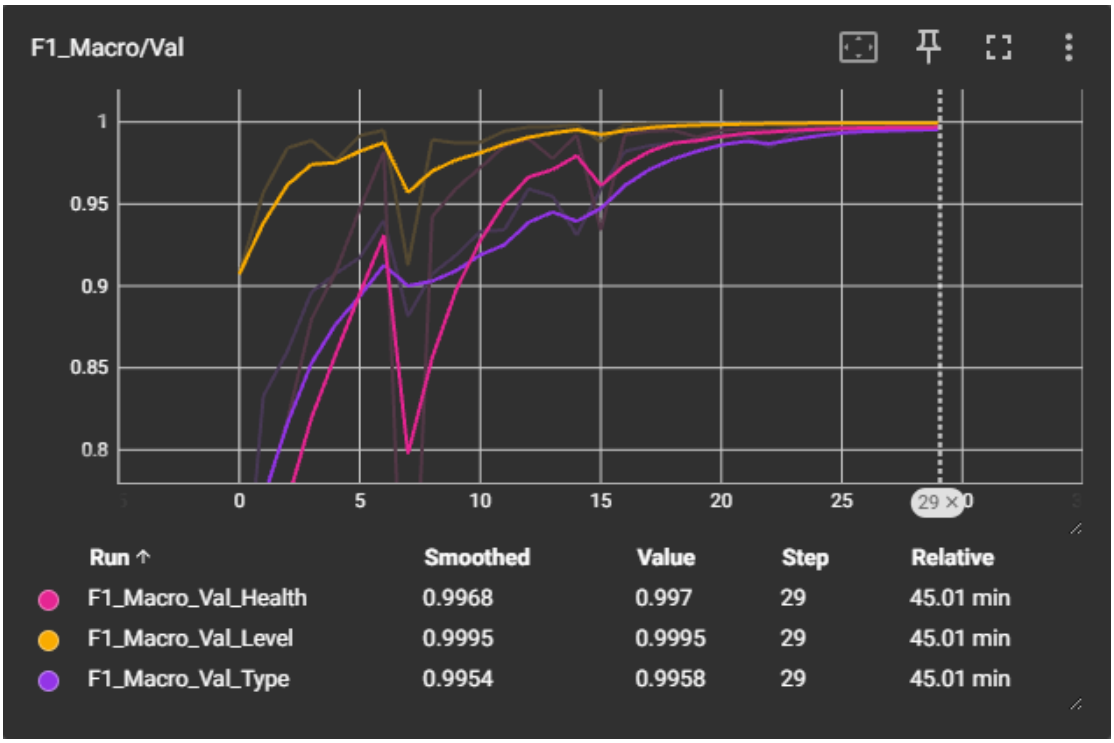


Figure 28. Hybrid with 128x128 Validation F1-Score

The comparison of F1 scores reveals that increasing the input resolution to 128x128 enhances the Hybrid model's ability to classify complex forest attributes, particularly in the more difficult categories. Specifically, the 128x128 model achieved a near-perfect smoothed score of 0.9995 for Forest Level, distinctively surpassing the 0.9952 peak achieved by the 64x64 configuration. Crucially, the larger input size also improved performance in the challenging Forest Health category, raising the final F1 score from 0.986 to 0.997. Although the 128x128 model exhibited distinct early volatility in the Forest Health metric, both resolutions maintained the characteristic robustness of the Hybrid architecture, rapidly correcting initial fluctuations to achieve high convergence. Ultimately, the 128x128 configuration proves to be the superior choice, as the added spatial detail allows for more accurate differentiation of critical variables like health and level.

Figure 29 displays the validation IoU Macro scores for the CNN model utilizing 64x64 input patches, tracking performance across the three target variables over 24 steps. The metrics for Forest Type, shown in grey, and Forest Level, in orange, demonstrate strong reliability, climbing to achieve superior final smoothed scores of 0.9554 and 0.9674, respectively. In contrast, the Forest Health metric (green) exhibits significant difficulty, characterized by a low starting baseline and persistent lag behind the other categories throughout the training. However, the model manages a steady upward progression despite this volatility, eventually stabilizing to reach a moderate final score of 0.7968. This trajectory confirms that while the 64x64 CNN model effectively captures structural and species features, it faces limitations in resolving the complex spatial nuances required for accurate health assessment.

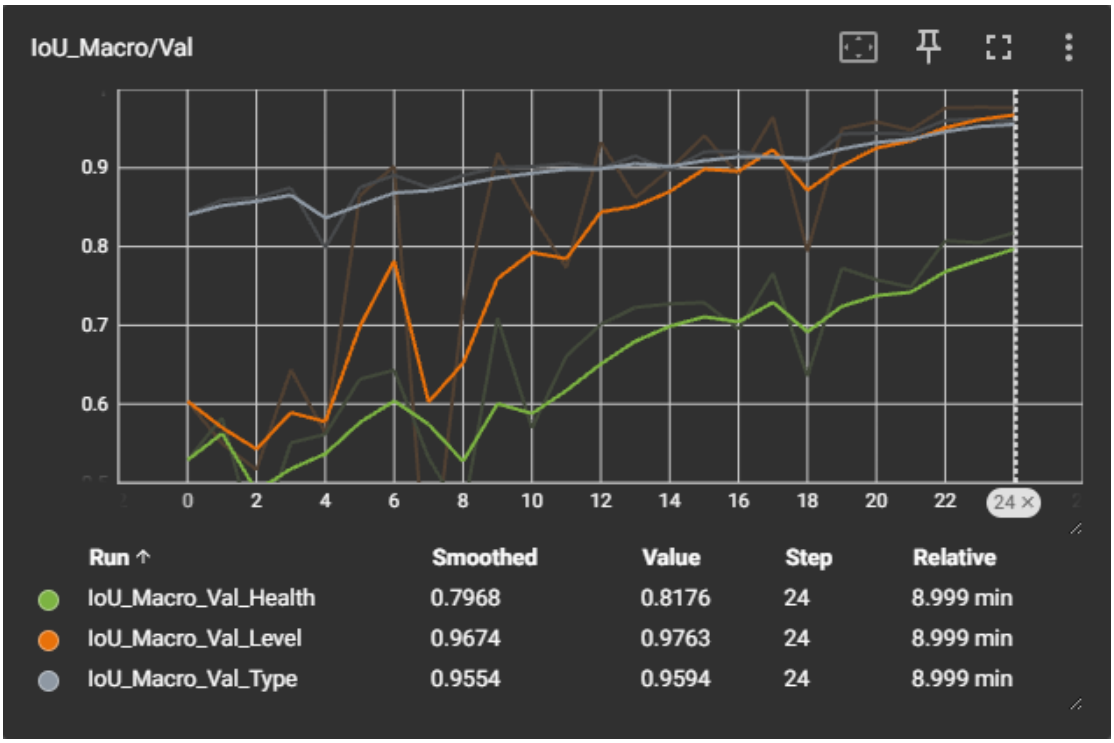


Figure 29. CNN with 64x64 Validation IoU

Figure 30 displays the validation IoU Macro scores for the CNN model utilizing 128x128 input patches, tracking performance across the three target variables over 24 steps. The metrics for Forest Type, shown in grey, and Forest Level, in orange, demonstrate strong initial acceleration, climbing rapidly to achieve robust final smoothed scores of 0.9731 and 0.9667. In contrast, the Forest Health metric (green) exhibits a steeper learning curve and distinct late-stage volatility, marked by a sharp systemic drop shared by all classes near step 20. However, the model rapidly corrects this instability, executing a steep upward recovery to reach a highly impressive final smoothed score of 0.9476. This trajectory confirms that while the 128x128 CNN model is susceptible to transient fluctuations, the expanded spatial context ultimately drives significantly superior convergence compared to the smaller input configuration.



Figure 30. CNN with 128x128 Validation IoU

The comparison of IoU scores reveals that increasing the input resolution to 128x128 drastically enhances the CNN's ability to segment complex forest attributes, effectively solving the performance bottleneck seen in the smaller model. Specifically, the 128x128 model achieved a massive improvement in the challenging Forest Health category, elevating the final smoothed score to 0.9476, whereas the 64x64 configuration plateaued significantly lower at 0.7968. Crucially, the larger input size also refined the Forest Type classification, raising the score from 0.9554 to 0.9731, while maintaining consistent high performance for Forest Level. Although the 128x128 model exhibited a distinct systemic volatility near step 20, it demonstrated superior recoverability, rapidly rebounding to surpass the stable but limited baselines of the 64x64 model. Ultimately, the 128x128 configuration proves to be the superior choice, as the added spatial context is essential for resolving the subtle spectral differences required for accurate health segmentation.

Figure 31 describes the validation IoU performance for the Vision Transformer (ViT) model utilizing 64x64 input patches across 29 training steps. The Forest Level metric, illustrated in yellow, demonstrates the most robust learning trajectory, ascending consistently from an initial baseline of 0.7 to achieve a superior final smoothed score of 0.96. In comparison, the Forest Type variable (purple) exhibits a moderate performance plateau, rising steadily before stabilizing at a final value of 0.8891. Conversely, the Forest Health metric (pink) presents a significant bottleneck for the ViT architecture, emerging from extremely low initial values to reach a modest final score of only 0.7028. This substantial performance gap highlights a limitation in the 64x64 ViT

configuration, suggesting that the model struggles to capture the fine-grained spectral dependencies required for accurate health assessment within such a constrained spatial window.

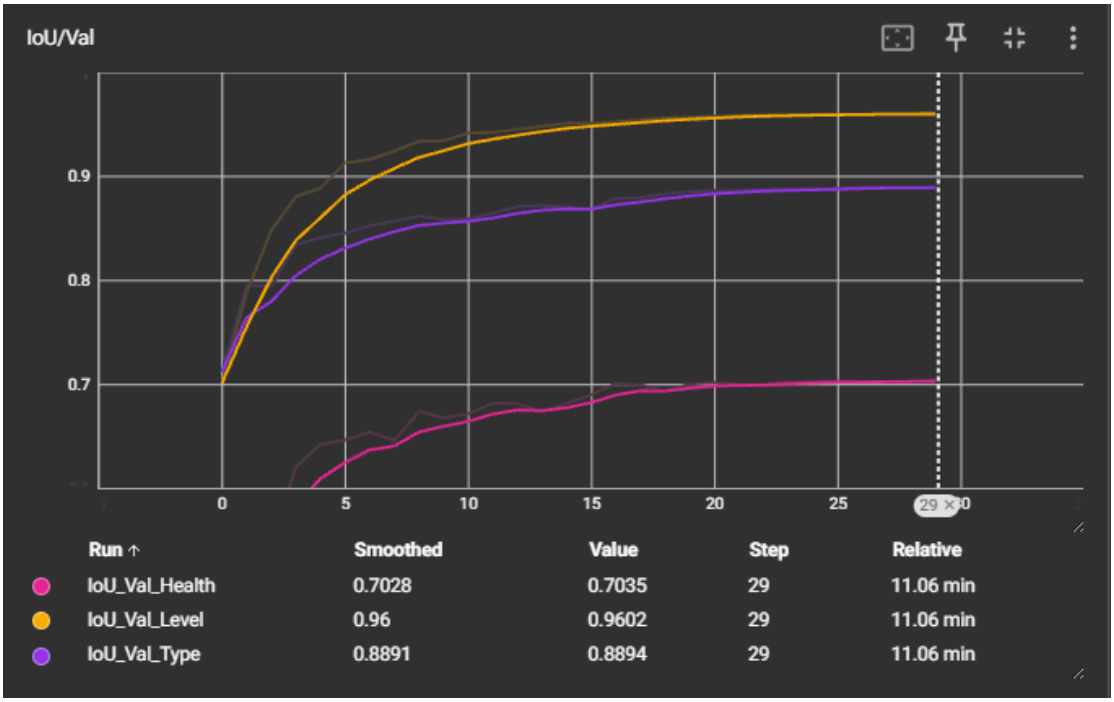


Figure 31. ViT with 64x64 Validation IoU

Figure 32 shows the validation IoU Macro performance for the Vision Transformer (ViT) architecture utilizing 128x128 input patches, illustrating distinct learning behaviors across the three target variables over 29 steps. The Forest Level metric, represented in yellow, exhibits exceptional stability and rapid convergence, initiating with a strong baseline and ascending smoothly to achieve a dominant final smoothed score of 0.983. In contrast, the Forest Type classification (purple) displays moderate early-stage volatility, particularly distinct fluctuations between steps five and eight, before stabilizing into a steady plateau with a final value of 0.8808. The Forest Health metric (pink) remains the most challenging parameter, characterized by a slow, gradual ascent from a low starting point to a modest final smoothed score of 0.738. This

performance stratification suggests that while the expanded 128x128 spatial context significantly benefits density classification, the ViT architecture still faces substantial difficulties in resolving the intricate spectral features necessary for high-precision health segmentation.

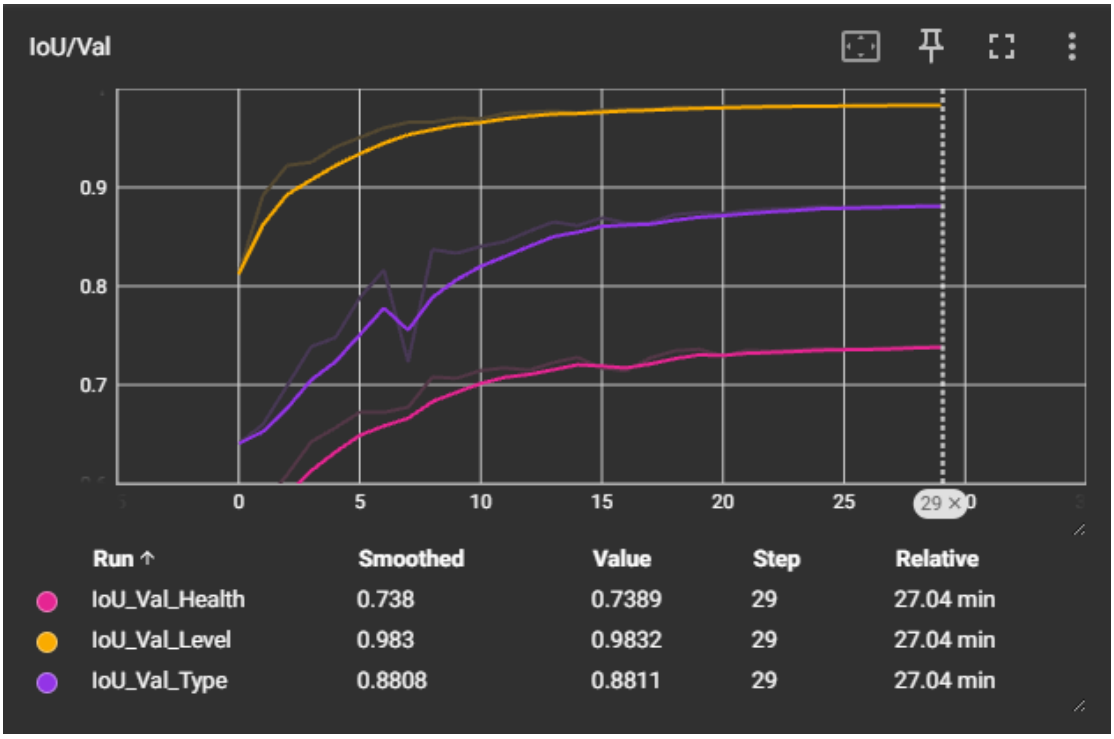


Figure 32. ViT with 128x128 Validation IoU

The comparison of IoU scores reveals that increasing the input resolution to 128x128 provides targeted enhancements to the Vision Transformer's ability to segment structural attributes, though it yields diminishing returns for species classification. Specifically, the 128x128 model achieved a distinct improvement in the Forest Level category, elevating the final smoothed score to 0.983, whereas the 64x64 configuration plateaued at a lower baseline of 0.96. Crucially, the larger input size also yielded a marginal gain in the challenging Forest Health category, raising the score from 0.7028 to 0.738, though the Forest Type classification surprisingly showed no benefit, stagnating near 0.88 across both resolutions. Although the 128x128 model exhibited

increased volatility in the Type metric during early epochs, it successfully stabilized, confirming that the larger patch size does not compromise training convergence. Ultimately, the 128x128 configuration proves to be the superior choice, as the added spatial context drives superior performance in density detection, even if the architecture remains constrained in resolving the finest health and species details.

Figure 33 shows the validation IoU Macro performance for the Hybrid model utilizing 64x64 input patches across 29 training steps. The metrics for Forest Type, shown in grey, and Forest Level, in orange, demonstrate immediate and robust reliability, climbing consistently to achieve near-perfect final smoothed scores of 0.9931 and 0.9904, respectively. In contrast, the Forest Health metric (green) exhibits significant initial difficulty, starting with extremely low values and struggling with erratic fluctuations during the first eight steps. However, the model rapidly corrects this volatility, executing a steep, continuous upward convergence to eventually reach a highly impressive final score of 0.9727. This trajectory confirms that while the 64x64 Hybrid model requires a brief warm-up period for complex health features, it ultimately attains exceptional segmentation balance across all categories.

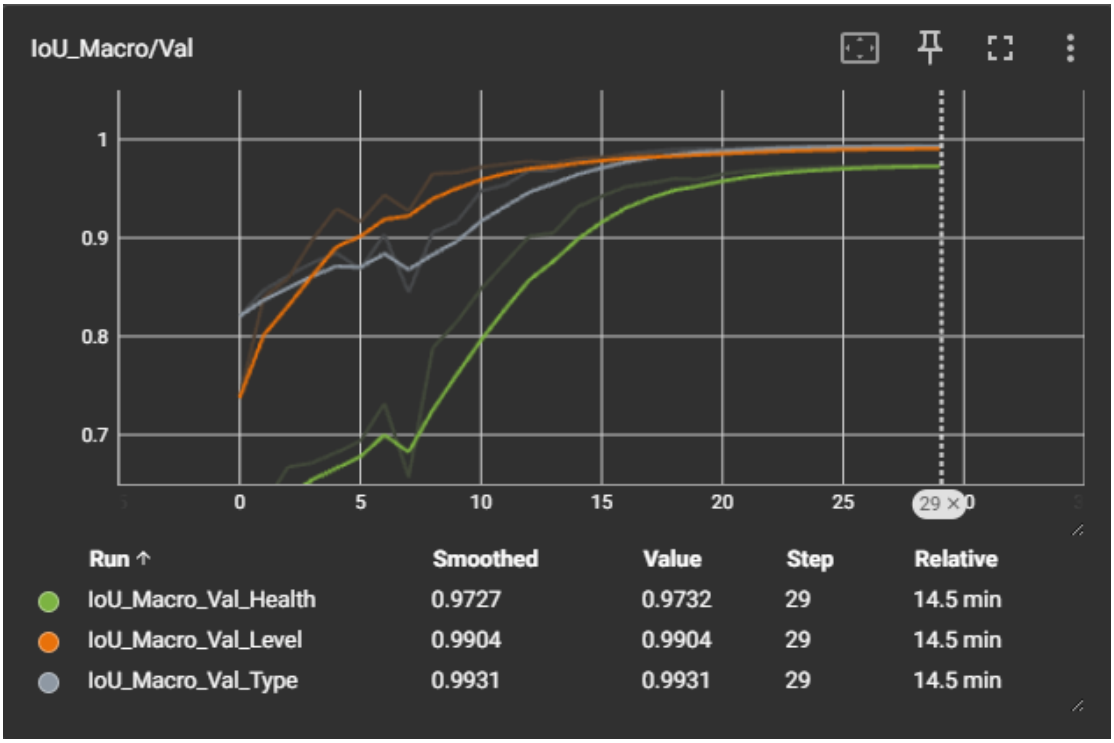


Figure 33. Hybrid with 64x64 Validation IoU

Figure 34 illustrates the validation IoU Macro performance for the Hybrid model utilizing larger 128x128 input patches over the course of 29 training steps. The Forest Level metric, depicted in orange, maintains superior dominance throughout, recovering quickly from early fluctuations to achieve a near-perfect final smoothed score of 0.9989, while Forest Type (grey) follows a consistent upward trajectory to reach 0.9909. Conversely, the Forest Health variable (green) demonstrates significant early instability, characterized by a sharp, deep decline near step seven and distinct transient noise around step fifteen. Following this volatility, the model effectively stabilizes all parameters, with the Health metric rebounding aggressively to reach an exceptional final value of 0.9937. This performance indicates that while the larger 128x128 input size introduces transient noise during the learning phase, it ultimately facilitates superior feature convergence and near-perfect segmentation accuracy.

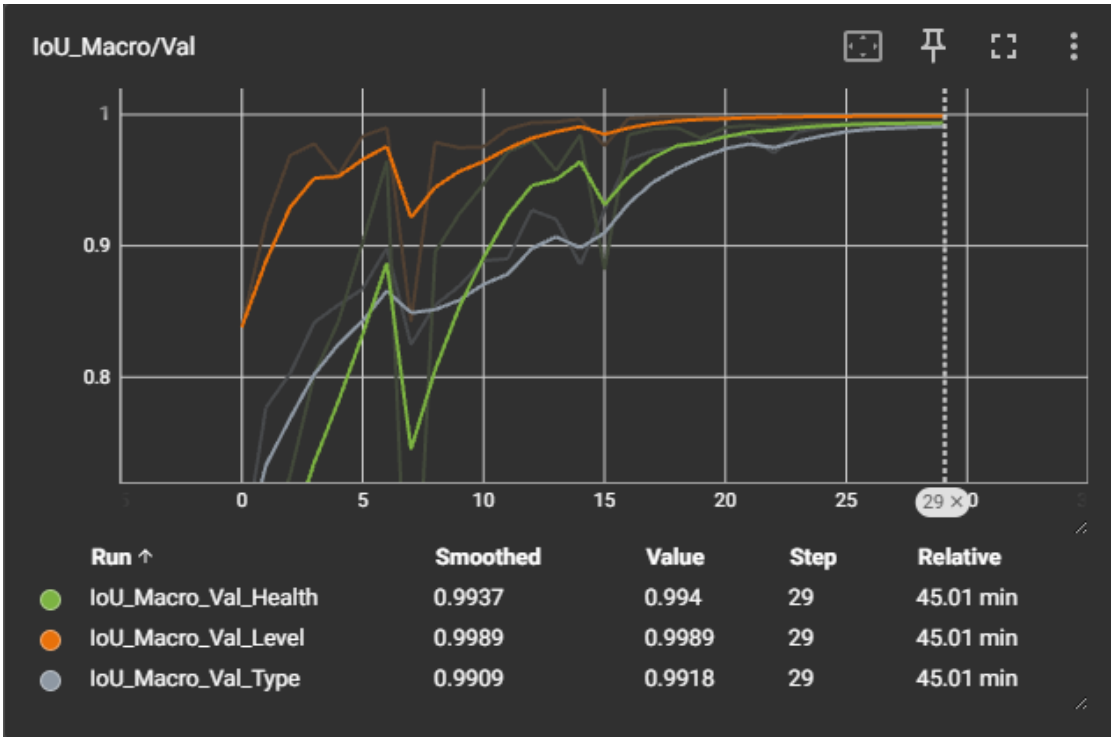


Figure 34. Hybrid with 128x128 Validation IoU

The comparison of IoU scores reveals that increasing the input resolution to 128x128 further refines the Hybrid model's segmentation capabilities, particularly for the most challenging attributes. Specifically, the 128x128 model achieved a near-perfect smoothed score of 0.9989 for Forest Level, noticeably surpassing the 0.9904 peak achieved by the 64x64 configuration. Crucially, the larger input size also boosted performance in the difficult Forest Health category, elevating the final smoothed score to 0.9937 compared to the 0.9727 baseline of the smaller model. Although the 128x128 model exhibited distinct mid-training volatility, which is marked by sharp fluctuations in the Health metric near step seven, it demonstrated superior recoverability compared to the slower, albeit steadier, ascent of the 64x64 model. Ultimately, the 128x128 configuration proves to be the superior choice, as the added spatial context drives the highest possible convergence for critical degradation indicators without compromising the model's overall stability.

Model Testing and Accuracy Assessment

Following the hyperparameter tuning and performance monitoring in the validation phase, this section presents the final evaluation of the trained models using an independent test dataset. Unlike the validation set, which influenced model optimization, the test set consists of completely unseen data, providing an unbiased measure of the models' true generalization capabilities. The evaluation is conducted across two distinct temporal contexts: the reserved split of the 2020 dataset and the subsequent 2021 dataset, allowing for an assessment of both immediate accuracy and temporal robustness. Key metrics, including Overall Accuracy, Mean Intersection over Union (mIoU), and F1-Score, are analyzed to determine the precision of each architecture in classifying Forest Type, Level, and Health. Ultimately, these results serve as the definitive basis for selecting the optimal model to power the tree species recommendation system.

Figure 35 presents the test results for the CNN model with 64x64 inputs on the 2020 dataset, revealing exceptional performance in identifying broad forest categories. The Forest Type classification achieved near-perfect precision with an overall accuracy of 99%, demonstrating the model's ability to distinguish forest areas from non-forest elements with negligible error. Similarly, the Forest Level assessment displayed robust generalization, securing an accuracy of 99% and a high Mean IoU of 98% across all elevation zones. While the confusion matrix shows minor misclassifications between Lowland and Lower Montane regions, the strong diagonal distribution confirms that the model effectively retained the spatial features learned during training. These results mirror the stability observed in the validation phase, confirming

that the 64x64 architecture is highly reliable for general structural forest segmentation.

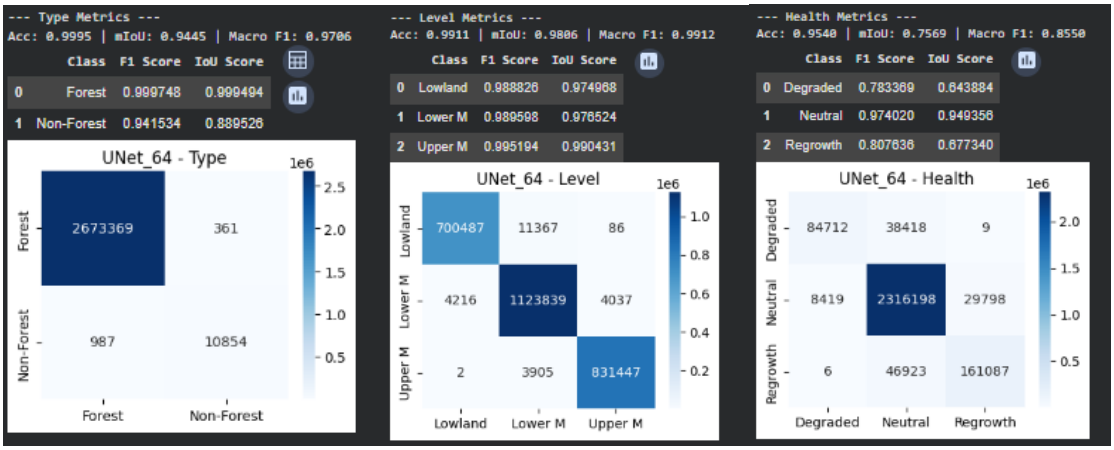


Figure 35. CNN with 64x64 Test Accuracy 2020

In contrast to the structural categories, the Forest Health classification proved to be the most challenging task for the model, yielding a significantly lower Mean IoU of 75% despite a decent overall accuracy of 95%. The class-wise metrics reveal a sharp disparity, where the dominant "Neutral" class achieved a high F1 score of 97%, while the "Degraded" and "Regrowth" categories dropped to IoU scores of roughly 64% and 67% respectively. Visual evidence from the confusion matrix indicates a recurring issue where the model frequently misclassifies subtle degradation and regrowth patterns as the stable "Neutral" class. This tendency suggests that the limited 64x64 spatial resolution struggles to capture the fine-grained spectral nuances required to differentiate early-stage forest recovery or damage from healthy vegetation. Consequently, while the model is structurally sound, these test results confirm the validation finding that the standalone CNN architecture faces limitations in precise health assessment. The detailed performance metrics regarding these structural strengths and health-based limitations are summarized in Table 26.

Table 26. CNN with 64x64 Test Set 2020 Table

Target Variable	Accuracy	Mean IoU (mIoU)	Macro F1 Score
Forest Type	0.9995	0.9445	0.9706
Forest Level	0.9911	0.9806	0.9912
Forest Health	0.9540	0.7569	0.8550

Figure 36 illustrates the test performance of the 64x64 CNN model on the 2021 dataset, highlighting a sharp decline in generalization capabilities compared to the previous year. While the Forest Type classification maintained a high overall accuracy of 99.59%, the ability to correctly identify non-forest areas deteriorated significantly, with the Non-Forest IoU dropping to 40.25%. The Forest Level classification suffered a more severe setback, achieving an accuracy of only 51.53% and a Mean IoU of 32.21%, indicating major confusion between elevation zones. The confusion matrix reveals that a vast number of "Lowland" pixels were incorrectly classified as "Lower Montane," suggesting that the model failed to adapt to slight topographical or spectral shifts in the 2021 imagery. This drastic reduction in performance indicates that the 64x64 CNN architecture overfitted to the specific spatial distributions of the 2020 data and lacks the robust features necessary for temporal transferability.

The Forest Health classification mirrored this downward trend, exhibiting a critical failure to detect specific degradation patterns in the new dataset. The overall accuracy dropped to 69.91%, while the Mean IoU plummeted to 30.61%, driven by the model's inability to distinguish minority classes. The most alarming result was observed in the "Regrowth" category, which yielded a near-zero IoU of 1.60%, as the model overwhelmingly misclassified these areas as "Neutral" or "Degraded." The confusion matrix confirms that the model defaulted to predicting the majority "Neutral" class (which maintained a decent

IoU of 70.46%) whenever it encountered ambiguous spectral data. These results definitively prove that the standalone CNN with low-resolution 64x64 inputs is insufficient for monitoring year-to-year forest health changes, as it cannot resolve the complex spectral evolutions associated with recovering vegetation.

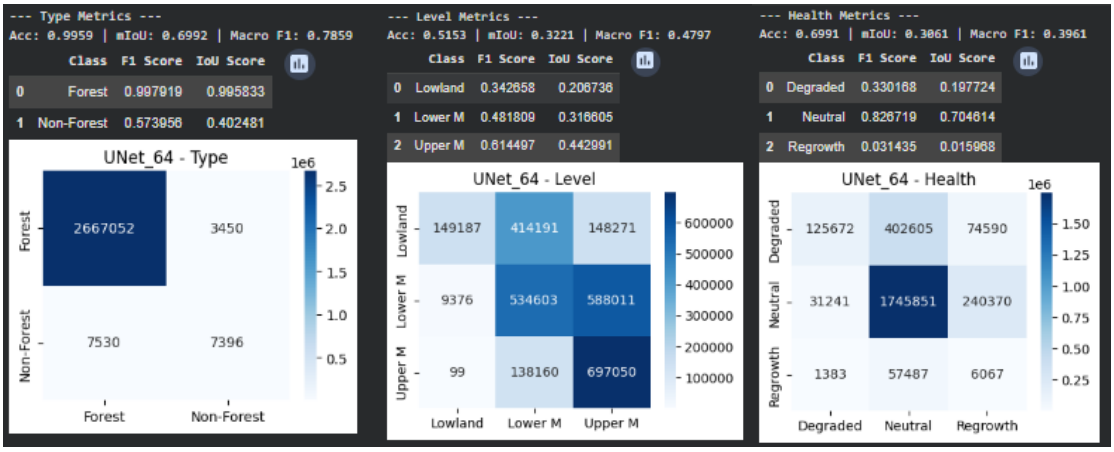


Figure 36. CNN with 64x64 Test Accuracy 2021

The temporal evaluation of the CNN 64x64 model reveals a catastrophic failure in generalization when applied to the 2021 dataset. While the model achieved a robust 0.9806 mIoU for Forest Level in 2020, this score plummeted by 67.15% to a mere 0.3221 in 2021. This confirms that the model did not learn generalized topographical features but rather overfitted to the specific spatial arrangement of the 2020 imagery. Furthermore, the Forest Health detection capabilities collapsed (-59.56%), with the specific IoU for the 'Regrowth' class falling to 0.0159 (1.6%), rendering the model effectively blind to vegetation recovery in the subsequent year.

Table 27. Temporal Stability Analysis: CNN (64x64) Performance (2020 vs. 2021)

Target Variable	2020 mIoU (Baseline)	2021 mIoU (Stress Test)	Performance Drop
Forest Type	0.9445	0.6992	-25.97%
Forest Level	0.9806	0.3221	-67.15%
Forest Health	0.7569	0.3061	-59.56%

Figure 37 shows the test performance of the CNN model using 128x128 input patches on the 2020 dataset, with near-perfect classification across all structural categories. The Forest Type classification achieved an impeccable overall accuracy of 100.00% and a Mean IoU of 97.80%, indicating that the expanded spatial context enabled the model to eliminate confusion between forest and non-forest areas virtually. Similarly, the Forest Level metrics showed exceptional precision, achieving 99.91% accuracy and a Mean IoU of 99.80%, with individual class scores for the Lowland and Montane regions all exceeding 99.6%. The confusion matrix reflects this dominance, showing extremely sparse off-diagonal errors, such as only 2,813 misclassified pixels out of over two million in the Lowland category. These results suggest that the 128x128 resolution provides sufficient contextual information for the CNN to fully resolve the spatial hierarchy of the forest terrain within the 2020 timeframe.

The most significant improvement provided by the larger input size is evident in the Forest Health classification, which previously struggled in the 64x64 configuration. The model achieved a remarkable overall accuracy of 99.75% and a Mean IoU of 98.37%, successfully differentiating complex vegetative states that usually confuse smaller models. Unlike the smaller architecture, the rare "Degraded" and "Regrowth" classes achieved high IoU

scores of 97.85% and 97.55% respectively, proving that the model can now distinguish subtle textural differences in vegetation health. The confusion matrix further validates this, as the interference between the stable "Neutral" class and the dynamic "Regrowth" class was minimized to negligible levels. However, given the validation phase's warning regarding instability and overfitting, these near-flawless scores on the same-year data suggest the model may have memorized these specific spatial patterns rather than learning purely generalized features. The detailed performance metrics regarding these structural strengths and health-based limitations are summarized in Table 27.

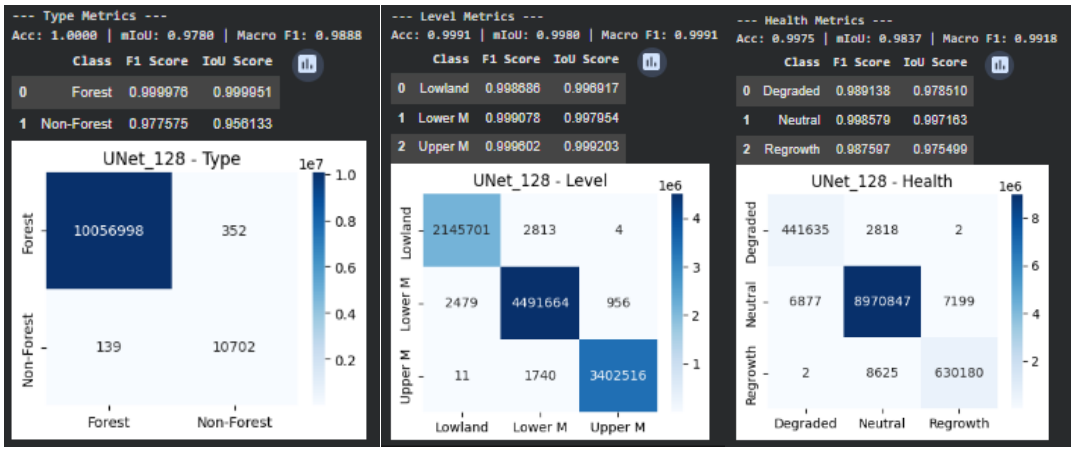


Figure 37. CNN with 128x128 Test Accuracy 2020

In contrast to the structural categories, the Forest Health classification proved to be the most challenging task for the model, yielding a significantly lower Mean IoU of 75% despite a decent overall accuracy of 95%. The class-wise metrics reveal a sharp disparity, where the dominant "Neutral" class achieved a high F1 score of 97%, while the "Degraded" and "Regrowth" categories dropped to IoU scores of roughly 64% and 67% respectively. Visual evidence from the confusion matrix indicates a recurring issue where the model frequently misclassifies subtle degradation and regrowth patterns as the stable

"Neutral" class. This tendency suggests that the limited 64x64 spatial resolution struggles to capture the fine-grained spectral nuances required to differentiate early-stage forest recovery or damage from healthy vegetation. Consequently, while the model is structurally sound, these test results confirm the validation finding that the standalone CNN architecture faces limitations in precise health assessment. The detailed performance metrics regarding these structural strengths and health-based limitations are summarized in Table 26.

Table 28. CNN with 128x128 Test Set 2020 Table

Target Variable	Accuracy	Mean IoU (mIoU)	Macro F1 Score
Forest Type	0.9991	0.9980	0.9991
Forest Level	1.0000	0.9780	0.9888
Forest Health	0.9975	0.9837	0.9918

Figure 38 illustrates the test performance of the 128x128 CNN model on the 2021 dataset, revealing catastrophic generalization failure despite near-perfect scores on the 2020 data. While the Forest Type accuracy is high at 99.76%, the Non-Forest IoU is low at 25.60%, indicating the model is biased toward predicting "Forest" regardless of the ground truth. The Forest Level classification suffered an even steeper decline, recording an accuracy of only 52.65% and a Mean IoU of 31.32%, which suggests the model completely lost its spatial bearing. The confusion matrix highlights a massive systemic error where over 2.4 million pixels belonging to the "Lower Montane" class were incorrectly labeled as "Upper Montane." This distinct shift proves that the model memorized the specific topographical layout of the 2020 training set and could not adapt to the slight temporal or spectral variations present in the 2021 imagery.

The Forest Health classification results provide definitive evidence of overfitting, with overall accuracy dropping to 74.20% and Mean IoU falling to 35.69%. While the majority of the "Neutral" class maintained a passable IoU of 73.45%, the model proved nearly incapable of identifying dynamic forest changes in the new dataset. The "Regrowth" category was virtually ignored, resulting in a negligible IoU of 2.71%, as the confusion matrix shows the model overwhelmingly misclassified these recovering areas as "Neutral." Similarly, the "Degraded" class achieved a poor IoU of 30.91%, further emphasizing the model's inability to distinguish stress signals from healthy vegetation. These disparities confirm that while the larger 128x128 input size improved spatial context, the standalone CNN architecture lacks the robustness to perform temporal transfer learning across different years reliably.

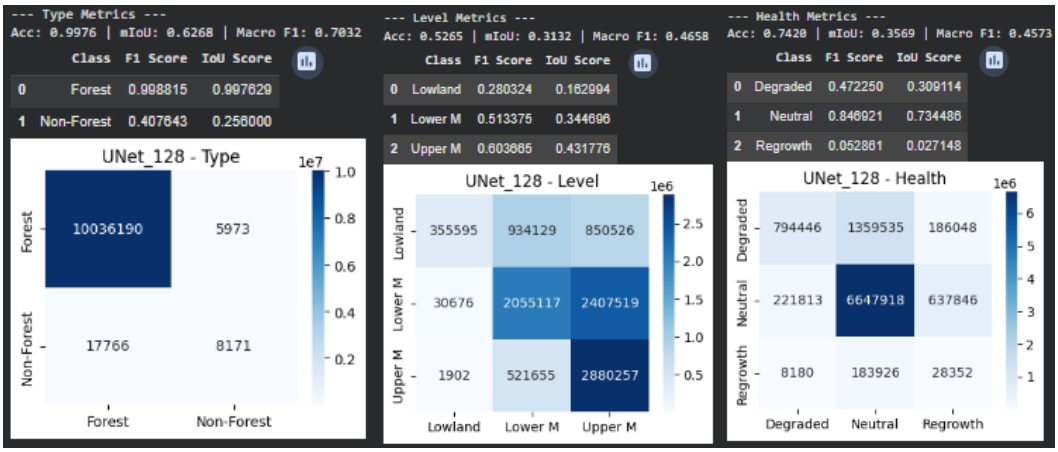


Figure 38. CNN with 128x128 Test Accuracy 2021

Table 29 illustrates the dramatic performance degradation of the CNN 128x128 architecture under the temporal stress test. Despite achieving near-saturated benchmarks on the baseline data, the model exhibited a massive 68.61% drop in Forest Level mIoU and a 63.72% drop in Forest Health mIoU when applied to the 2021 dataset. This sharp contrast reveals a critical

limitation: while the expanded 128x128 spatial context allowed the model to memorize the topographical and vegetative signatures of 2020 perfectly, it failed to learn generalized features that persist over time. The collapse of the 'Regrowth' detection capabilities (dropping to an IoU of 0.027) indicates that the model's high baseline precision relied heavily on specific, static spectral conditions that did not carry over to the subsequent year.

Table 29. Temporal Stability Analysis: CNN (128x128) Performance (2020 vs. 2021)

Target Variable	2020 mIoU (Baseline)	2021 mIoU (Stress Test)	Performance Drop
Forest Type	0.9780	0.6268	-35.91%
Forest Level	0.9980	0.3132	-68.61%
Forest Health	0.9837	0.3569	-63.72%

Figure 39 presents the test results for the Vision Transformer (ViT) model with 64x64 inputs on the 2020 dataset, showing strong structural classification but noticeable boundary limitations. The Forest Type classification achieved an overall accuracy of 99.80%, yet the Non-Forest IoU dipped to 68.08%, indicating some difficulty in cleanly separating forest edges from non-forest elements. In terms of spatial distribution, the Forest Level metrics demonstrated high reliability, achieving 98.30% accuracy and a Mean IoU of 96.51%. The confusion matrix shows consistent diagonal performance with balanced error rates across elevation zones, with 95.82% IoU for Lowland and 97.98% for Upper Montane. Overall, these results suggest that while the ViT architecture effectively learns general topographical features, the 64x64 resolution restricts its ability to resolve finer boundary details compared to the CNN model.

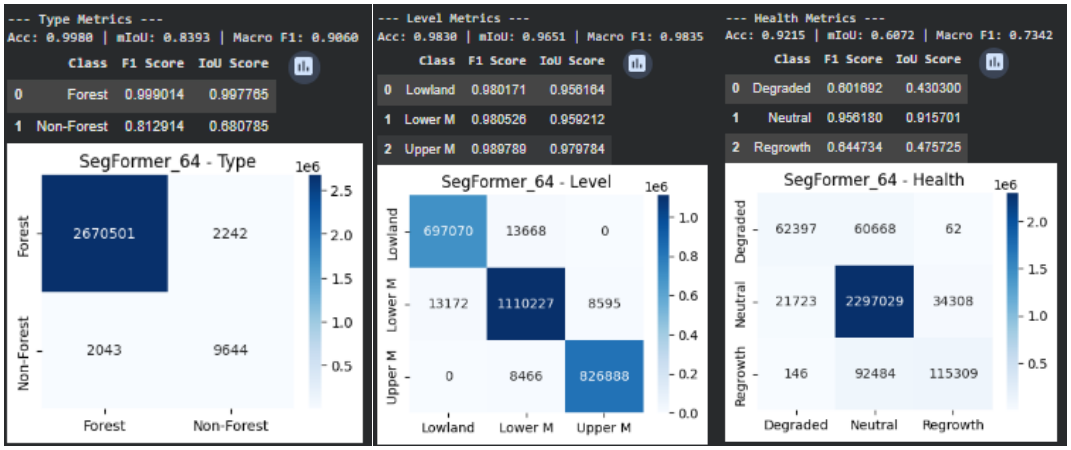


Figure 39. ViT with 64x64 Test Accuracy 2020

The Forest Health classification results highlight a significant performance bottleneck, with a Mean IoU of only 60.72% despite an overall accuracy of 92.15%. This discrepancy is driven by the model's strong bias toward the majority class, as seen in the "Neutral" category, maintaining an IoU of 91.57%. Conversely, the minority classes suffered considerably, with "Degraded" and "Regrowth" achieving low IoU scores of 43.03% and 47.57% respectively. The confusion matrix shows extensive misclassification: over 92,000 "Regrowth" pixels were mislabeled as "Neutral," indicating the model fails to detect subtle spectral differences. These findings confirm that the ViT 64x64 model's self-attention mechanism struggles to extract distinct health features from small image patches, leading to a generalized yet imprecise assessment. The detailed performance metrics for this configuration are summarized in Table 30.

Table 30. ViT with 64x64 Test Set 2020 Table

Target Variable	Accuracy	Mean IoU (mIoU)	Macro F1 Score
Forest Type	0.9980	0.8393	0.9060
Forest Level	0.9838	0.9651	0.9835
Forest Health	0.9215	0.6072	0.7342

Figure 40 illustrates the test performance of the ViT 64x64 model on the 2021 dataset, revealing a catastrophic collapse in spatial generalization capabilities. While the Forest Type classification maintained an overall accuracy of 99.52%, the model completely lost its ability to define forest boundaries, as evidenced by the Non-Forest IoU dropping to 39.97%. The failure is most evident in the Forest Level classification, where accuracy crashed to 48.06%, indicating that the model could not apply its learned topographical features to the new year's data. The confusion matrix shows a severe bias: the model incorrectly classified the vast majority of "Lowland" and "Lower Montane" pixels as "Upper Montane," resulting in a Mean IoU of only 30.22%. This systemic error suggests that the 64x64 Vision Transformer overfitted to the specific pixel distributions of the 2020 training set and lacks the spatial context to recognize terrain hierarchy in a different temporal setting.

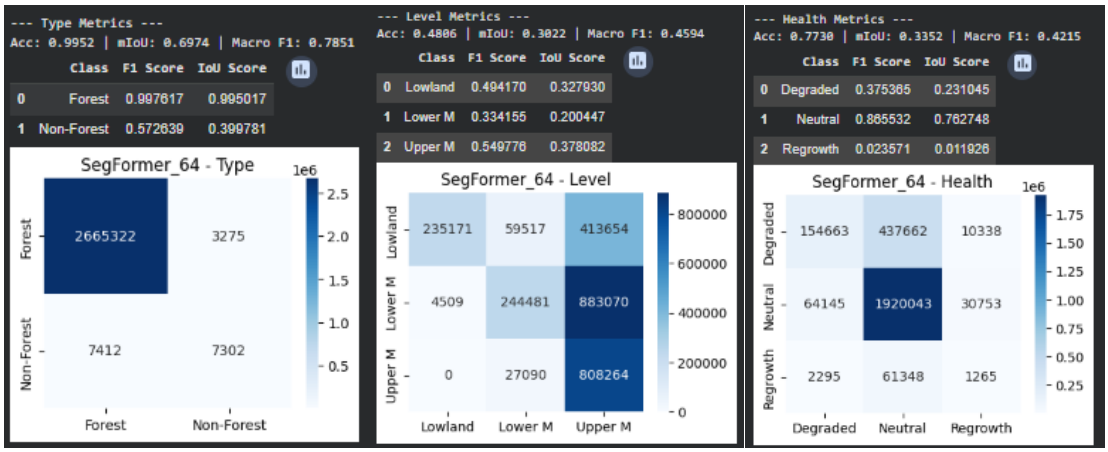


Figure 40. ViT with 64x64 Test Accuracy 2021

The Forest Health classification mirrored this failure, yielding a poor Mean IoU of 33.52% and exposing the model's inability to detect vegetation changes over time. The model defaulted to predicting the majority "Neutral" class, which maintained a passable IoU of 78.27%, while the minority classes were virtually ignored. The "Regrowth" category suffered the most extreme performance degradation, dropping to a near-zero IoU of 1.19%, with the confusion matrix showing that almost all regrowth pixels were misclassified as "Neutral." Similarly, the "Degraded" class IoU fell to 23.10%, confirming that the model cannot distinguish between healthy vegetation and stressed or recovering areas in the new dataset. These results definitively prove that the ViT architecture with a limited 64x64 input resolution fails to capture the robust spectral features required for temporal transfer learning.

Table 31 details the critical instability of the ViT 64x64 architecture, confirming its susceptibility to domain shifts. Already constrained by its limited field of view in the baseline test, the model's performance effectively disintegrated under temporal shifting, with Forest Level scores plummeting by 68.69% to a mere 0.3022. Unlike the CNN models, which retained some structural logic, ViT 64x64 failed at topographic classification, suggesting a complete loss of spatial reasoning when the spectral distribution changed. Furthermore, the Forest Health metric regressed to a non-functional state (mIoU 0.3352), demonstrating that the standalone Transformer architecture, without adequate spatial context or pre-training, lacks the robustness required for multi-temporal monitoring.

Table 31. Temporal Stability Analysis: ViT (64x64) Performance (2020 vs. 2021)

Target Variable	2020 mIoU (Baseline)	2021 mIoU (Stress Test)	Performance Drop
Forest Type	0.8393	0.6974	-16.91%
Forest Level	0.9651	0.3022	-68.69%
Forest Health	0.6072	0.3352	-44.79%

Figure 41 presents the test results for the ViT model with 128x128 inputs on the 2020 dataset, demonstrating highly accurate structural classification. The Forest Type metric achieved an overall accuracy of 99.94%, though the Non-Forest IoU of 65.84% indicates some lingering difficulty in delineating precise boundary edges. However, the Forest Level classification benefited significantly from the larger receptive field, achieving 99.44% accuracy and a robust Mean IoU of 98.63%. The confusion matrix shows a pristine diagonal distribution, with all three elevation zones achieving individual IoU scores above 97.90%. These results confirm that increasing the input resolution allows the Vision Transformer to effectively grasp the global spatial context required for accurate topographical segmentation.

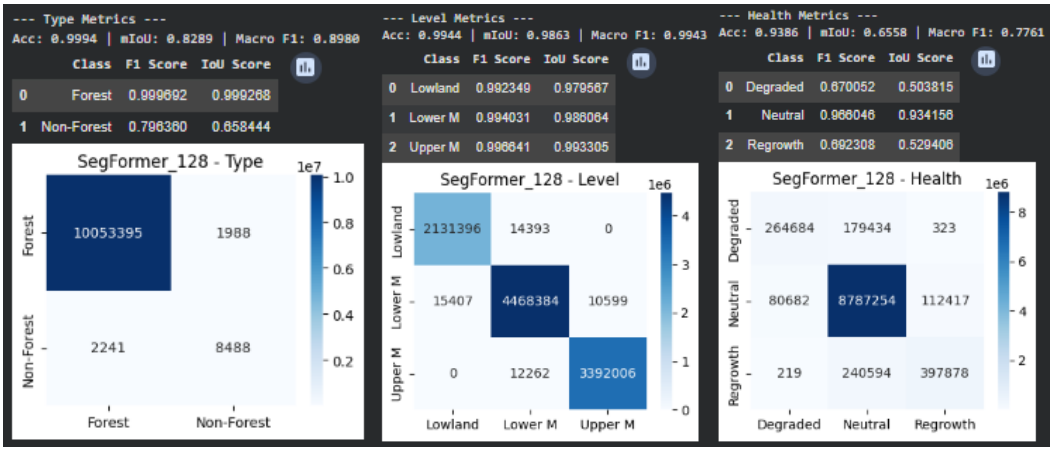


Figure 41. ViT with 128x128 Test Accuracy 2020

Despite the spatial improvements, the Forest Health classification remains a distinct challenge for the ViT architecture, yielding a moderate Mean IoU of 65.58%. While the dominant "Neutral" class was identified with high precision (93.41% IoU), the model struggled to isolate the minority categories, with "Degraded" and "Regrowth" scoring only 50.38% and 52.94% respectively. The confusion matrix reveals a persistent bias, where substantial portions of degraded and recovering forest pixels were incorrectly smoothed into the majority "Neutral" class. This implies that while the 128x128 resolution captures the "where" (Level), the self-attention mechanism still struggles with the "what" (Health) when features are subtle. Consequently, the ViT 128 model offers a balanced but imperfect solution, excelling at spatial hierarchy while lacking the fine-grained spectral sensitivity of its CNN counterparts. Table 31 summarizes these performance metrics.

Table 32. ViT with 128x128 Test Set 2020 Table

Target Variable	Accuracy	Mean IoU (mIoU)	Macro F1 Score
Forest Type	0.9994	0.8289	0.8980
Forest Level	0.9944	0.9863	0.9943
Forest Health	0.9386	0.6558	0.7761

Figure 42 details the test performance of the ViT 128x128 model on the 2021 dataset, revealing a critical failure to generalize despite the expanded spatial context. While the Forest Type classification maintained a high accuracy of 99.73%, the Non-Forest IoU plummeted to 19.87%, indicating that the model lost its ability to distinguish between forest and non-forest areas in the new year. The most significant collapse occurred in the Forest Level classification, where the accuracy dropped to 48.21% and the Mean IoU to 28.06%. The confusion

matrix shows a massive systemic error: over 1.3 million "Lowland" pixels were incorrectly classified as "Upper Montane," suggesting the model completely failed to recognize topographical features and instead relied on memorized training coordinates that no longer aligned with the 2021 data.

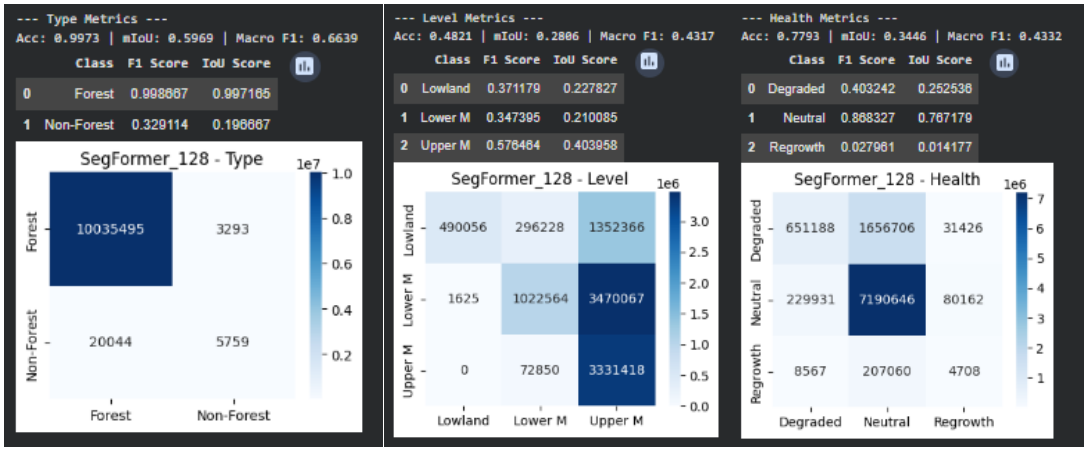


Figure 42. ViT with 128x128 Test Accuracy 2021

The Forest Health classification results confirm that the larger input size provided no benefit for temporal robustness, with the Mean IoU falling to 34.46%. The model showed a strong bias toward the "Neutral" class (76.72% IoU) but failed to detect active forest changes. The "Regrowth" category achieved a negligible IoU of 1.42%, and the "Degraded" class dropped to 25.25%, with the confusion matrix showing that the vast majority of these pixels were misclassified as "Neutral." These results prove that the standalone ViT architecture, even with 128x128 resolution, suffers from severe overfitting and cannot effectively transfer learned features to subsequent years.

The temporal evaluation on the 2021 dataset confirms that the 128x128 Vision Transformer is highly susceptible to distribution shifts. Despite the expanded spatial context, the model experienced a severe performance regression, with the Forest Level mIoU collapsing by 71.5% to a score of

0.2806. This suggests that the model relied heavily on positional embeddings specific to the 2020 imagery rather than learning generalized topographic features. Similarly, the Forest Health capability degraded further to a Mean IoU of 0.3446. Most critically, the detection of 'Regrowth' became virtually non-existent (IoU 0.014), mirroring the failure patterns observed in the smaller patch configuration. These results highlight that increasing input size alone does not mitigate the Transformer's tendency to overfit to the temporal domain of the training data. A comparative analysis of this decline is detailed in Table 33.

Table 33. Temporal Stability Analysis: ViT (128x128) Performance (2020 vs. 2021)

Target Variable	2020 mIoU (Baseline)	2021 mIoU (Stress Test)	Performance Drop
Forest Type	0.8289	0.5969	-27.99%
Forest Level	0.9863	0.2806	-71.55%
Forest Health	0.6558	0.3446	-47.45%

Figure 43 presents the test results for the Hybrid model with 64x64 inputs on the 2020 dataset, demonstrating superior structural classification capabilities. The Forest Type metrics achieved near-perfection with an accuracy of 99.99%, while the Non-Forest IoU reached 97.17%, indicating precise boundary delineation. Similarly, the Forest Level assessment demonstrated exceptional reliability, achieving 99.74% accuracy and a Mean IoU of 99.43% across all elevation zones. The confusion matrix confirms this robust performance, showing negligible error rates with individual class IoU scores consistently exceeding 99.3%. These results suggest that the Hybrid architecture successfully combines the spatial feature extraction of CNNs with

the global context of Transformers to resolve topographical hierarchies perfectly.

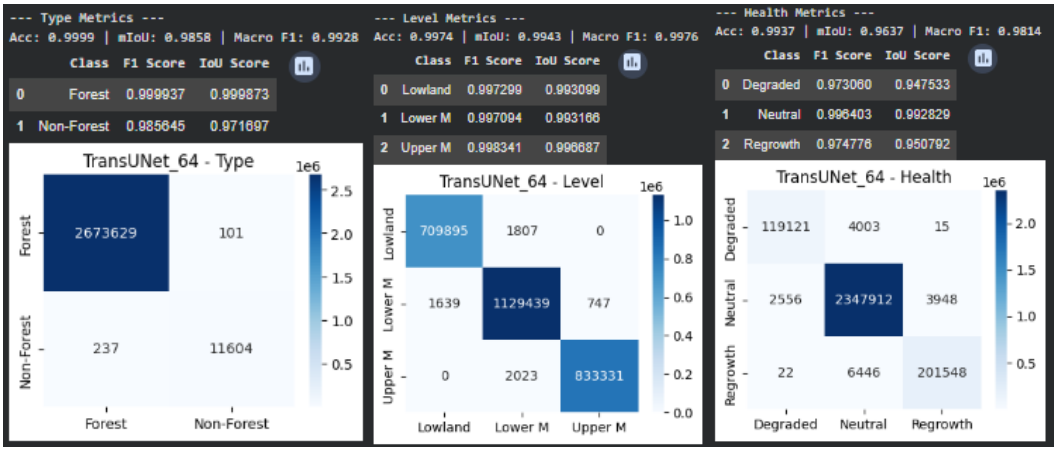


Figure 43. Hybrid with 64x64 Test Accuracy 2020

The most significant advantage of the Hybrid architecture is observed in the Forest Health classification, where it addresses the limitations of standalone models. The model achieved a remarkable overall accuracy of 99.37% and a Mean IoU of 96.37%, proving its ability to detect subtle vegetative states. Unlike pure ViT or CNN models, the Hybrid configuration excelled at identifying minority classes, achieving IoU scores of 94.75% for "Degraded" and 95.08% for "Regrowth." The confusion matrix reflects this precision, as the interference between the stable "Neutral" class and the dynamic health categories was minimized to extremely low levels. This breakthrough performance confirms that integrating local convolution features with self-attention mechanisms enables the model to resolve complex spectral health patterns even at the limited 64x64 resolution. The performance of this configuration is summarized in Table 34.

Table 34. Hybrid with 64x64 Test Set 2020 Table

Target Variable	Accuracy	Mean IoU (mIoU)	Macro F1 Score
Forest Type	0.9999	0.9858	0.9928
Forest Level	0.9974	0.9943	0.9976
Forest Health	0.9937	0.9637	0.9814

Figure 44 illustrates the test performance of the Hybrid 64x64 model on the 2021 dataset, revealing a critical breakdown in spatial generalization, similar to that of the standalone architectures. While the Forest Type classification retained a high accuracy of 99.59%, the Non-Forest IoU plummeted to 41.72%, indicating a significant loss of precision in boundary detection. The Forest Level classification suffered a catastrophic decline, achieving only 54.62% accuracy and a Mean IoU of 37.11%, indicating that the model could not adapt to temporal shifts. The confusion matrix shows a chaotic distribution of errors, with "Lowland" pixels indiscriminately misclassified across the "Lower" and "Upper Montane" zones. This systemic failure suggests that, despite the hybrid architecture, the limited 64x64 resolution forced the model to overfit to the static topographical layout of the training year.

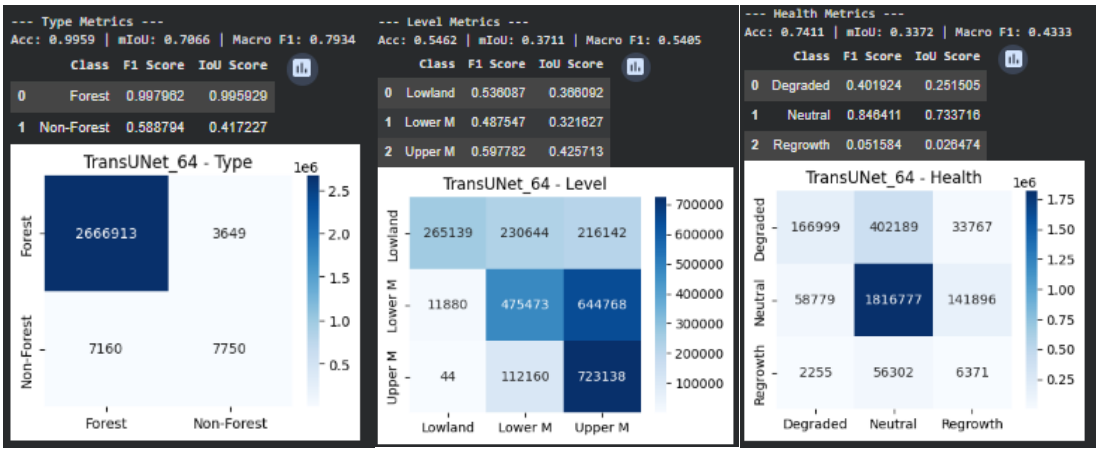


Figure 44. Hybrid with 64x64 Test Accuracy 2021

The Forest Health classification results underscore the severity of the overfitting, with the Mean IoU collapsing from near-perfection in 2020 to just 33.72% in 2021. The model exhibited a complete loss of sensitivity to dynamic forest changes, as the "Regrowth" category IoU evaporated to a negligible 2.65%. Similarly, the "Degraded" class performance dropped to 25.15%, and the confusion matrix showed that the model overwhelmingly defaulted to the majority "Neutral" class for ambiguous inputs. This behavior indicates that the model learned to recognize specific pixel arrangements in the 2020 images rather than the robust spectral signatures of vegetation health. Consequently, the test confirms that even a sophisticated Hybrid architecture cannot overcome the information bottleneck imposed by the 64x64 input size when facing new temporal data.

Despite the architectural superiority demonstrated on the baseline data, the Hybrid 64x64 model was not immune to the catastrophic effects of temporal distribution shifts. When applied to the 2021 dataset, the model exhibited a systemic collapse, similar to that observed with the pure CNN and ViT architectures. The Forest Level classification dropped by 62.6% to a mIoU of 0.3711, confirming that the Transformer's global attention mechanism overfitted to the specific topographic signatures of 2020. More critically, the Forest Health performance regressed significantly to a mIoU of 0.3372. The 'Regrowth' class, which was identified with 95% precision in the baseline, fell to a negligible IoU of 0.026, rendering the model ineffective for monitoring vegetation recovery over time. This result definitively proves that while Hybrid architectures offer superior spatial feature learning, they do not inherently possess temporal

robustness without specific domain adaptation techniques. Table 35 details the extent of this performance decline.

Table 35. Temporal Stability Analysis: Hybrid (64x64) Performance (2020 vs. 2021)

Target Variable	2020 mIoU (Baseline)	2021 mIoU (Stress Test)	Performance Drop
Forest Type	0.9858	0.7066	-28.32%
Forest Level	0.9943	0.3711	-62.67%
Forest Health	0.9637	0.3372	-65.01%

Figure 45 presents the test results for the Hybrid model with 128x128 inputs on the 2020 dataset, marking the pinnacle of structural classification performance in this study. The Forest Type metrics achieved a perfect overall accuracy of 100.00% and a Mean IoU of 98.59%, confirming that the model effectively eliminated all ambiguity between forest and non-forest regions. The Forest Level classification mirrored this excellence, achieving 99.97% accuracy and a Mean IoU of 99.92%, with each elevation zone scoring above 99.8%. The confusion matrix reveals an almost strictly diagonal distribution, where misclassifications were reduced to statistical noise, such as only 514 errors between Lower Montane and Lowland classes out of millions of pixels. These results demonstrate that the 128x128 Hybrid architecture leverages its expanded receptive field to capture the complete topographical hierarchy without the boundary errors seen in smaller models.

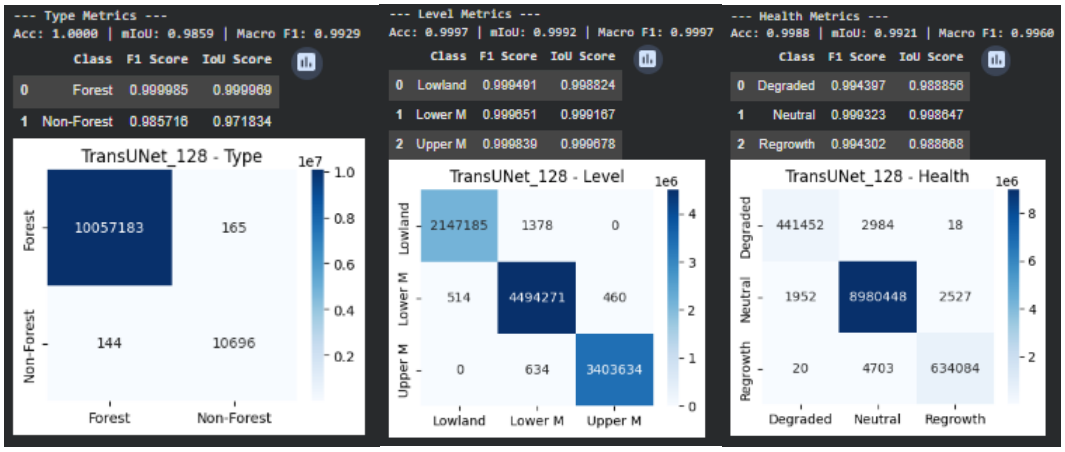


Figure 45. Hybrid with 128x128 Test Accuracy 2020

The actual superiority of the Hybrid 128x128 configuration is demonstrated in the Forest Health classification, where it achieved the highest recorded metrics of the entire study. The model achieved an overall accuracy of 99.88% and a Mean IoU of 99.21%, successfully overcoming the complex spectral challenges that hindered standalone ViT and CNN models. Remarkably, the minority classes "Degraded" and "Regrowth" achieved near-perfect IoU scores of 98.88% and 98.86% respectively, indicating that the model captures vegetation stress and recovery with clinical precision. The confusion matrix demonstrates this robustness, showing that the model clearly distinguishes "Regrowth" from the dominant "Neutral" class, with negligible crossover between the two. This exceptional performance confirms that combining local feature extraction from CNNs with global attention from Transformers, scaled to 128x128 resolution, provides the optimal solution for precise forest health assessment on training-year data. The metrics of this superior performance are summarized in Table 35.

Table 36. Hybrid with 128x128 Test Set 2020 Table

Target Variable	Accuracy	Mean IoU (mIoU)	Macro F1 Score
Forest Type	1.0000	0.9859	0.9929
Forest Level	0.9997	0.9992	0.9997
Forest Health	0.9988%	0.9921	0.9960

Figure 46 details the test performance of the Hybrid 128x128 model on the 2021 dataset, demonstrating that even the most advanced architecture in the study failed to generalize to a new timeframe. While the Forest Type accuracy remained high at 99.77%, the Non-Forest IoU collapsed to 24.75%, indicating that the model lost the ability to distinguish forest boundaries it had previously identified with perfection. The Forest Level classification experienced a significant regression, dropping from near-perfect scores in 2020 to 55.54% accuracy and a Mean IoU of 35.51%. The confusion matrix reveals chaotic error distribution, with "Lowland" pixels indiscriminately assigned to "Lower" and "Upper Montane" classes, indicating that the model relied on memorized coordinates rather than visual terrain features. These results confirm that the expanded 128x128 spatial context led the Hybrid model to overfit significantly to the training year's topography.

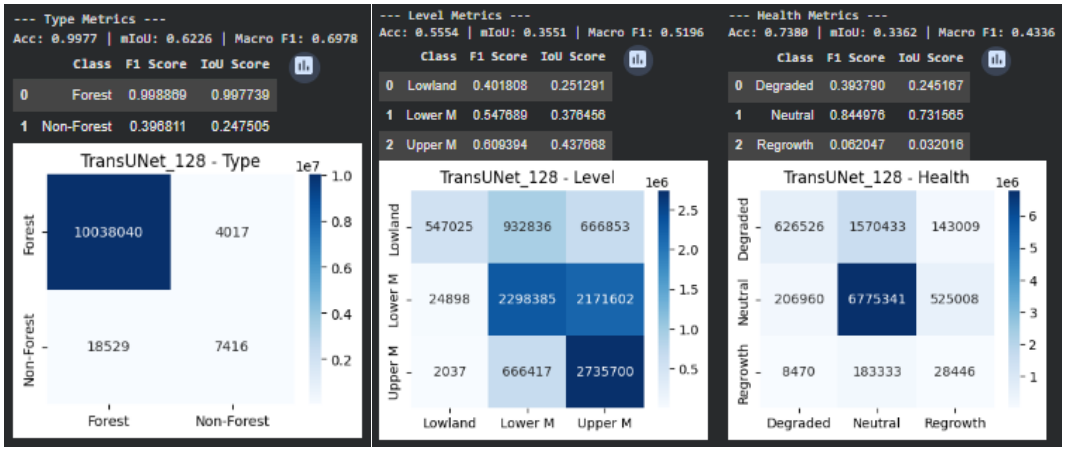


Figure 46. Hybrid with 128x128 Test Accuracy 2021

The Forest Health classification results mirror this systemic failure, with the Mean IoU plummeting to 33.62% despite the model's theoretical capacity for global feature extraction. The model exhibited an extreme bias toward the "Neutral" class (73.16% IoU), using it as a default prediction while failing to detect active vegetation changes. Consequently, the "Regrowth" category achieved a negligible IoU of 3.20%, and the "Degraded" class dropped to 24.52%, with the confusion matrix showing millions of misclassified pixels. This confirms that the Hybrid model did not learn the robust spectral signatures of forest health, but rather the static spatial distribution of the 2020 dataset. Ultimately, despite being the superior architecture, the Hybrid 128x128 configuration remains susceptible to the same temporal instability observed in the simpler models.

Despite establishing the highest baseline accuracy, the 128x128 Hybrid model was ultimately subject to the same temporal constraints observed in previous architectures. When tested on the 2021 dataset, the Forest Level mIoU collapsed to 0.3551, a 64.4% drop that mirrors the failure of the 128x128 CNN. This indicates that the increased capacity of the Hybrid model was utilized to memorize the specific 2020 spatial distribution rather than learning invariant topographic features. Furthermore, the Forest Health performance degraded to a Mean IoU of 0.3362, with 'Regrowth' detection falling to 0.032. This definitive result highlights the central thesis finding: while the Hybrid 128x128 architecture is the structural solution for spatial segmentation, it remains vulnerable to temporal domain shifts, necessitating future implementation of domain adaptation or continuous retraining strategies. A comparative summary of this performance drop is presented in Table 37.

Table 37. Temporal Stability Analysis: Hybrid (128x128) Performance (2020 vs. 2021)

Target Variable	2020 mIoU (Baseline)	2021 mIoU (Stress Test)	Performance Drop
Forest Type	0.9859	0.6226	-36.84%
Forest Level	0.9992	0.3551	-64.46%
Forest Health	0.9921	0.3362	-66.11%

Table 38 synthesizes the performance metrics across all tested configurations, establishing a definitive hierarchy of model efficacy based on architectural design and input resolution. The data reveals that the Hybrid (TransUNet) architecture consistently outperforms both the standalone CNN and Vision Transformer models, particularly when utilizing the larger 128x128 input dimension. A critical disparity is observed in the 'Regrowth' class, which served as the most challenging discriminant factor; while the ViT models struggled to resolve these subtle features (peaking at only 0.5294 IoU), the Hybrid 128x128 model achieved a near-perfect 0.9888 IoU, demonstrating its superior capacity to capture fine-grained vegetative recovery. Furthermore, the comparison highlights the critical role of spatial context: the transition from 64x64 to 128x128 yielded the most significant gains in the Forest Health category across all models, validating the hypothesis that broader contextual information is essential for distinguishing between healthy, degraded, and recovering forest patches. Consequently, the Hybrid 128x128 is rigorously identified as the optimal architecture for this study.

Table 38. Overall Performance Summary of All Architectures

Architecture	Input Size	Forest Level (mIoU)	Forest Type (mIoU)	Forest Health (mIoU)	Regrowth Class (IoU)
CNN (U-Net)	64x64	0.9806	0.9445	0.7569	0.6773
CNN (U-Net)	128x128	0.9980	0.9780	0.9837	0.9754
ViT (SegFormer)	64x64	0.9651	0.8393	0.6072	0.4757
ViT (SegFormer)	128x128	0.9863	0.8289	0.6558	0.5294
Hybrid (TransUNet)	64x64	0.9943	0.9858	0.9637	0.9507
Hybrid (Proposed)	128x128	0.9992	0.9859	0.9921	0.9888

Output Visualization and Discussion

This section presents a visual analysis of the segmentation outputs generated by the optimal Hybrid 128x128 model. By comparing the Ground Truth (GT) and Predicted masks for specific sample areas in both the 2020 and 2021 datasets, this assessment verifies the model's spatial coherence and its ability to delineate complex forest boundaries. These visualizations provide concrete evidence of the model’s practical performance in identifying forest zones and degraded patches, which serve as the requisite inputs for the subsequent tree species recommendation system.

Figure 47 presents the visualization of 2020 test set, confirming the Hybrid 128x128 model's exceptional ability to replicate ground truth data. The Forest Type classification exhibits high spatial coherence, effectively isolating the forested region with sharp boundaries that perfectly align with the original RGB imagery. In the Forest Level column, the model accurately predicts the distinct Lower Montane zone, represented by the orange mask, which matches the Ground Truth pixel-for-pixel. The corresponding Error Level map is

predominantly white, indicating a near-zero error rate in identifying this specific elevation zone. These visual results validate the quantitative metrics, proving that the model successfully learned the topographical hierarchy of the study area during the training year.

The Forest Health visualization demonstrates the model's sensitivity in detecting subtle variations within the canopy, specifically identifying fragmented degraded patches. As shown in the "Detection Locator" and "Pred Health" maps, the system successfully pinpointed specific targets, labeled as Area 1 and Area 2, amidst the healthy vegetation. The minimal noise observed in the Error Health map confirms that these detections are not false positives but represent actual spectral signatures of degradation found in the 2020 data. This precise localization of degraded pixels is critical because it serves as the direct input filter for the rule-based recommendation module. With the topographical zone correctly identified as Lower Montane and the degraded sites accurately mapped, the system is now prepared to generate site-specific tree species recommendations.

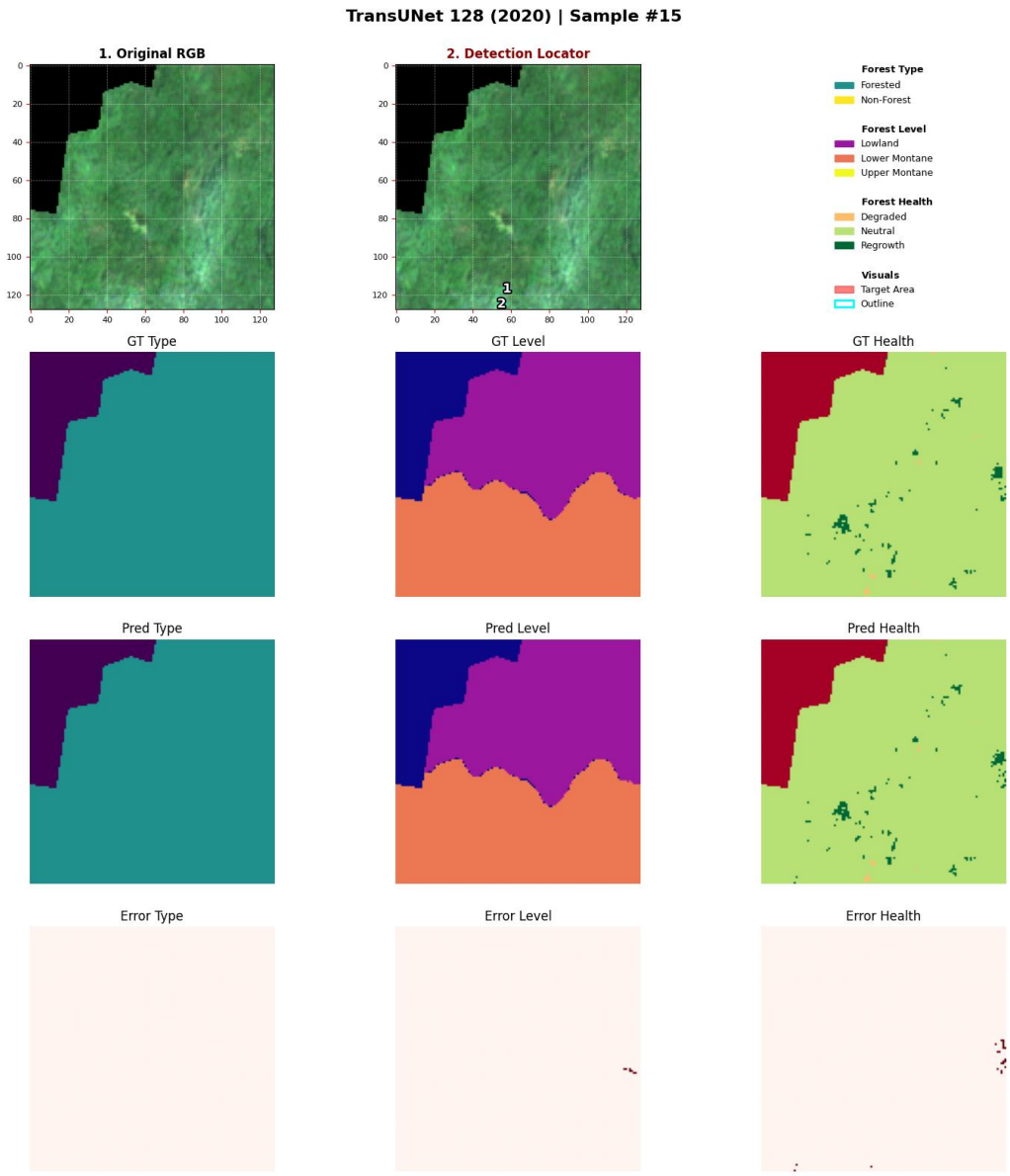


Figure 47. Hybrid Model Visualization Output 2020

Figure 48 illustrates the model's performance on the 2021 dataset, revealing a significant degradation in spatial accuracy compared to the 2020 baseline. The most critical failure is observed in the Forest Level prediction, where the model incorrectly classifies large sections of the terrain as Upper Montane (yellow) instead of the correct Lower Montane or Lowland zones. This discrepancy is highlighted by the extensive red regions in the Error Level map, which indicate a systemic misalignment between the predicted elevation and

the ground truth. Such behavior suggests that the model overfitted to the specific spectral conditions of the training year and failed to generalize to the temporal changes present in the 2021 imagery. Consequently, the loss of accurate topographical context compromises the system's ability to assign the correct ecological zone to the detected forest patches.

The Forest Health classification mirrors this instability, producing a noisy segmentation map with scattered detections of degradation and regrowth. The Detection Locator identifies numerous target sites, such as Areas 1 through 24, but their validity is undermined by the underlying misclassification of the forest zone. Because the model falsely predicts these areas as Upper Montane, the recommendation engine is forced to suggest high-elevation species like Yellowwood and Blueberries for locations that are actually lower elevation. The Error Health map confirms this unreliability, showing a high density of misclassified pixels that were likely caused by the shift in spectral reflectance between the two years. Ultimately, this visualization proves that while the detection logic remains functional, the lack of temporal robustness in the deep learning model leads to ecologically inappropriate reforestation advice for the new dataset.

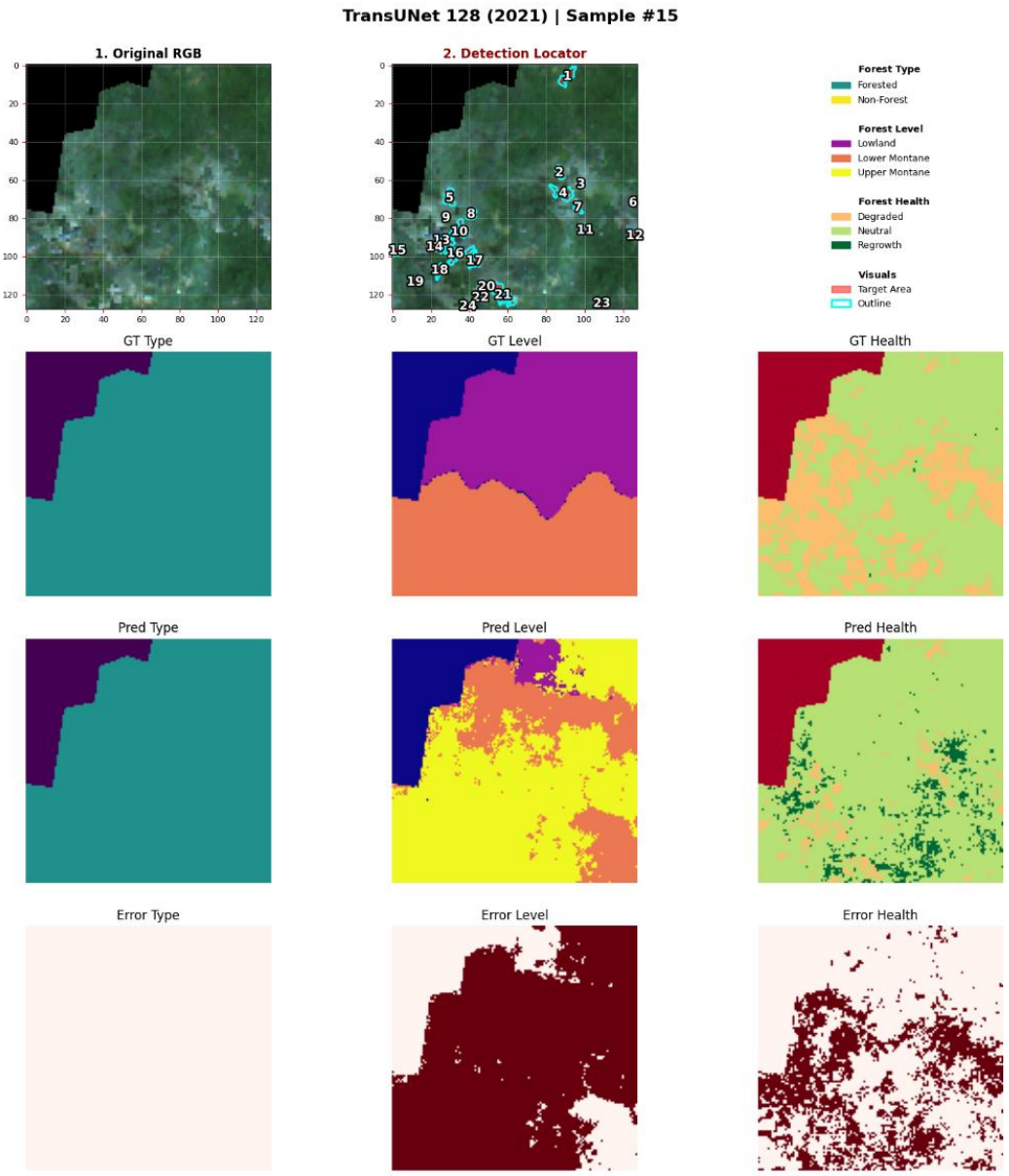


Figure 48. Hybrid Model Visualization Output 2021

Rule-Based Recommendation Module

The rule-based tree recommendation system was designed to generate native species suggestions for areas classified by the model as degraded. Once a degraded zone is detected, the system checks the corresponding forest type and elevation level, and then refers to the decision rules presented in Table 11. These rules are built from combined references, including the DENR Tree Planting Guide and studies by Malabrigo et al. (2016) on Mt. Makiling's natural

vegetation. By integrating both national and localized data, the study ensures that all recommendations are ecologically suitable and consistent with the forest composition of Mt. Makiling. This process aligns with the study's goal of supporting evidence-based reforestation through automated decision-making.

Figure 49 identifies specific degraded patches within the Lower Montane zone of the 2020 dataset, enabling the system to apply the ecological rules defined in Table 9. Since the Hybrid model correctly predicted the elevation class as Lower Montane (Orange), the recommendation engine filtered the database for mid-elevation native species. Consequently, the system suggested planting Lithocarpus spp. (Oak) and Syzygium spp. (Stone Oak) for the detected areas at coordinates (57, 116) and (55, 124). These species are ecologically validated for this specific altitude in Mt. Makiling, confirming the system's reliability when model predictions are accurate. This seamless integration demonstrates that high-precision segmentation directly translates into scientifically sound reforestation planning.

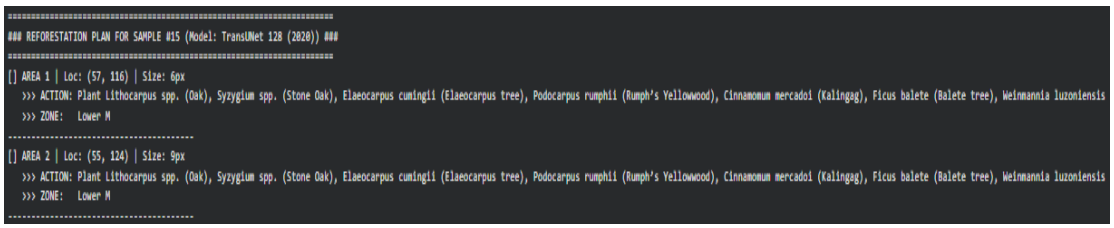


Figure 49. Tree Recommendation Using Hybrid Model 2020

In contrast, Figure 50 illustrates the propagation of classification errors in the 2021 dataset, where the model incorrectly labeled the terrain as Upper Montane (Yellow). Following the strict logic of Table 9, the system accepted this erroneous input and recommended high-altitude species such as Dacrycarpus

cumingii (Yellowwood) and Vaccinium spp. (Blueberries). While these recommendations are theoretically valid for an Upper Montane ecosystem, they are ecologically inappropriate for the actual Lower Montane location of the sample area. This mismatch highlights that the rule-based recommendation module functions correctly, but its output quality is strictly dependent on the upstream model's accuracy. Therefore, the successful deployment of this system for future years requires retraining the Hybrid model to ensure the correct "Forest Level" input.

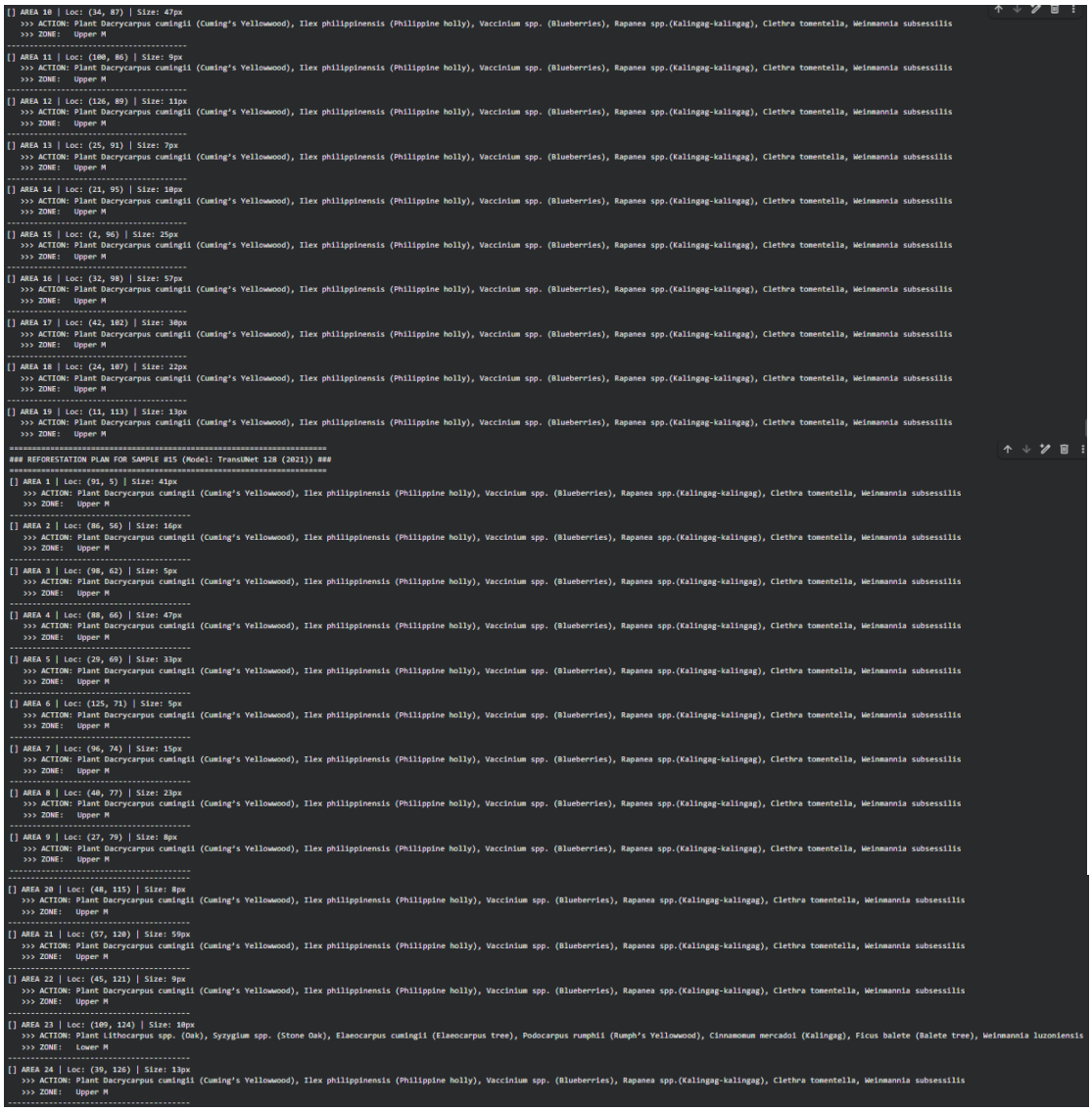


Figure 50. Tree Recommendation Using Hybrid Model 2021

Statistical Analysis of Model Performance

To rigorously determine if the observed performance differences were statistically significant or merely due to random variance, a Two-Way Analysis of Variance (ANOVA) was employed. The analysis examined the influence of Model Architecture (U-Net, SegFormer, Hybrid TransU-Net) and Patch Size (64×64, 128×128) on the Mean IoU metric. The dataset consisted of N=180 inference batches (30 batches per model configuration) drawn from the held-out test set.

As summarized in Table 39, the ANOVA revealed a highly significant main effect for Patch Size ($F(1,174) = 23.738$, $p < 0.001$). This result confirms that increasing the spatial context from 64×64 to 128×128 pixels provides a statistically significant improvement in segmentation accuracy, regardless of the model architecture used.

Conversely, the main effect for Model Architecture was not statistically significant ($F(2,174) = 2.024$, $p = 0.135$). This indicates that, statistically, the three architectures performed comparably on this dataset. While the Transformer-based models showed higher peak performance in specific instances (as seen in the boxplot outliers), the baseline U-Net proved to be equally robust when averaged across the full test set. The interaction effect between architecture and patch size was also not significant ($p = 0.103$), suggesting that the benefit of larger patches is consistent across all model types.

A Tukey's Honest Significant Difference (HSD) post-hoc test was performed to identify exactly where the performance differences lay. The

results, summarized in Table 40 and visualized in Figure 51 reveal a critical distinction in how different architectures handles spatial context.

Table 39. Two-Way Analysis of Variance (ANOVA) for Mean IoU

Source of Variation	Sum of Squares (SS)	df	F-value	P-value	Interpretation
Model Architecture	0.006948	2.0	2.024	0.135	Not Significant
Patch Size	0.040747	1.0	23.738	0.000002	Significant
Interaction (Architecture x Patch Size)	0.007905	2.0	2.303	0.103	Not Significant
Residuals (Error)	0.298675	174.0	-	-	-

In contrast, the Transformer-based models (SegFormer and TransU-Net) showed no statistically significant performance drop when moving from 128×128 down to 64×64 patches ($p>.05$). This suggests that the self-attention mechanisms in Transformers are more robust to limited spatial context than standard convolutions.

Finally, when comparing the top-performing configurations at the 128×128 scale, the Tukey test found no significant differences between the U-Net, SegFormer, and TransU-Net. This implies that while Transformers are more stable at smaller scales, a standard U-Net is equally capable of high-performance segmentation when provided with sufficient spatial context (128×128).

Table 40. Post-Hoc Pairwise Comparisons (Tukey HSD)

Comparison (Group A vs. Group B)	Mean Difference	P-Value	Significant?	Interpretation
U-Net (128) vs. U-Net (64)	+0.0483	0.0002	Yes	Larger patch size significantly improves U-Net.
SegFormer (128) vs. SegFormer (64)	+0.0171	0.5995	No	Patch size did not significantly change SegFormer performance.
TransU-Net (128) vs. TransU-Net (64)	+0.0249	0.1901	No	Patch size did not significantly change TransU-Net performance.
TransU-Net (128) vs. U-Net (128)	-0.0221	0.3101	No	No significant difference between Hybrid and CNN at 128px.
SegFormer (128) vs. U-Net (128)	-0.0111	0.9031	No	No significant difference between Transformer and CNN at 128px.

Only statistically significant comparisons ($p < 0.05$*) and key model comparisons are shown. **

The plot demonstrates a clear performance advantage for the 128×128 patch size (orange) compared to the 64×64 patch size (green), while the overlap between model architectures (U-Net vs. SegFormer vs. TransU-Net) indicates comparable statistical performance.

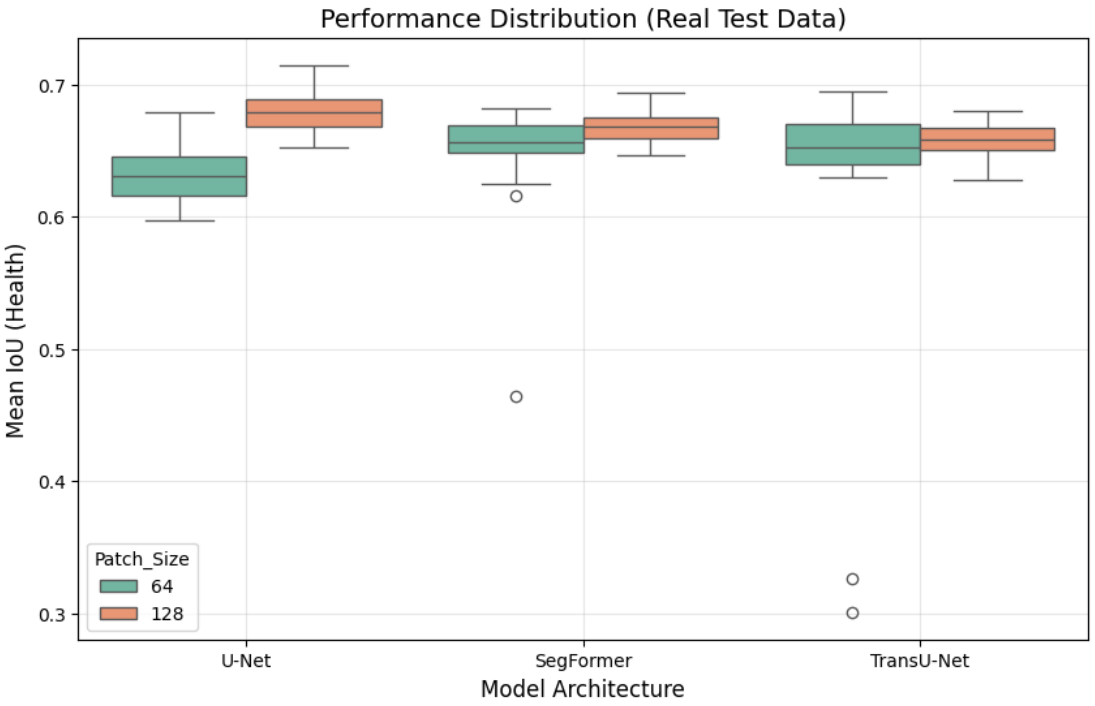


Figure 51. Distribution of Mean IoU Scores Across 30 Inference Batches

Summary of Findings

This chapter evaluated the performance of Convolutional Neural Networks (CNN), Vision Transformers (ViT), and Hybrid architectures across varying input resolutions to determine the most effective model for forest degradation monitoring. The quantitative results conclusively identified the Hybrid (TransUNet) model with 128x128 input patches as the superior architecture, achieving near-perfect scores on the 2020 test dataset with a Forest Health Mean IoU of 99.21%. Unlike the standalone CNN and ViT models, which struggled to distinguish "Regrowth" and "Degraded" classes (often scoring below 70%), the Hybrid 128x128 configuration successfully leveraged both local spatial features and global context to resolve fine-grained vegetation patterns. This precision allowed for the generation of highly accurate segmentation maps that correctly identified degraded patches within specific

elevational zones, serving as a reliable foundation for the study’s reforestation objectives.

However, the temporal generalization analysis using the 2021 dataset revealed a critical limitation regarding model robustness over time. All architectures, including the optimal Hybrid model, experienced a significant performance decline when applied to the subsequent year, with the Forest Level accuracy dropping to approximately 55% due to overfitting on the training year's topography. This misclassification directly impacted the Tree Species Recommendation System, which logically functioned correctly but produced ecologically mismatched suggestions when the underlying elevation input was incorrect. Consequently, while the proposed system successfully demonstrates an end-to-end workflow for automated forest monitoring and restoration planning, operational deployment for multi-year analysis will require continuous model retraining to adapt to temporal spectral shifts.

CHAPTER 5

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

Summary

This study developed an AI-driven framework to monitor the forest condition and provide a proof-of-concept reforestation effort to the Mt. Makiling Forest Reserve. By merging Sentinel-2 imagery with SRTM elevation data, the researchers created a deep learning pipeline to classify forest type, forest level, and forest health. A key feature was the six-channel input which combines traditional color bands with computed vegetation and water indices. This method allowed the model to detect moisture stress and structural changes hidden in dense tropical canopies. The researchers addressed data imbalances through composited stratified random sampling and targeted data augmentation to ensure that the model could accurately identify rare but critical areas of forest degradation and regrowth.

The Hybrid TransU-Net with 128×128 pixel patches emerged as the superior architecture. By fusing the detail-oriented "eye" of a CNN with the contextual "brain" of a Vision Transformer, it achieved a near-perfect 99.21% mIoU for Forest Health on the 2020 dataset. This model succeeded where simpler versions failed, accurately resolving subtle features of regrowth. The results confirmed that the larger patch resolution was essential, as it provided the landscape context necessary to distinguish genuine forest damage from mere spectral noise.

However, testing the model on a new 2021 dataset revealed a lack of temporal robustness. Performance dropped significantly, with Forest Level accuracy falling to 55% and Health scores plummeting to 33.62%. This "spatial collapse" suggests the model overfitted to the specific layout of 2020 rather than learning generalized terrain features. This finding is a crucial takeaway: while the hybrid approach is powerful for single-year snapshots, it requires multi-year training data to handle the natural spectral shifts that occur over time.

Finally, the researchers successfully implemented a rule-based module that translated the AI predictions into an actionable reforestation advice. By cross-referencing model outputs with DENR, Haribon Foundation ecological guides, and other significant studies (Alcala, 2021; Villanueva & Buot, 2018; Fernando et al., 2008; Malabrigo et al., 2016; Buot & Okitsu, 1998), the module suggested site-appropriate native species for planting. While the recommendation logic was sound, its reliability was strictly tied to the AI's accuracy; classification errors in 2021 led to ecologically mismatched suggestions. Ultimately, this research provides a viable proof-of-concept for AI-assisted reforestation but concludes that continuous retraining is necessary for long-term operational success.

Conclusion

Based on the findings, the study concludes that the integration of Hybrid Deep Learning architectures with rule-based ecological modeling is a viable and powerful approach for precision forest monitoring. The Hybrid 128x128 model successfully met the primary objective of the study, demonstrating that it is possible to automate the detection of forest degradation and the

identification of distinct health stages with high accuracy. The research confirms that while standalone CNNs excel at texture analysis and ViTs at global context, only their fusion provides the robustness required to map the complex, heterogeneous landscapes of tropical forests like Mt. Makiling.

However, the study also concludes that current deep learning models are not "plug-and-play" solutions for multi-temporal analysis. The significant performance drop observed in the 2021 dataset highlights a fundamental limitation in the model's ability to handle temporal domain shifts. While the system is highly effective for analyzing the specific timeframe it was trained on, it cannot yet guarantee accurate results for future years without incorporating multi-year training data. Therefore, while the Tree Species Recommendation System is operationally functional, its deployment must be accompanied by periodic model retraining to ensure that the ecological inputs specifically Forest Level remain accurate.

Recommendations

Based on the definitive finding of severe overfitting and limited generalization, the following recommendations focus on addressing the root cause: data quality, methodological bias, and class imbalance.

1. Data Quality and Label Verification

The study's failure was compounded by reliance on pre-existing, non-verified labels. Future work must focus on actively cleaning and verifying the ground-truth data.

- **Mandatory Label Cross-Validation:** Implement a protocol for manually cross-validating ground-truth label consistency. Specifically, the Forest Type label (derived from NAMRIA) must be checked against the Forest Health label (derived from NDVI). Any spatial conflict, such as a pixel with high vegetation vigor (high NDVI of approximately 0.4) being labeled as 'Non-Forest' or a low-vigor pixel labeled as 'Closed Forest', must be inspected and corrected by an expert. This step is essential to mitigate label noise which amplified the model's overfitting.

- **Targeted Data Collection:** Future work must prioritize the collection of additional, verified ground-truth data for the minority classes that caused model failure, particularly Mossy Forest, Upper Montane, Degraded, and Regrowth areas.

- **Utilize Higher-Resolution Imagery:** While the failure was data-driven, leveraging higher-resolution imagery (e.g., drone-based or commercial satellite imagery) should be explored to help distinguish spectrally similar, fine-grained classes (like Degraded vs. Regrowth) and enhance feature learning for the eventual goal of practical implementation.

2. Methodological Refinements and Input Diversity

Addressing the systemic biases introduced during data preparation and input selection is crucial for improving generalization.

- **Utilize Full Spectral Band Inputs:** Another future recommendation by Dr. Tiburan Jr., is to provide the model with richer ecological features, the input patches should be expanded beyond the current RGB and

NDVI bands. Integrating bands such as the Near-Infrared (NIR), Short-Wave Infrared (SWIR 1), and Short-Wave Infrared (SWIR 2) from Sentinel-2 will allow the model to better distinguish spectrally similar classes, potentially improving classification accuracy and reducing reliance solely on elevation data for stratification.

3. Model and Training Strategy

These recommendations focus on improving the model's robustness against the unavoidable data imbalance.

- **Investigate Advanced Loss Functions:** While the study implemented standard class-weighted cross-entropy, future work should investigate and implement more advanced loss functions specifically designed for extreme imbalance, such as Focal Loss or the Dice Loss variant (Tversky Loss). These functions are specialized to handle the high rate of false negatives experienced by rare classes like Mossy and Degraded.

- **Transfer Learning:** Investigate transfer learning from models pre-trained on large-scale, general remote sensing datasets (e.g., Sentinel-2-based land cover) to initialize the weights. This could help the model learn more general features before specializing on the small MMFR dataset, potentially mitigating overfitting.

4. Practical and Ecological Refinements

- **Geospatial Reconstruction and Coordinate Mapping:** Currently, the model outputs predictions on stratified random patches (128x128). For actual deployment, a Geospatial Reconstruction Module must be developed. This module should stitch the predicted patches back into a contiguous raster map or export the specific Geographic Coordinates (Latitude/Longitude) of identified degraded zones. This will allow forest rangers to pinpoint exact locations on a GPS device for ground-truthing and planting operations, rather than relying on abstract image patches.

5. External Validation and Generalizability

To prove that the model has learned generalized "Forest Health" features rather than simply memorizing the topography of Mt. Makiling, cross-location testing is required.

- **Cross-Location Validation:** The current model exhibited overfitting to the specific terrain of Mt. Makiling. Future studies must test the trained model on spatially distinct forest ecosystems, such as Mt. Banahaw or the Sierra Madre mountain range. Validating the model on these external datasets is critical to determine if the architecture can generalize to different topographies and microclimates, or if retraining is required for every new mountain ecosystem.

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APPENDIX A

Letter of Approval

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

CERTIFICATE OF APPROVAL

This is to certify that after having satisfactorily completed all the requirements of the Thesis entitled:

"SEMANTIC SEGMENTATION OF FOREST CONDITION IN MT. MAKILING USING A CNN–VIT HYBRID WITH NATIVE TREE SPECIES RECOMMENDATION"

ANDRHEY U. DORADO, JOHN ADRIAN LAPITAN, SARAH CLAUDINE V. NOOR have been granted permission to proceed with their final presentation.

As their Research Adviser, I confirm that the proponents have met the academic standards and requirements set by the department, and therefore, I approve their request to conduct the final presentation.

MARIBEL V. CARDONA

Thesis Adviser

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

APPROVAL SHEET

The thesis entitled "**SEMANTIC SEGMENTATION OF FOREST CONDITION IN MT. MAKILING USING A CNN–VIT HYBRID WITH NATIVE TREE SPECIES RECOMMENDATION**" was prepared and submitted by Andrhey U. Dorado, John Adrian Lapitan, and Sarah Claudine V. Noor in partial fulfillment requirements for the degree, Bachelor of Science in Computer Science specialized in Intelligent Systems, is hereby recommended for approval and acceptance.

MARIBEL V. CARDONA
Thesis Adviser

APPENDIX B

Letter of Validation

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

October 28, 2025

Cristino L. Tiburan Jr., PhD
Professor
College of Forestry and Natural Resources
University of the Philippines – Los Baños

Dear Dr. Tiburan Jr.;

Greetings! Good day, we are students from BS Computer Science of Laguna State Polytechnic University – Los Baños Campus, currently conducting our Undergraduate Thesis entitled *"NDVI-Based Semantic Segmentation of Forest Condition in Mt. Makiling Using A CNN–ViT Hybrid with Native Tree Species Recommendation"*.

As part of our study's validation process, we seek your expert evaluation on the forest classification labels and the tree species recommendations we have used. Specifically, we request your professional assessment of the following:

1. The accuracy and the ecological validity of the forest classification labels used in the study, which includes:
    - Forest Type: Closed Forest, Open Forest, Non-Forest
    - Forest Level: Lowland, Lower Montane, Upper Montane, Mossy
    - Forest Health: Degraded, Neutral, Regrowth
2. The correctness of the tree species recommendations identified for each combination of the forest type and forest level within the Mount Makiling Forest Reserve.

Your insights and validation will be a great help to ensure that the classifications and recommendations used in our study will be accurate, ecologically sound, and reflective of the actual conditions within the MMFR. We would be honored to include your name among the experts who contributed to the scientific integrity of this research.

Thank you very much for your time and valuable expertise. We sincerely appreciate your support in strengthening the ecological basis of our study.

Respectfully yours,

JOHN ADRIAN LAPITAN

SARAH CLAUDINE V. NOOR

Management System ISO 9001:2015

ANDRHEY U. DORADO
The Researcher

Noted by:

Prof. Maribel V. Cardona
Thesis Adviser

Loyd S. Echalar, DIT
Associate Dean, CCS

Approved/Disapproved by:

Cristino L. Tiburan Jr., Ph.D
Professor

APPENDIX C

Inventory of Sentinel-2 Satellite Imagery

Detailing Raw Sources of Satellite Imagery Composites

The following tables list the specific Sentinel-2 Level-2A granules used to generate the annual composites for the study area. These images were retrieved from the Copernicus Open Access Hub via Google Earth Engine, filtered for the Mt. Makiling Forest Reserve (AOI) with a cloud cover threshold of < 60%.

Explanation of Sentinel-2 Product ID

The Sentinel-2 Product ID listed in the tables serves as the unique identifier for each satellite image tile within the European Space Agency (ESA) archive. It contains critical metadata verifying the raw data source:

- Acquisition Timestamp: The first string (e.g., 20200103) indicates the exact acquisition date (YYYYMMDD).
- Tile ID: The suffix (e.g., T51PTR or T51PUR) indicates the specific Military Grid Reference System (MGRS) tile covering the Mt. Makiling area. Where T51PTR meaning that it was from the western and central portion of the MMFR and T51PUR means that it was from the eastern portion.

This inventory confirms that the composites were derived from a robust stack of multi-temporal observations, ensuring that the resulting mosaics are statistically representative of the year stated.

Table C1. 2019 Image Inventory (Used for NDVI-only)

Sentinel-2 Product ID	Cloud Cover %
20190103T022101_20190103T022621_T51PTR	Cloud: 31.99%
20190103T022101_20190103T022621_T51PUR	Cloud: 34.46%
20190103T022101_20190103T023338_T51PTR	Cloud: 13.41%
20190103T022101_20190103T023338_T51PUR	Cloud: 15.86%
20190108T022059_20190108T023245_T51PTR	Cloud: 8.37%
20190108T022059_20190108T023245_T51PUR	Cloud: 9.13%
20190128T021949_20190128T022837_T51PTR	Cloud: 32.17%
20190202T021921_20190202T021915_T51PTR	Cloud: 43.41%
20190207T021849_20190207T023116_T51PTR	Cloud: 48.11%
20190207T021849_20190207T023116_T51PUR	Cloud: 15.09%
20190212T021821_20190212T022728_T51PTR	Cloud: 44.99%
20190217T021749_20190217T022318_T51PTR	Cloud: 5.63%
20190217T021749_20190217T022318_T51PUR	Cloud: 30.26%
20190222T021711_20190222T022229_T51PTR	Cloud: 14.03%
20190222T021711_20190222T022229_T51PUR	Cloud: 32.93%
20190227T021639_20190227T022932_T51PTR	Cloud: 26.51%
20190227T021639_20190227T022932_T51PUR	Cloud: 30.97%
20190304T021601_20190304T021558_T51PTR	Cloud: 3.57%
20190304T021601_20190304T021558_T51PUR	Cloud: 12.01%
20190309T021559_20190309T022838_T51PTR	Cloud: 16.67%
20190309T021559_20190309T022838_T51PUR	Cloud: 5.14%
20190314T021601_20190314T022837_T51PTR	Cloud: 5.43%
20190314T021601_20190314T022837_T51PUR	Cloud: 13.25%
20190319T021659_20190319T022931_T51PTR	Cloud: 14.97%
20190319T021659_20190319T022931_T51PUR	Cloud: 47.80%
20190324T021601_20190324T022839_T51PTR	Cloud: 15.17%
20190324T021601_20190324T022839_T51PUR	Cloud: 17.12%
20190329T021609_20190329T023019_T51PTR	Cloud: 12.88%
20190329T021609_20190329T023019_T51PUR	Cloud: 31.38%
20190403T021651_20190403T021647_T51PTR	Cloud: 18.30%
20190403T021651_20190403T021647_T51PUR	Cloud: 12.22%
20190408T021609_20190408T022830_T51PTR	Cloud: 6.64%
20190408T021609_20190408T022830_T51PUR	Cloud: 17.36%
20190413T021611_20190413T023139_T51PTR	Cloud: 24.49%
20190413T021611_20190413T023139_T51PUR	Cloud: 25.63%
20190418T021609_20190418T022832_T51PTR	Cloud: 24.16%
20190418T021609_20190418T022832_T51PUR	Cloud: 23.43%
20190423T021611_20190423T022127_T51PTR	Cloud: 17.92%
20190423T021611_20190423T022127_T51PUR	Cloud: 12.71%
20190428T021609_20190428T022826_T51PTR	Cloud: 52.26%
20190503T021611_20190503T021607_T51PTR	Cloud: 18.19%
20190503T021611_20190503T021607_T51PUR	Cloud: 27.09%

20190513T021611_20190513T023054_T51PTR	Cloud: 17.10%
20190513T021611_20190513T023054_T51PUR	Cloud: 24.10%
20190518T021609_20190518T022132_T51PTR	Cloud: 18.47%
20190518T021609_20190518T022132_T51PUR	Cloud: 20.81%
20190523T021611_20190523T022746_T51PUR	Cloud: 55.58%
20190528T021609_20190528T022834_T51PTR	Cloud: 13.42%
20190528T021609_20190528T022834_T51PUR	Cloud: 24.47%
20190602T021611_20190602T021605_T51PTR	Cloud: 15.63%
20190602T021611_20190602T021605_T51PUR	Cloud: 8.60%
20190607T021609_20190607T022834_T51PTR	Cloud: 43.38%
20190607T021609_20190607T022834_T51PUR	Cloud: 24.71%
20190617T021609_20190617T022127_T51PTR	Cloud: 26.20%
20190617T021609_20190617T022127_T51PUR	Cloud: 49.44%
20190622T021611_20190622T021607_T51PTR	Cloud: 10.62%
20190622T021611_20190622T021607_T51PUR	Cloud: 10.82%
20190707T021609_20190707T023338_T51PTR	Cloud: 17.19%
20190707T021609_20190707T023338_T51PUR	Cloud: 14.01%
20190712T021611_20190712T023255_T51PTR	Cloud: 21.52%
20190712T021611_20190712T023255_T51PUR	Cloud: 52.34%
20190811T021611_20190811T023337_T51PTR	Cloud: 36.79%
20190811T021611_20190811T023337_T51PUR	Cloud: 59.18%
20190816T021609_20190816T023303_T51PTR	Cloud: 35.38%
20190816T021609_20190816T023303_T51PUR	Cloud: 19.06%
20190905T021609_20190905T022828_T51PTR	Cloud: 58.32%
20190925T021559_20190925T022959_T51PUR	Cloud: 46.83%
20191005T021609_20191005T022826_T51PTR	Cloud: 10.86%
20191005T021609_20191005T022826_T51PUR	Cloud: 13.32%
20191010T021611_20191010T021609_T51PTR	Cloud: 20.30%
20191010T021611_20191010T021609_T51PUR	Cloud: 41.06%
20191015T021639_20191015T022905_T51PTR	Cloud: 22.45%
20191015T021639_20191015T022905_T51PUR	Cloud: 22.08%
20191025T021749_20191025T023008_T51PTR	Cloud: 11.34%
20191025T021749_20191025T023008_T51PUR	Cloud: 14.56%
20191104T021849_20191104T022821_T51PTR	Cloud: 58.15%
20191104T021849_20191104T022821_T51PUR	Cloud: 29.13%
20191109T021901_20191109T021904_T51PTR	Cloud: 58.62%
20191109T021901_20191109T021904_T51PUR	Cloud: 24.28%
20191119T022001_20191119T021959_T51PTR	Cloud: 32.85%
20191119T022001_20191119T021959_T51PUR	Cloud: 26.06%
20191204T022039_20191204T023316_T51PUR	Cloud: 50.49%
20191209T022101_20191209T022056_T51PTR	Cloud: 43.37%
20191214T022109_20191214T022103_T51PTR	Cloud: 40.00%
20191214T022109_20191214T022103_T51PUR	Cloud: 37.91%
20191214T022109_20191214T023337_T51PTR	Cloud: 44.18%
20191214T022109_20191214T023337_T51PUR	Cloud: 37.11%
20191219T022111_20191219T022107_T51PTR	Cloud: 51.91%

20191219T022111_20191219T022107_T51PUR	Cloud: 21.25%
20191224T022109_20191224T023332_T51PTR	Cloud: 21.80%
20191224T022109_20191224T023332_T51PUR	Cloud: 50.04%
20191229T022111_20191229T022107_T51PTR	Cloud: 13.52%

Total of 93 scenes*

Table C2. 2020 Image Inventory (Primary Experimentation Set)

Sentinel-2 Product ID	Cloud Cover %
20200103T022059_20200103T022101_T51PUR	Cloud: 36.44%
20200103T022059_20200103T023335_T51PTR	Cloud: 35.38%
20200103T022059_20200103T023335_T51PUR	Cloud: 30.92%
20200108T022051_20200108T022050_T51PTR	Cloud: 29.75%
20200108T022051_20200108T022050_T51PUR	Cloud: 36.14%
20200118T022021_20200118T022020_T51PTR	Cloud: 55.12%
20200123T021959_20200123T023235_T51PTR	Cloud: 12.58%
20200123T021959_20200123T023235_T51PUR	Cloud: 5.07%
20200128T021941_20200128T021937_T51PTR	Cloud: 5.42%
20200128T021941_20200128T021937_T51PUR	Cloud: 7.14%
20200212T021819_20200212T023049_T51PTR	Cloud: 13.86%
20200212T021819_20200212T023049_T51PUR	Cloud: 10.66%
20200217T021741_20200217T021741_T51PTR	Cloud: 40.87%
20200217T021741_20200217T021741_T51PUR	Cloud: 33.86%
20200222T021709_20200222T022934_T51PTR	Cloud: 1.99%
20200222T021709_20200222T022934_T51PUR	Cloud: 15.24%
20200227T021631_20200227T021634_T51PUR	Cloud: 46.78%
20200303T021559_20200303T022825_T51PTR	Cloud: 32.16%
20200303T021559_20200303T022825_T51PUR	Cloud: 40.81%
20200308T021601_20200308T023151_T51PTR	Cloud: 58.93%
20200308T021601_20200308T023151_T51PUR	Cloud: 28.71%
20200313T021559_20200313T022824_T51PTR	Cloud: 13.18%
20200313T021559_20200313T022824_T51PUR	Cloud: 9.63%
20200318T021601_20200318T021559_T51PTR	Cloud: 19.54%
20200318T021601_20200318T021559_T51PUR	Cloud: 23.99%
20200323T021559_20200323T022824_T51PTR	Cloud: 1.63%
20200323T021559_20200323T022824_T51PUR	Cloud: 6.85%
20200328T021601_20200328T021559_T51PTR	Cloud: 3.89%
20200328T021601_20200328T021559_T51PUR	Cloud: 4.69%
20200402T021559_20200402T022823_T51PTR	Cloud: 15.40%
20200402T021559_20200402T022823_T51PUR	Cloud: 15.93%
20200407T021601_20200407T023152_T51PTR	Cloud: 12.15%
20200407T021601_20200407T023152_T51PUR	Cloud: 25.17%
20200422T021559_20200422T023323_T51PTR	Cloud: 30.55%

20200422T021559_20200422T023323_T51PUR	Cloud: 32.56%
20200427T021611_20200427T021605_T51PUR	Cloud: 57.37%
20200502T021559_20200502T022822_T51PTR	Cloud: 26.44%
20200502T021559_20200502T022822_T51PUR	Cloud: 27.33%
20200507T021611_20200507T023210_T51PTR	Cloud: 38.48%
20200507T021611_20200507T023210_T51PUR	Cloud: 25.63%
20200512T021559_20200512T022824_T51PTR	Cloud: 9.92%
20200512T021559_20200512T022824_T51PUR	Cloud: 35.12%
20200522T021609_20200522T022826_T51PTR	Cloud: 45.58%
20200527T021611_20200527T021609_T51PTR	Cloud: 10.37%
20200527T021611_20200527T021609_T51PUR	Cloud: 15.53%
20200601T021609_20200601T022827_T51PTR	Cloud: 46.60%
20200606T021611_20200606T023201_T51PTR	Cloud: 15.95%
20200606T021611_20200606T023201_T51PUR	Cloud: 17.28%
20200616T021611_20200616T021609_T51PTR	Cloud: 23.04%
20200616T021611_20200616T021609_T51PUR	Cloud: 15.87%
20200621T021609_20200621T022825_T51PTR	Cloud: 13.13%
20200621T021609_20200621T022825_T51PUR	Cloud: 20.04%
20200701T021609_20200701T022810_T51PTR	Cloud: 12.04%
20200701T021609_20200701T022810_T51PUR	Cloud: 7.52%
20200701T021609_20200701T023342_T51PTR	Cloud: 11.52%
20200701T021609_20200701T023342_T51PUR	Cloud: 6.76%
20200706T021611_20200706T023159_T51PTR	Cloud: 13.05%
20200706T021611_20200706T023159_T51PUR	Cloud: 15.14%
20200716T021611_20200716T021608_T51PUR	Cloud: 50.06%
20200721T021609_20200721T022827_T51PTR	Cloud: 13.07%
20200721T021609_20200721T022827_T51PUR	Cloud: 45.08%
20200726T021611_20200726T021608_T51PTR	Cloud: 27.37%
20200726T021611_20200726T021608_T51PUR	Cloud: 28.66%
20200825T021611_20200825T021608_T51PTR	Cloud: 22.28%
20200825T021611_20200825T021608_T51PUR	Cloud: 13.75%
20200904T021611_20200904T023207_T51PTR	Cloud: 21.86%
20200904T021611_20200904T023207_T51PUR	Cloud: 41.78%
20200914T021611_20200914T021607_T51PTR	Cloud: 41.55%
20200914T021611_20200914T021607_T51PUR	Cloud: 26.15%
20200919T021609_20200919T022509_T51PTR	Cloud: 36.31%
20200919T021609_20200919T022509_T51PUR	Cloud: 39.07%
20200929T021609_20200929T022847_T51PUR	Cloud: 42.78%
20201004T021611_20201004T022814_T51PTR	Cloud: 41.12%
20201029T021819_20201029T023057_T51PTR	Cloud: 23.07%
20201029T021819_20201029T023057_T51PUR	Cloud: 37.76%
20201103T021851_20201103T023052_T51PTR	Cloud: 33.70%
20201103T021851_20201103T023052_T51PUR	Cloud: 39.47%
20201113T021941_20201113T023144_T51PTR	Cloud: 33.07%
20201113T021941_20201113T023144_T51PUR	Cloud: 39.18%
20201118T021959_20201118T022903_T51PTR	Cloud: 22.62%

20201118T021959_20201118T022903_T51PUR	Cloud: 57.38%
20201123T022021_20201123T022633_T51PTR	Cloud: 57.21%
20201213T022111_20201213T023311_T51PTR	Cloud: 27.40%
20201213T022111_20201213T023311_T51PUR	Cloud: 44.58%
20201223T022111_20201223T022811_T51PTR	Cloud: 22.58%
20201223T022111_20201223T022811_T51PUR	Cloud: 20.25%

Total of 86 scenes*

Table C3. 2021 Image Inventory (External Test Set)

Sentinel-2 Product ID	Cloud Cover %
20210107T022059_20210107T023336_T51PTR	Cloud: 49.42%
20210201T021921_20210201T023126_T51PTR	Cloud: 36.63%
20210211T021821_20210211T023026_T51PTR	Cloud: 24.12%
20210211T021821_20210211T023026_T51PUR	Cloud: 25.39%
20210216T021749_20210216T022613_T51PTR	Cloud: 17.85%
20210216T021749_20210216T022613_T51PUR	Cloud: 42.66%
20210221T021721_20210221T022406_T51PTR	Cloud: 50.17%
20210226T021639_20210226T022921_T51PTR	Cloud: 23.05%
20210226T021639_20210226T022921_T51PUR	Cloud: 21.07%
20210303T021611_20210303T022811_T51PTR	Cloud: 15.59%
20210303T021611_20210303T022811_T51PUR	Cloud: 24.23%
20210318T021559_20210318T022832_T51PTR	Cloud: 19.22%
20210318T021559_20210318T022832_T51PUR	Cloud: 26.37%
20210328T031559_20210328T022831_T51PTR	Cloud: 56.67%
20210328T031559_20210328T022831_T51PUR	Cloud: 43.23%
20210402T021601_20210402T022805_T51PTR	Cloud: 6.85%
20210402T021601_20210402T022805_T51PUR	Cloud: 9.66%
20210407T021559_20210407T022830_T51PTR	Cloud: 29.99%
20210407T021559_20210407T022830_T51PUR	Cloud: 53.76%
20210412T021601_20210412T022814_T51PTR	Cloud: 48.62%
20210422T021601_20210422T022803_T51PTR	Cloud: 56.94%
20210422T021601_20210422T022803_T51PUR	Cloud: 34.51%
20210502T021601_20210502T022806_T51PTR	Cloud: 13.15%
20210502T021601_20210502T022806_T51PUR	Cloud: 26.43%
20210512T021601_20210512T022807_T51PTR	Cloud: 13.80%
20210512T021601_20210512T022807_T51PUR	Cloud: 17.08%
20210517T021559_20210517T022503_T51PTR	Cloud: 12.20%
20210517T021559_20210517T022503_T51PUR	Cloud: 18.45%
20210522T021601_20210522T023336_T51PTR	Cloud: 44.48%
20210522T021601_20210522T023336_T51PUR	Cloud: 24.90%
20210527T021609_20210527T022833_T51PTR	Cloud: 22.04%
20210527T021609_20210527T022833_T51PUR	Cloud: 22.87%

20210601T021601_20210601T022809_T51PTR	Cloud: 57.75%
20210601T021601_20210601T022809_T51PUR	Cloud: 47.06%
20210616T021609_20210616T022418_T51PTR	Cloud: 18.50%
20210616T021609_20210616T022418_T51PUR	Cloud: 27.82%
20210626T021609_20210626T022833_T51PTR	Cloud: 22.41%
20210626T021609_20210626T022833_T51PUR	Cloud: 15.04%
20210701T021611_20210701T022810_T51PTR	Cloud: 19.42%
20210701T021611_20210701T022810_T51PUR	Cloud: 20.11%
20210711T021611_20210711T022811_T51PTR	Cloud: 36.98%
20210726T021609_20210726T022835_T51PTR	Cloud: 47.28%
20210726T021609_20210726T022835_T51PUR	Cloud: 50.68%
20210820T021611_20210820T022256_T51PTR	Cloud: 57.21%
20210820T021611_20210820T022256_T51PUR	Cloud: 19.59%
20210825T021559_20210825T022806_T51PTR	Cloud: 35.57%
20210904T021559_20210904T022829_T51PTR	Cloud: 41.78%
20210904T021559_20210904T022829_T51PUR	Cloud: 36.01%
20210909T021601_20210909T022808_T51PUR	Cloud: 44.97%
20210914T021559_20210914T022500_T51PTR	Cloud: 27.09%
20210914T021559_20210914T022500_T51PUR	Cloud: 21.45%
20210924T021559_20210924T022831_T51PTR	Cloud: 28.65%
20210924T021559_20210924T022831_T51PUR	Cloud: 44.53%
20211019T021721_20211019T022407_T51PTR	Cloud: 15.53%
20211019T021721_20211019T022407_T51PUR	Cloud: 21.84%
20211029T021821_20211029T023022_T51PTR	Cloud: 12.08%
20211029T021821_20211029T023022_T51PUR	Cloud: 52.43%
20211103T021839_20211103T022714_T51PTR	Cloud: 29.14%
20211108T021901_20211108T023014_T51PTR	Cloud: 47.94%
20211123T022009_20211123T023253_T51PTR	Cloud: 54.95%
20211203T022049_20211203T023324_T51PTR	Cloud: 34.47%
20211203T022049_20211203T023324_T51PUR	Cloud: 56.79%
20211208T022101_20211208T023303_T51PTR	Cloud: 47.12%
20211208T022101_20211208T023303_T51PUR	Cloud: 59.66%
20211223T022109_20211223T022109_T51PTR	Cloud: 13.82%
20211223T022109_20211223T022109_T51PUR	Cloud: 39.78%

Total of 66 scenes*

APPENDIX D

Visualizations and Tree Recommendation Output

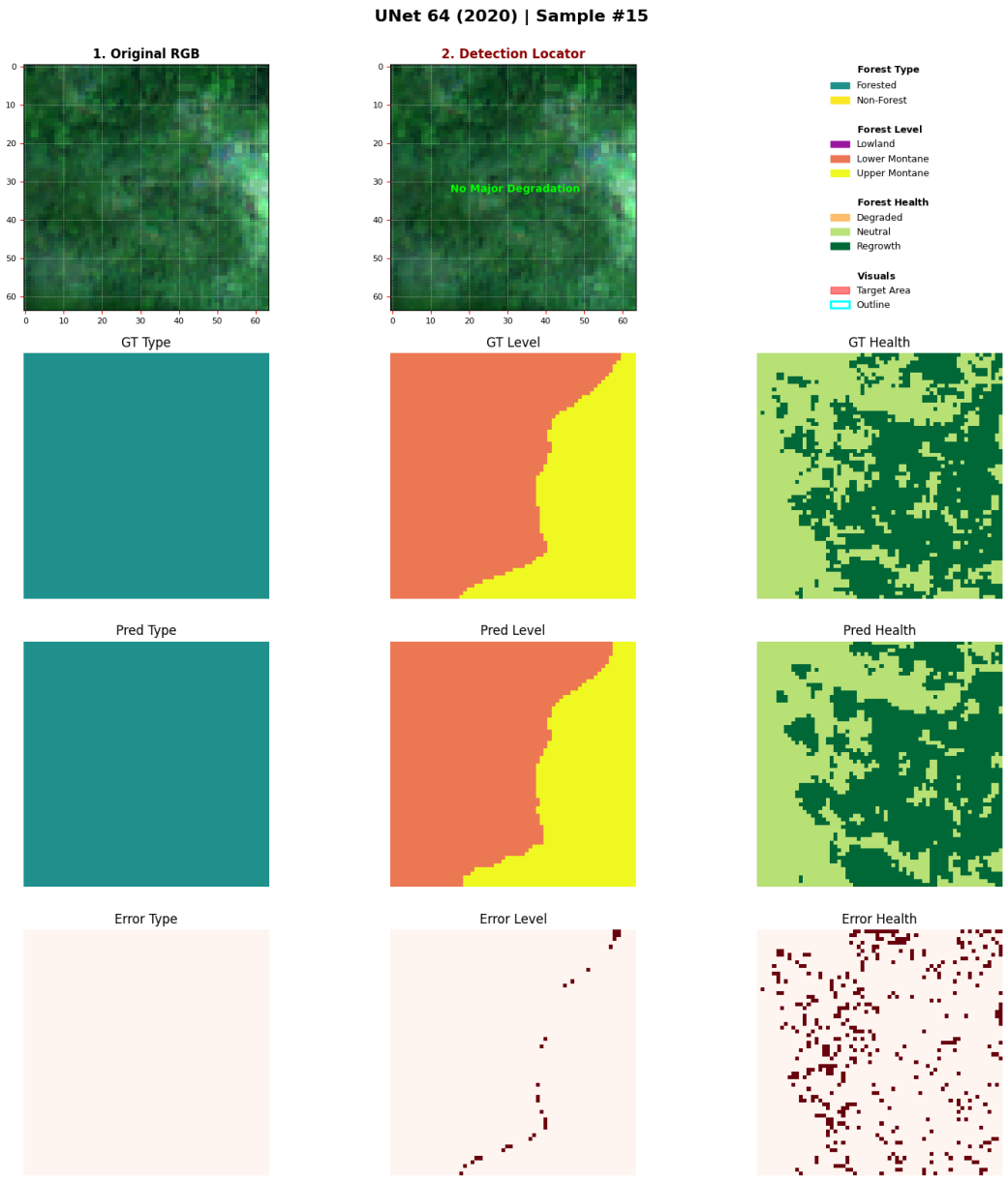


Figure D1.1. U-Net 64x64 on 2020 Test Set - Visualization

```
================================================================
### REFORESTATION PLAN FOR SAMPLE #15 (Model: UNet 64 (2020)) ###
================================================================
--- Status: Healthy / No Major Action Needed ---
```

Figure D1.2. U-Net 64x64 on 2020 Test Set – Tree Recommendation

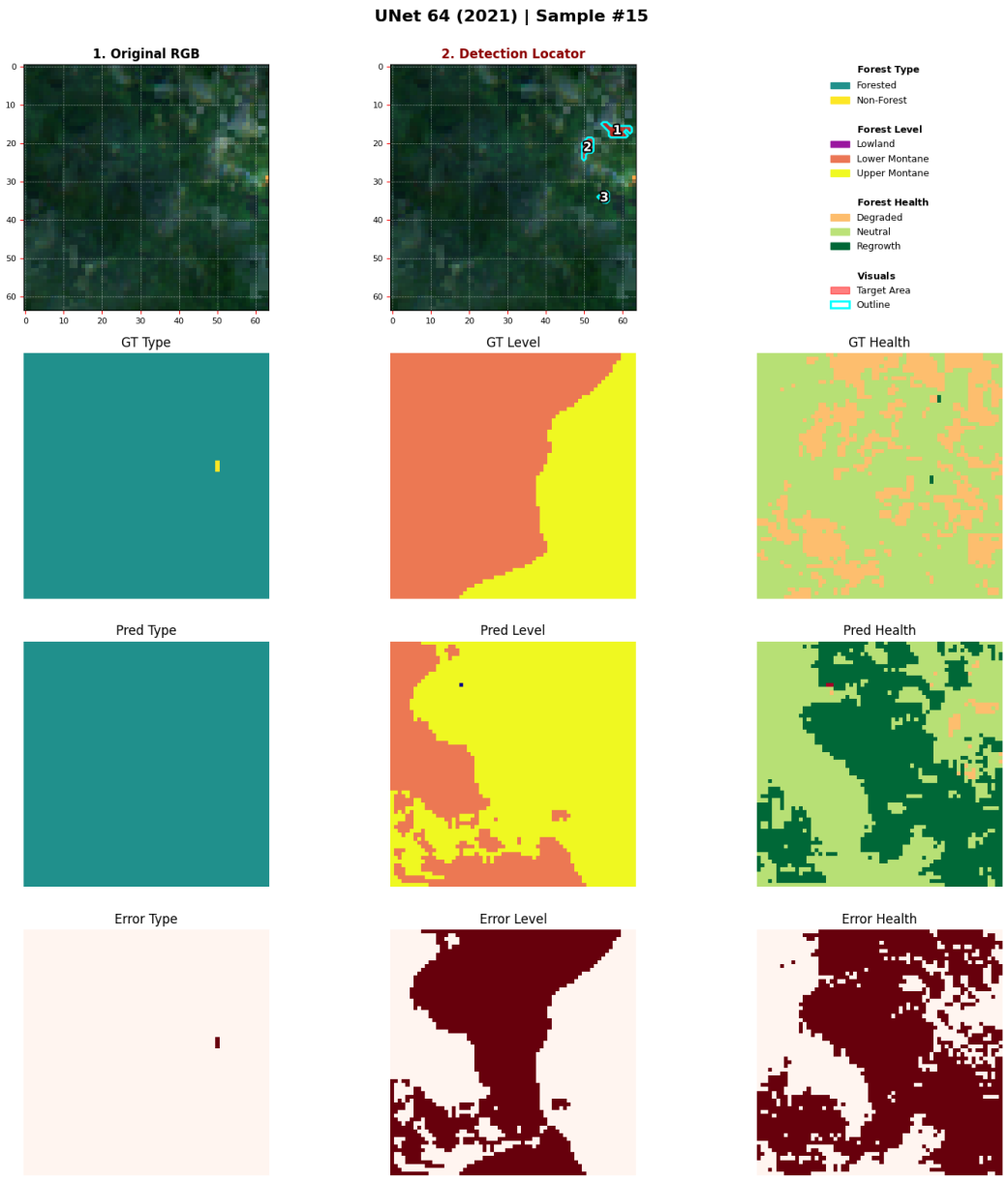


Figure D2.1. U-Net 64x64 on 2021 Test Set – Visualization

```
================================================================
### REFORESTATION PLAN FOR SAMPLE #15 (Model: UNet 64 (2021)) ###
================================================================
[] AREA 1 | Loc: (58, 16) | Size: 19px
   >>> ACTION: Plant Dacrycarpus cumingii (Cuming's Yellowwood), Ilex philippinensis (Philippine holly), Vaccinium spp. (Blueberries), Rapanea spp.(Kalingag-kalingag), Clethra tomentella, Weinmannia subsessilis
   >>> ZONE:   Upper M
----------------------------------------
[] AREA 2 | Loc: (50, 21) | Size: 13px
   >>> ACTION: Plant Dacrycarpus cumingii (Cuming's Yellowwood), Ilex philippinensis (Philippine holly), Vaccinium spp. (Blueberries), Rapanea spp.(Kalingag-kalingag), Clethra tomentella, Weinmannia subsessilis
   >>> ZONE:   Upper M
----------------------------------------
[] AREA 3 | Loc: (55, 34) | Size: 6px
   >>> ACTION: Plant Dacrycarpus cumingii (Cuming's Yellowwood), Ilex philippinensis (Philippine holly), Vaccinium spp. (Blueberries), Rapanea spp.(Kalingag-kalingag), Clethra tomentella, Weinmannia subsessilis
   >>> ZONE:   Upper M
----------------------------------------
```

Figure D2.2. U-Net 64x64 on 2021 Test Set – Tree Recommendation

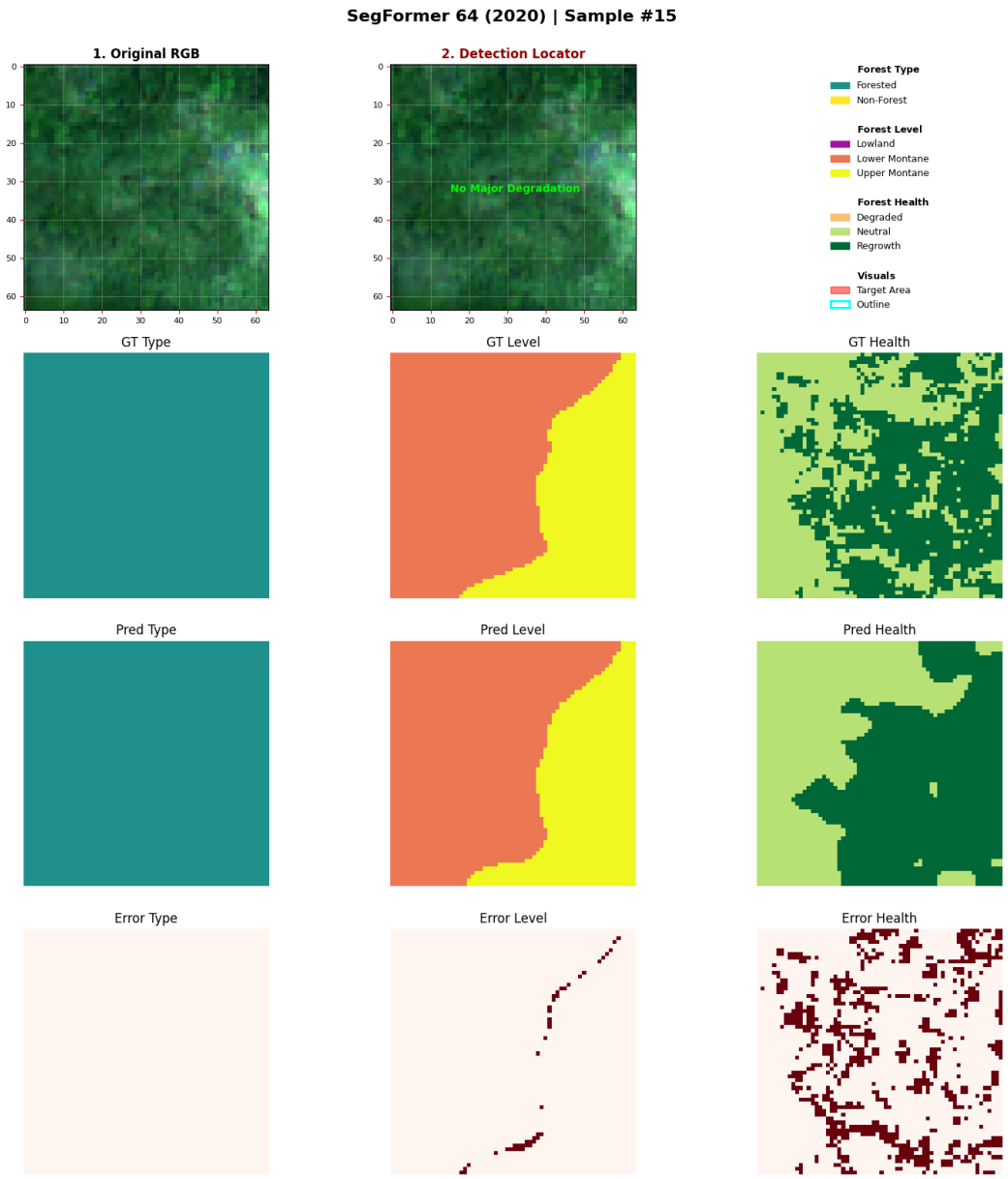


Figure D3.1. SegFormer 64x64 on 2020 Test Set – Visualization

```
================================================================
### REFORESTATION PLAN FOR SAMPLE #15 (Model: SegFormer 64 (2020)) ###
================================================================
--- Status: Healthy / No Major Action Needed ---
```

Figure D3.2. U-Net 64x64 on 2020 Test Set – Tree Recommendation

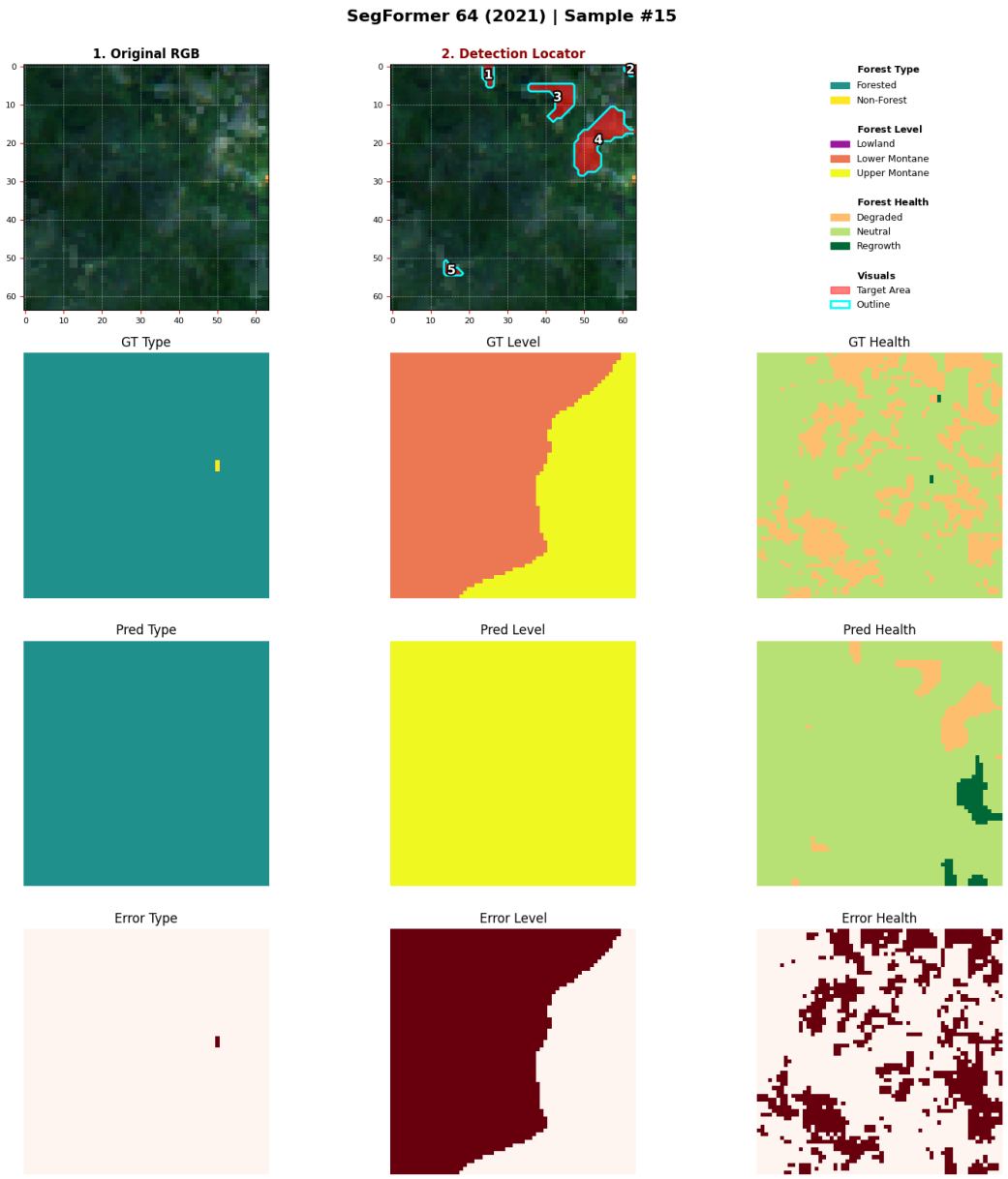


Figure D4.1. SegFormer 64x64 on 2021 Test Set – Visualization

```
================================================================
### REFORESTATION PLAN FOR SAMPLE #15 (Model: SegFormer 64 (2021)) ###
================================================================
[] AREA 1 | Loc: (25, 2) | Size: 16px
   >>> ACTION: Plant Dacrycarpus cumingii (Cuming's Yellowwood), Ilex philippinensis (Philippine holly), Vaccinium spp. (Blueberries), Rapanea spp.(Kalingag-kalingag), Clethra tomentella, Weinmannia subsessilis
   >>> ZONE:   Upper M
----------------------------------------
[] AREA 2 | Loc: (62, 0) | Size: 8px
   >>> ACTION: Plant Dacrycarpus cumingii (Cuming's Yellowwood), Ilex philippinensis (Philippine holly), Vaccinium spp. (Blueberries), Rapanea spp.(Kalingag-kalingag), Clethra tomentella, Weinmannia subsessilis
   >>> ZONE:   Upper M
----------------------------------------
[] AREA 3 | Loc: (43, 8) | Size: 58px
   >>> ACTION: Plant Dacrycarpus cumingii (Cuming's Yellowwood), Ilex philippinensis (Philippine holly), Vaccinium spp. (Blueberries), Rapanea spp.(Kalingag-kalingag), Clethra tomentella, Weinmannia subsessilis
   >>> ZONE:   Upper M
----------------------------------------
[] AREA 4 | Loc: (53, 19) | Size: 127px
   >>> ACTION: Plant Dacrycarpus cumingii (Cuming's Yellowwood), Ilex philippinensis (Philippine holly), Vaccinium spp. (Blueberries), Rapanea spp.(Kalingag-kalingag), Clethra tomentella, Weinmannia subsessilis
   >>> ZONE:   Upper M
----------------------------------------
[] AREA 5 | Loc: (15, 53) | Size: 11px
   >>> ACTION: Plant Dacrycarpus cumingii (Cuming's Yellowwood), Ilex philippinensis (Philippine holly), Vaccinium spp. (Blueberries), Rapanea spp.(Kalingag-kalingag), Clethra tomentella, Weinmannia subsessilis
   >>> ZONE:   Upper M
----------------------------------------
```

Figure D4.2. Segformer 64x64 on 2021 Test Set – Tree Recommendation

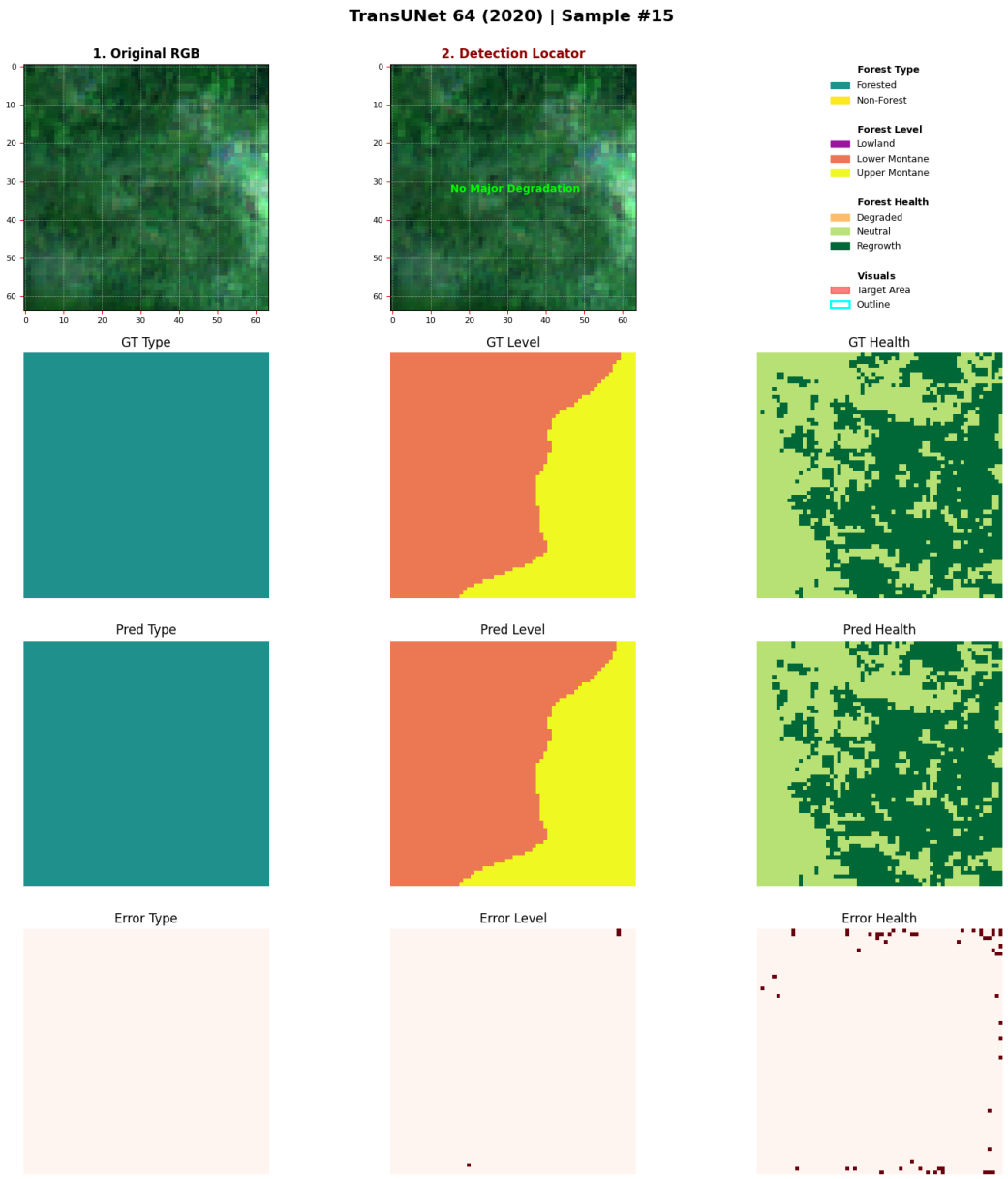


Figure D5.1. TransU-Net 64x64 on 2020 Test Set – Visualization

```
================================================================
### REFORESTATION PLAN FOR SAMPLE #15 (Model: TransUNet 64 (2020)) ###
================================================================
--- Status: Healthy / No Major Action Needed ---
```

Figure D5.2. TransU-Net 64x64 on 2020 Test Set – Tree Recommendation

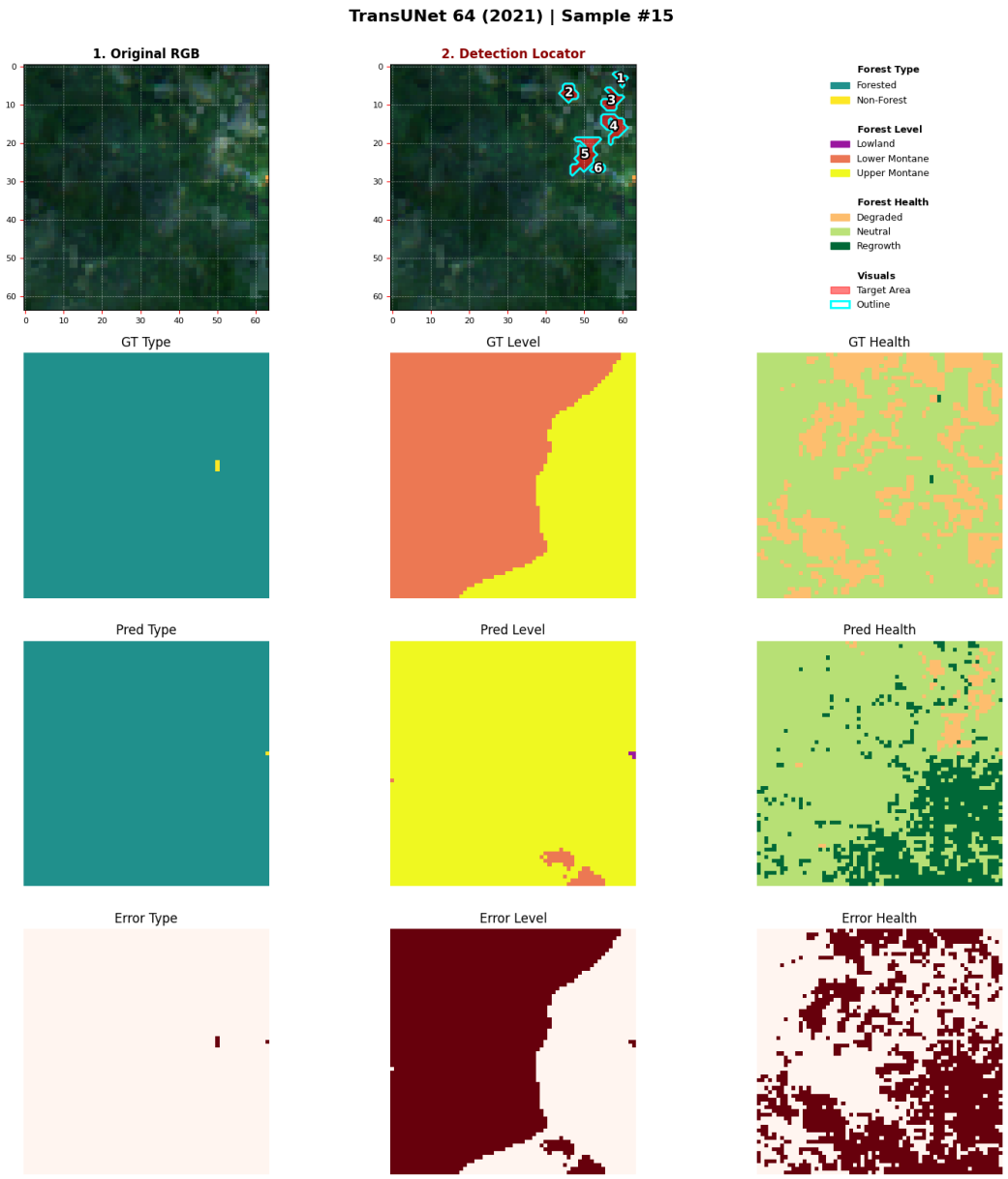


Figure D6.1. TransU-Net 64x64 on 2021 Test Set – Visualization

```
======================================================================
### REFORESTATION PLAN FOR SAMPLE #15 (Model: TransUNet 64 (2021)) ###
======================================================================
[] AREA 1 | Loc: (59, 3) | Size: 7px
   >>> ACTION: Plant Dacrycarpus cumingii (Cuming's Yellowwood), Ilex philippinensis (Philippine holly), Vaccinium spp. (Blueberries), Rapanea spp.(Kalingag-kalingag), Clethra tomentella, Weinmannia subsessilis
   >>> ZONE:   Upper M
--------------------------------------
[] AREA 2 | Loc: (46, 6) | Size: 16px
   >>> ACTION: Plant Dacrycarpus cumingii (Cuming's Yellowwood), Ilex philippinensis (Philippine holly), Vaccinium spp. (Blueberries), Rapanea spp.(Kalingag-kalingag), Clethra tomentella, Weinmannia subsessilis
   >>> ZONE:   Upper M
--------------------------------------
[] AREA 3 | Loc: (57, 8) | Size: 19px
   >>> ACTION: Plant Dacrycarpus cumingii (Cuming's Yellowwood), Ilex philippinensis (Philippine holly), Vaccinium spp. (Blueberries), Rapanea spp.(Kalingag-kalingag), Clethra tomentella, Weinmannia subsessilis
   >>> ZONE:   Upper M
--------------------------------------
[] AREA 4 | Loc: (57, 15) | Size: 28px
   >>> ACTION: Plant Dacrycarpus cumingii (Cuming's Yellowwood), Ilex philippinensis (Philippine holly), Vaccinium spp. (Blueberries), Rapanea spp.(Kalingag-kalingag), Clethra tomentella, Weinmannia subsessilis
   >>> ZONE:   Upper M
--------------------------------------
[] AREA 5 | Loc: (50, 22) | Size: 41px
   >>> ACTION: Plant Dacrycarpus cumingii (Cuming's Yellowwood), Ilex philippinensis (Philippine holly), Vaccinium spp. (Blueberries), Rapanea spp.(Kalingag-kalingag), Clethra tomentella, Weinmannia subsessilis
   >>> ZONE:   Upper M
--------------------------------------
[] AREA 6 | Loc: (53, 26) | Size: 7px
   >>> ACTION: Plant Dacrycarpus cumingii (Cuming's Yellowwood), Ilex philippinensis (Philippine holly), Vaccinium spp. (Blueberries), Rapanea spp.(Kalingag-kalingag), Clethra tomentella, Weinmannia subsessilis
   >>> ZONE:   Upper M
--------------------------------------
```

Figure D6.2. TransU-Net 64x64 on 2021 Test Set – Tree Recommendation

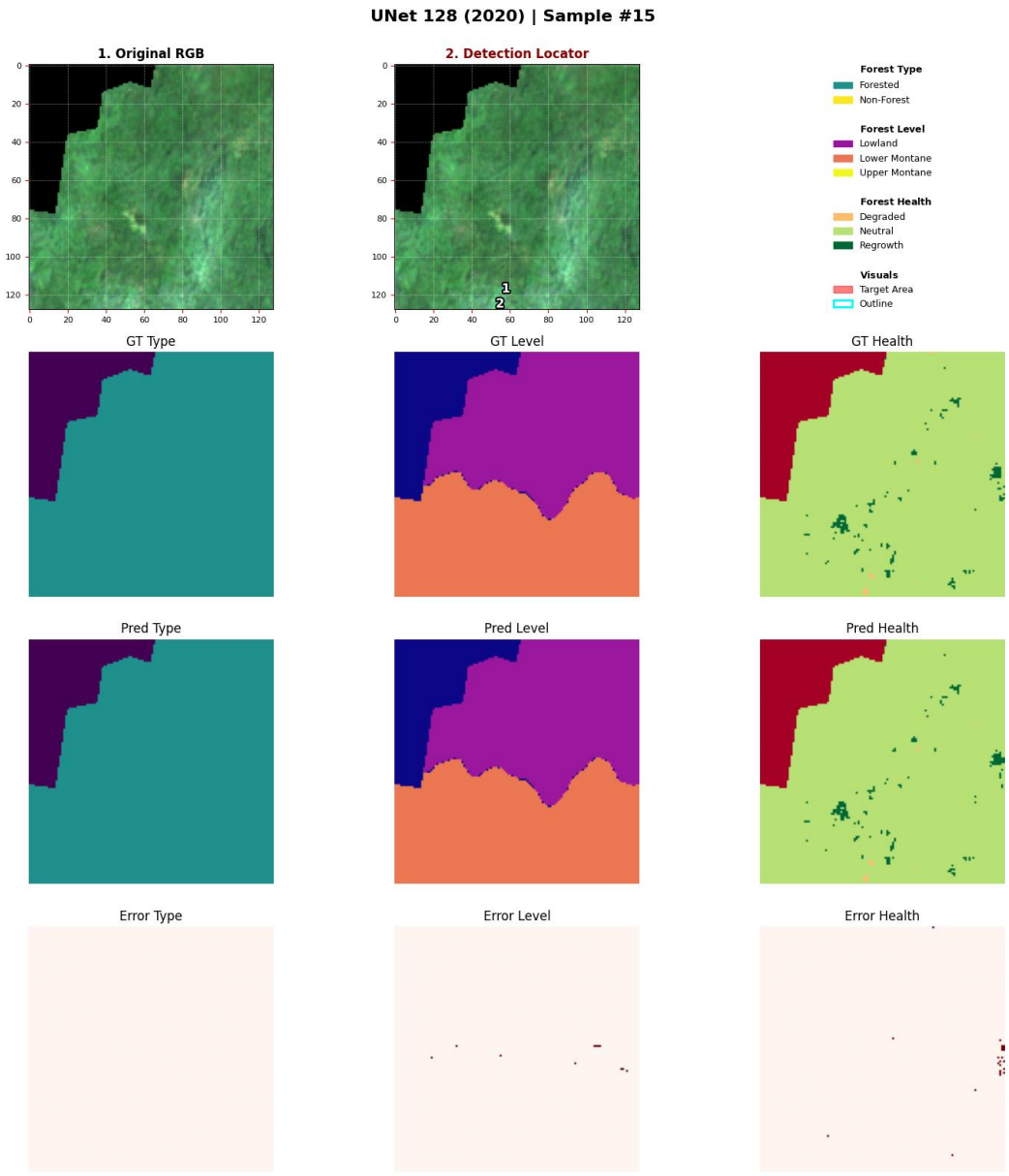


Figure D7.1. U-Net 128x128 on 2020 Test Set – Visualization

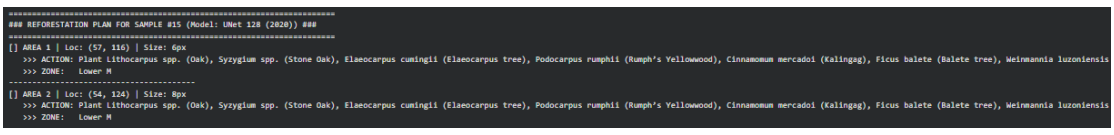


Figure D7.2. U-Net 128x128 on 2020 Test Set – Tree Recommendation

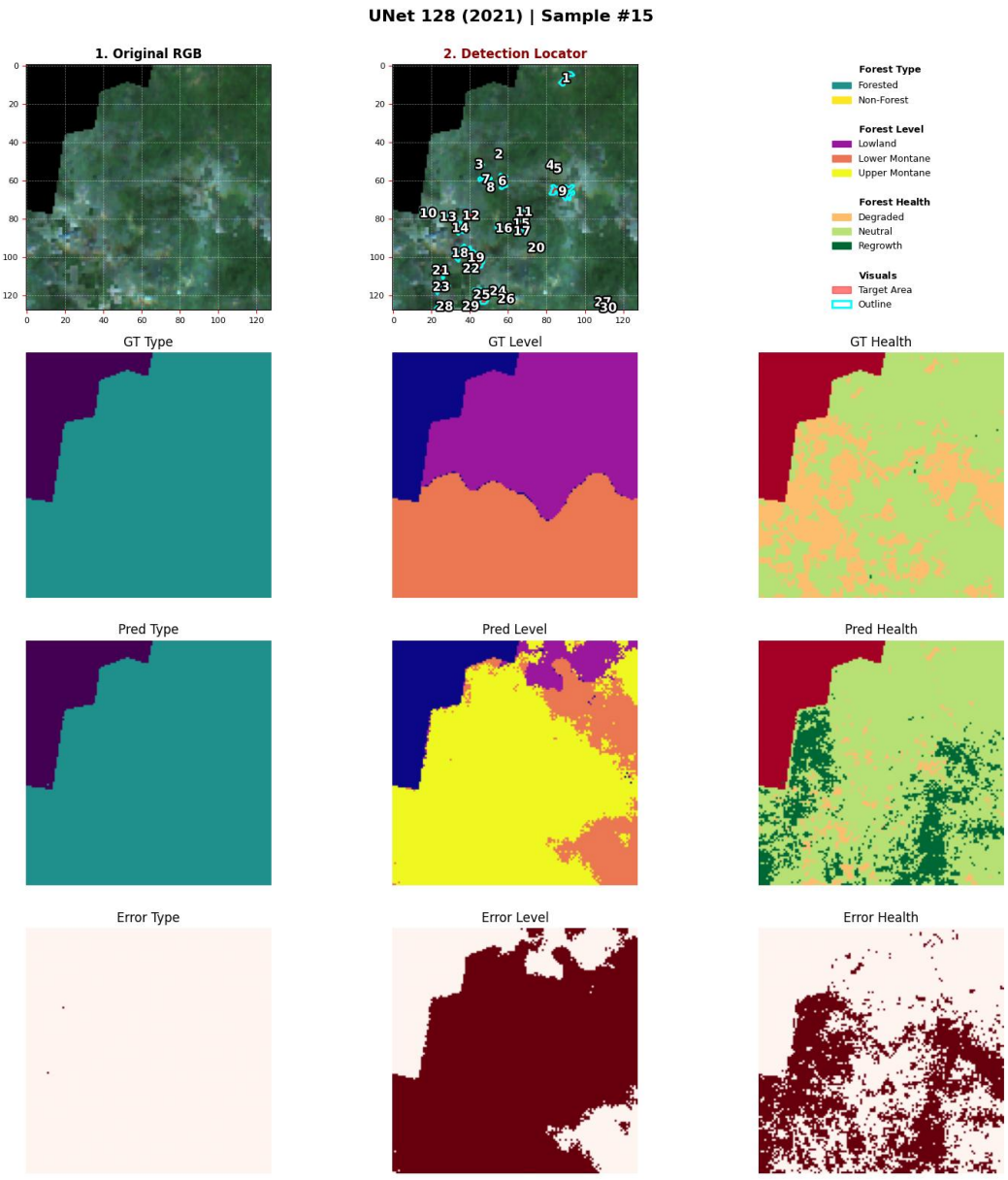


Figure D8.1. U-Net 128x128 on 2021 Test Set – Visualization

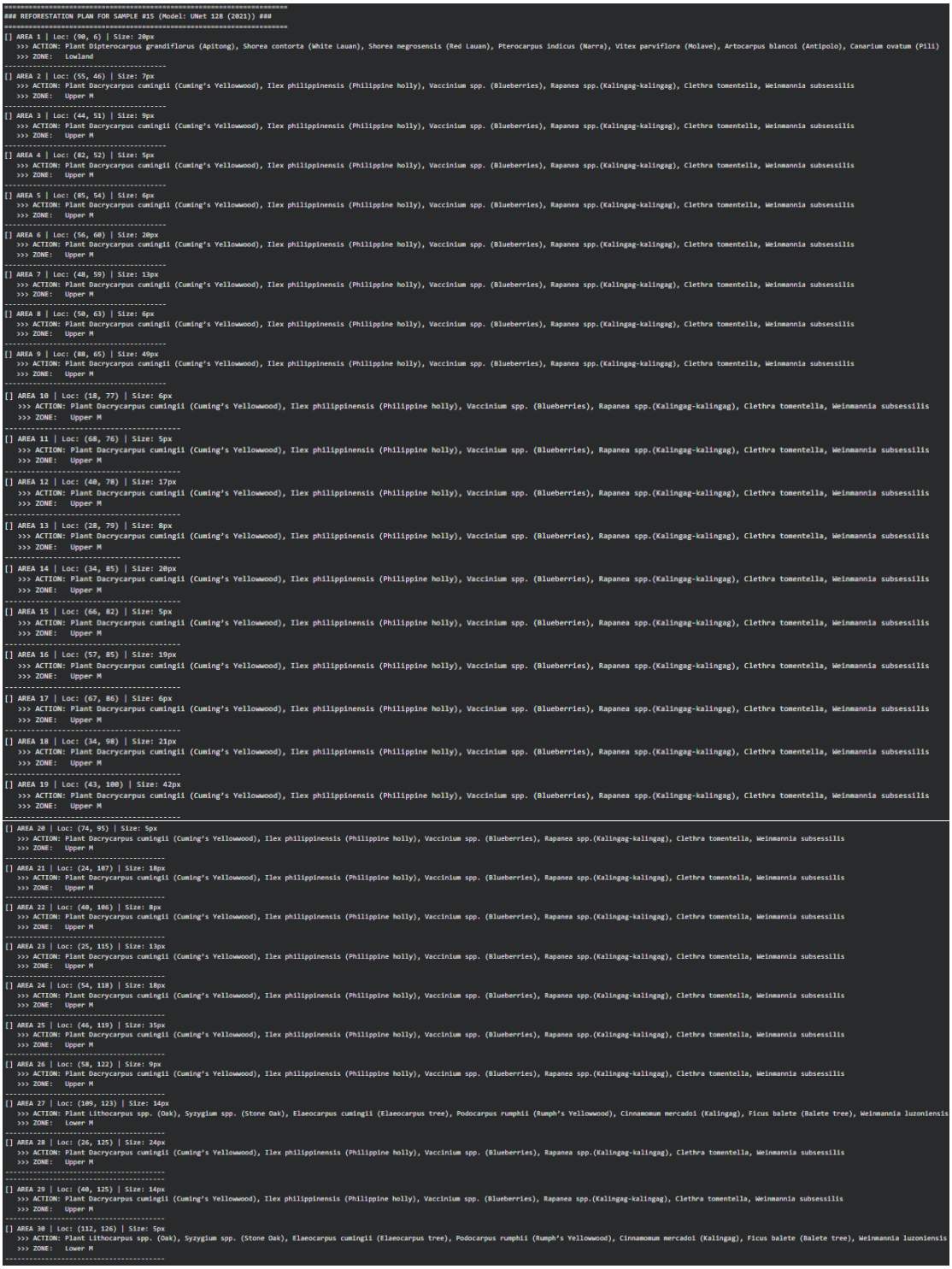


Figure D8.2. TransU-Net 128x128 on 2021 Test Set – Tree Recommendation

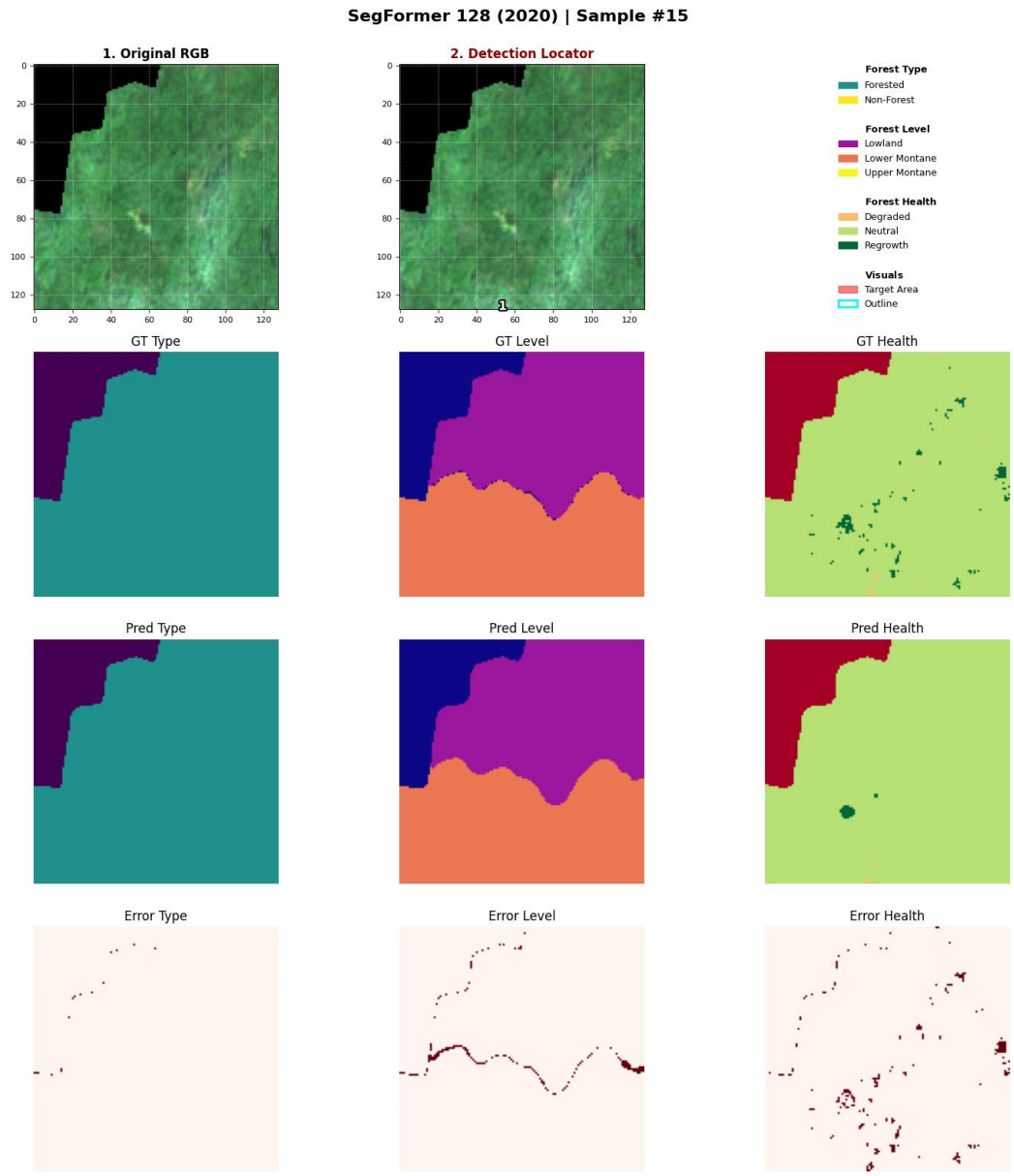


Figure D9.1. SegFormer128x128 on 2020 Test Set – Visualization



Figure D9.2. SegFormer 128x128 on 2020 Test Set – Tree Recommendation

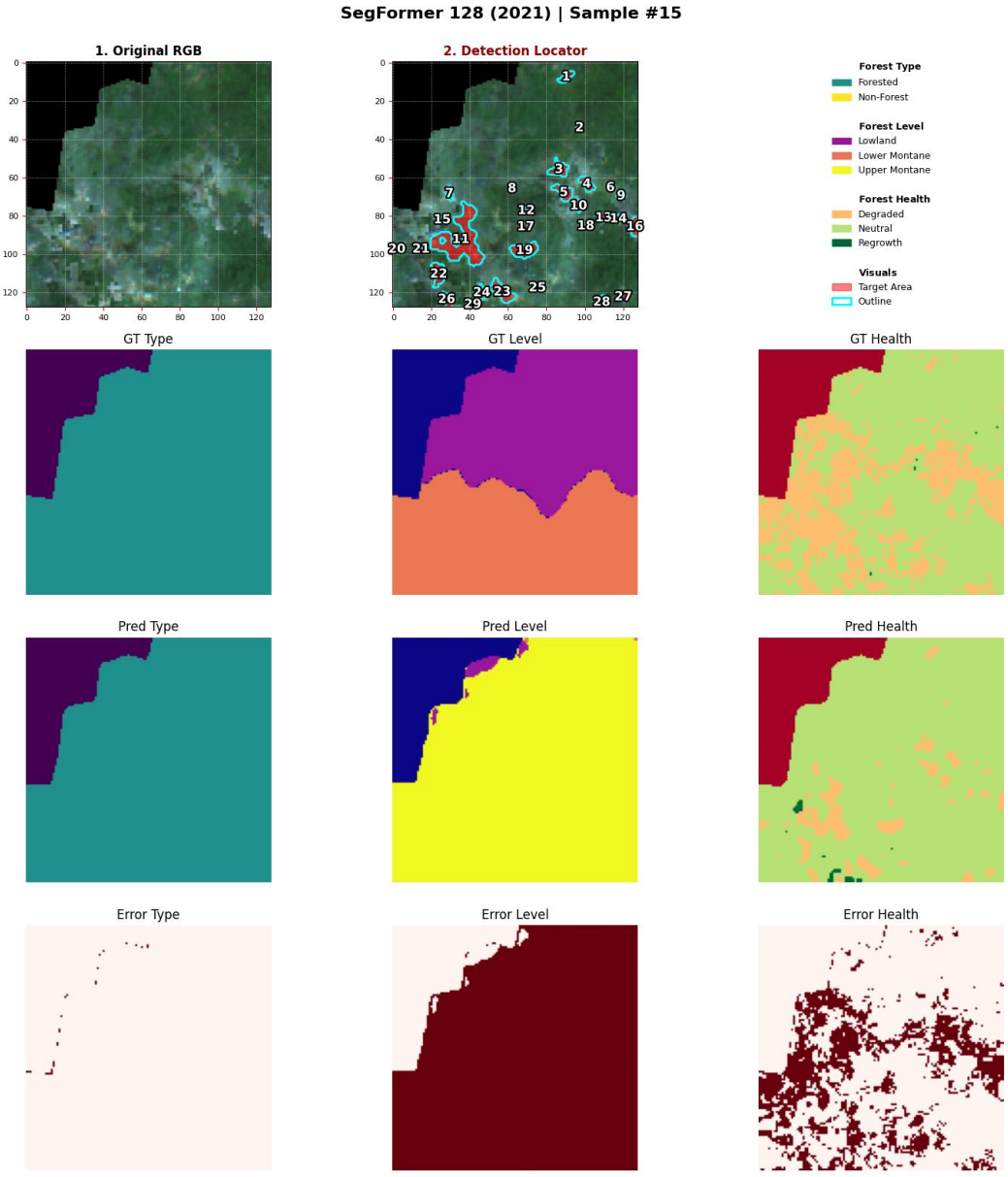


Figure D10.1. SegFormer 128x128 on 2021 Test Set – Visualization

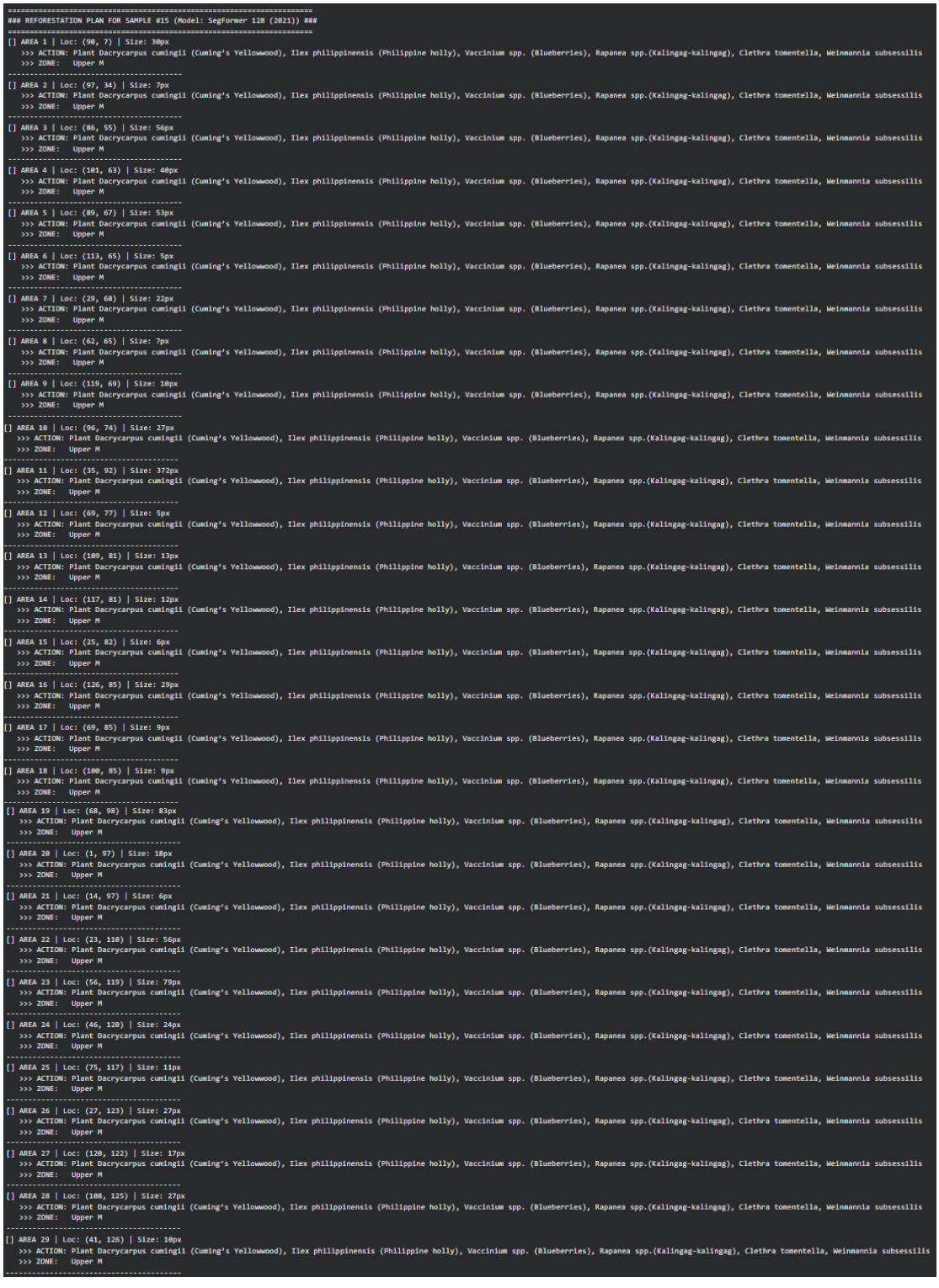


Figure D10.2. SegFormer 128x128 on 2021 Test Set – Tree Recommendation

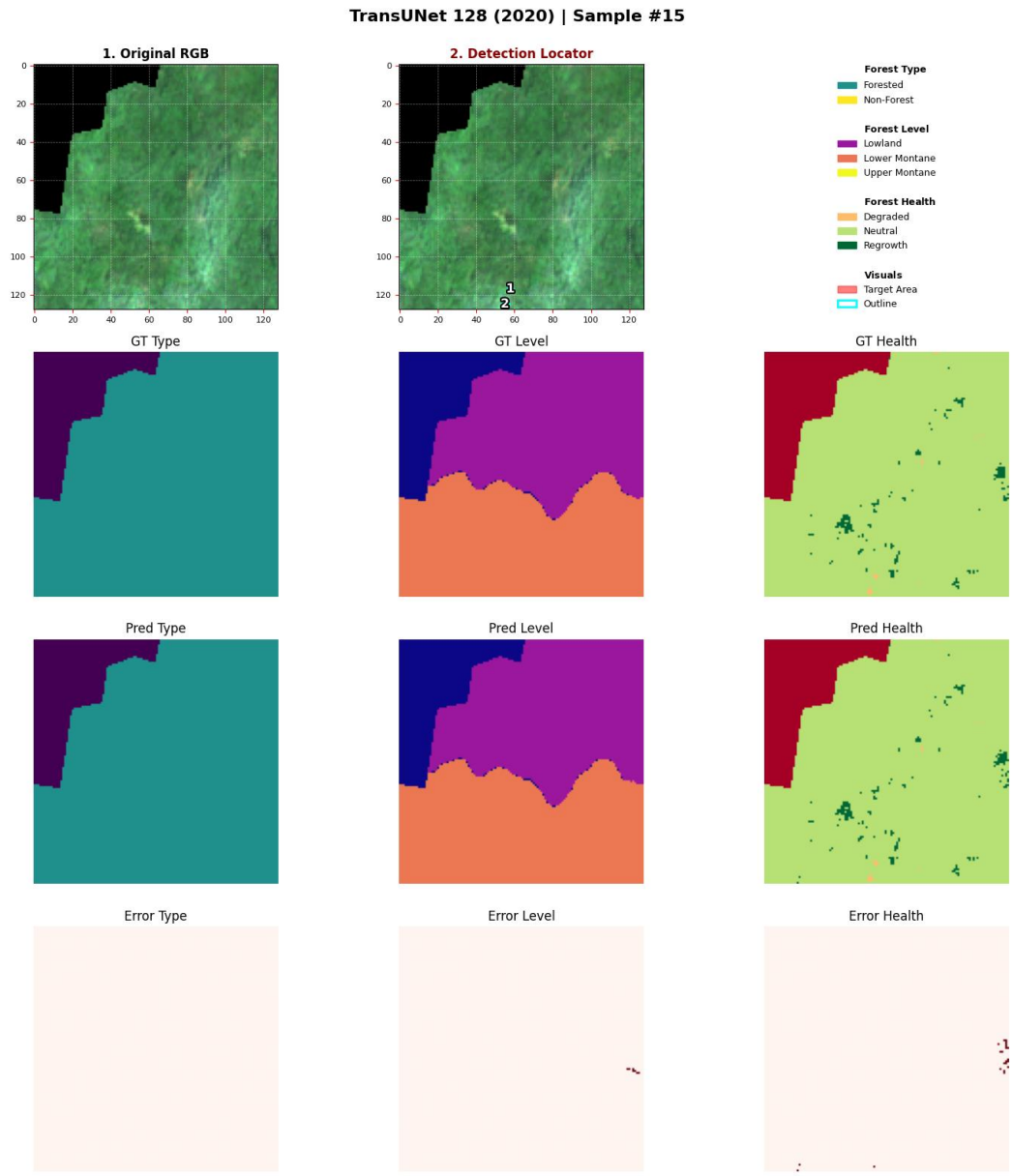


Figure D11.1. TransU-Net 128x128 on 2020 Test Set – Visualization

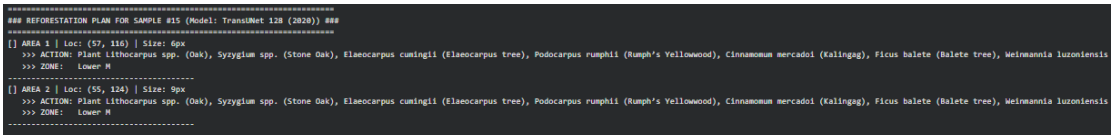


Figure D11.2. TransU-Net 128x128 on 2020 Test Set – Tree Recommendation

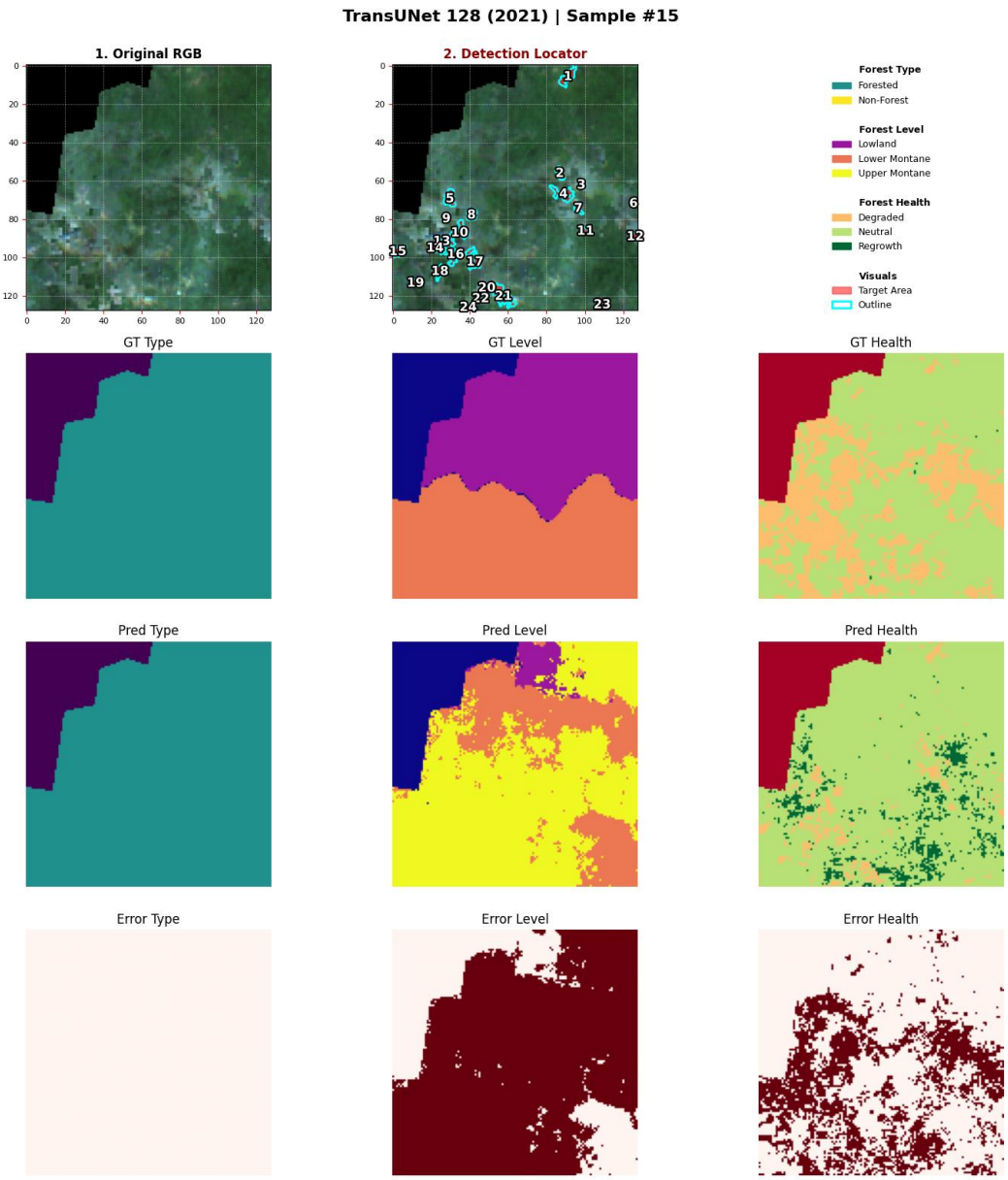


Figure D12.1. TransU-Net 128x128 on 2021 Test Set – Visualization

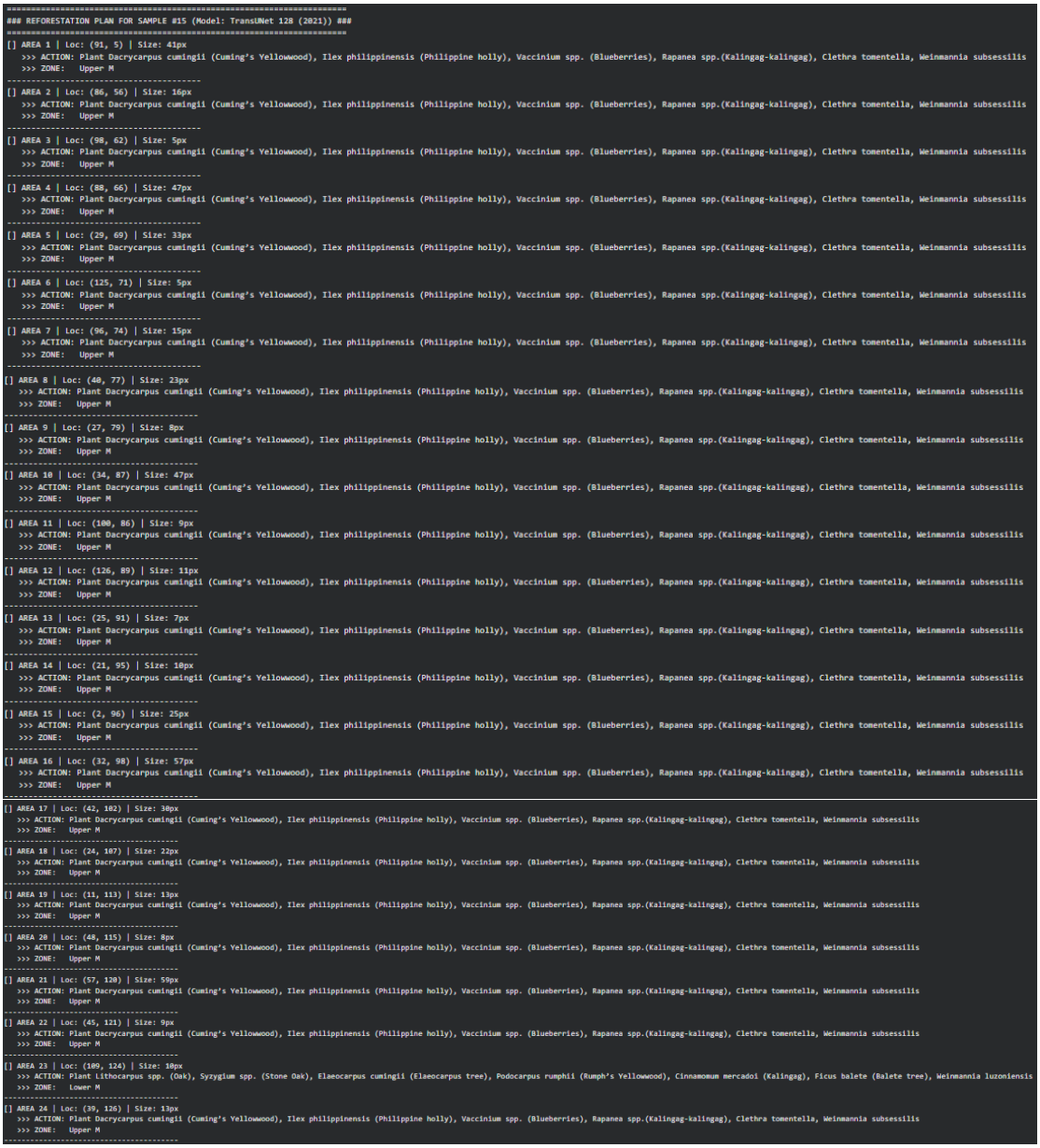


Figure D12.2. TransU-Net 128x128 on 2021 Test Set – Tree Recommendation

APPENDIX E

Google Earth Engine Scripts

Forest Type Label Collection

```
var AOI = ee.FeatureCollection('projects/makiling-v2-dataset/assets/Makiling-AOI');
var wc2020 = ee.Image('ESA/WorldCover/v100/2020').select('Map');
var wc2021 = ee.Image('ESA/WorldCover/v200/2021').select('Map');

function reclassifyWorldCoverTwoClasses(image) {
  var forestClasses = [10];
  var fromClasses = forestClasses;
  var toClasses = fillArray(forestClasses.length, 1);
  var remapped = image.remap(fromClasses, toClasses, 2);
  return remapped.rename('ForestType');
}

var wc2020Class = reclassifyWorldCoverTwoClasses(wc2020).clip(AOI).set('year', 2020).set('source', 'WorldCover 2020 (2 classes)');
var wc2021Class = reclassifyWorldCoverTwoClasses(wc2021).clip(AOI).set('year', 2021).set('source', 'WorldCover 2021 (2 classes)');
```

Forest Level Label Collection

```
var AOI = ee.FeatureCollection('projects/makiling-v2-dataset/assets/Makiling-AOI');

var srtm30 = ee.Image('USGS/SRTMGL1_003').clip(AOI);
var srtm10 = srtm30
  .resample('bilinear')
  .reproject({crs: 'EPSG:4326', scale: 10});

function classifyElevation(image) {
  return image.expression(
    "b('elevation') < 300 ? 1" +  // Lowland
    ": (b('elevation') >= 301 && b('elevation') <= 700) ? 2" +  // Lower Montane
    ": (b('elevation') >= 701 && b('elevation') <= 1114) ? 3" + // Upper Montane
    ": 0",
    {'elevation': image}
  ).rename('forest_level').clip(AOI);
}
```

Forest Health Label Collection

```
var AOI = ee.FeatureCollection('projects/makiling-v2-dataset/assets/Makiling-
          AOI');

function maskS2(image) {
  var scl = image.select('SCL');
  var mask = scl.neq(2)  // cloud shadow
```

```
              .and(scl.neq(3))
              .and(scl.neq(8))
              .and(scl.neq(9))
              .and(scl.neq(10));
  return image.updateMask(mask);
}

var s2_2019 = ee.ImageCollection('COPERNICUS/S2_SR')
  .filterBounds(AOI)
  .filterDate('2019-01-01', '2019-12-31')
  .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 40))
  .map(maskS2);

var ndvi2019 = s2_2019
  .select(['B8','B4'])
  .median()
  .normalizedDifference(['B8','B4'])
  .rename('NDVI')
  .clip(AOI);


function loadNDVI(assetPath) {
  return ee.Image(assetPath).select('NDVI').clip(AOI);
}

var ndvi2020 = loadNDVI('projects/makiling-v2-
          dataset/assets/Makiling_S2_Composite_2020');
var ndvi2021 = loadNDVI('projects/makiling-v2-
          dataset/assets/Makiling_S2_Composite_2021');

function computeNDVIDiff(newImg, oldImg) {
  return newImg.subtract(oldImg)
              .rename('NDVI_diff')
              .clip(AOI);
}

function classifyHealth(diff) {
  return diff.expression(
    "(d < -0.1) ? 1 : (d > 0.1) ? 3 : 2",
    {d: diff.select('NDVI_diff')}
  )
  .rename('forest_health')
  .toInt8()
  .clip(AOI);
}
```

APPENDIX F

Model and Other Relevant Source Code

Standalone CNN Model (Multi-Head U-Net) Definition

```
class DoubleConv(nn.Module):
    def __init__(self, in_ch, out_ch):
        super().__init__()
        self.double_conv = nn.Sequential(
            nn.Conv2d(in_ch, out_ch, 3, padding=1, bias=False),
            nn.BatchNorm2d(out_ch),
            nn.ReLU(inplace=True),
            nn.Conv2d(out_ch, out_ch, 3, padding=1, bias=False),
            nn.BatchNorm2d(out_ch),
            nn.ReLU(inplace=True)
        )
    def forward(self, x):
        return self.double_conv(x)

class UNetMultiHead(nn.Module):
    def __init__(self, in_channels=6, num_type=3, num_level=4,
num_health=4):
        super().__init__()
        self.inc = DoubleConv(in_channels, 64)
        self.down1 = nn.Sequential(nn.MaxPool2d(2), nn.Dropout2d(0.5),
DoubleConv(64, 128))
        self.down2 = nn.Sequential(nn.MaxPool2d(2), nn.Dropout2d(0.5),
DoubleConv(128, 256))
        self.down3 = nn.Sequential(nn.MaxPool2d(2), nn.Dropout2d(0.5),
DoubleConv(256, 512))

        self.up3 = nn.ConvTranspose2d(512, 256, kernel_size=2, stride=2)
        self.conv3 = DoubleConv(512, 256)
        self.up2 = nn.ConvTranspose2d(256, 128, kernel_size=2, stride=2)
        self.conv2 = DoubleConv(256, 128)
        self.up1 = nn.ConvTranspose2d(128, 64, kernel_size=2, stride=2)
        self.conv1 = DoubleConv(128, 64)

        self.head_type = nn.Conv2d(64, num_type, kernel_size=1)
        self.head_level = nn.Conv2d(64, num_level, kernel_size=1)
        self.head_health = nn.Conv2d(64, num_health, kernel_size=1)

    def forward(self, x):
        x1 = self.inc(x)
        x2 = self.down1(x1)
        x3 = self.down2(x2)
        x4 = self.down3(x3)

        x = self.up3(x4)
        x = torch.cat([x, x3], dim=1)
        x = self.conv3(x)
        x = self.up2(x)
        x = torch.cat([x, x2], dim=1)
```

```
        x = self.conv2(x)
        x = self.up1(x)
        x = torch.cat([x, x1], dim=1)
        x = self.conv1(x)

        return self.head_type(x), self.head_level(x), self.head_health(x)
```

Standalone ViT Model (SegFormer) Definition

```
class MultiTaskSegFormer(nn.Module):
    def __init__(self, pretrained_name="nvidia/mit-b0", in_channels=6,
                 num_type=3, num_level=4, num_health=4):
        super().__init__()

        self.backbone = SegformerModel.from_pretrained(pretrained_name)

        old_layer = self.backbone.encoder.patch_embeddings[0].proj

        new_layer = nn.Conv2d(
            in_channels=in_channels, # 6
            out_channels=old_layer.out_channels,
            kernel_size=old_layer.kernel_size,
            stride=old_layer.stride,
            padding=old_layer.padding
        )

        with torch.no_grad():
            new_layer.weight[:, :3, :, :] = old_layer.weight
            new_layer.weight[:, 3:, :, :] = torch.mean(old_layer.weight, dim=1,
keepdim=True)
            new_layer.bias = old_layer.bias

        self.backbone.encoder.patch_embeddings[0].proj = new_layer

        encoder_dims = self.backbone.config.hidden_sizes

        # Linear projection layers to unify channel depth to 256
        self.linear_c4 = nn.Conv2d(encoder_dims[3], self.embed_dim, 1)
        self.linear_c3 = nn.Conv2d(encoder_dims[2], self.embed_dim, 1)
        self.linear_c2 = nn.Conv2d(encoder_dims[1], self.embed_dim, 1)
        self.linear_c1 = nn.Conv2d(encoder_dims[0], self.embed_dim, 1)

        fused_dim = self.embed_dim * 4

        self.head_type = nn.Sequential(
            nn.Conv2d(fused_dim, 256, 3, padding=1, bias=False),
            nn.BatchNorm2d(256), nn.ReLU(),
            nn.Conv2d(256, num_type, 1)
        )
```

```
        self.head_level = nn.Sequential(
            nn.Conv2d(fused_dim, 256, 3, padding=1, bias=False),
            nn.BatchNorm2d(256), nn.ReLU(),
            nn.Conv2d(256, num_level, 1)
        )
        self.head_health = nn.Sequential(
            nn.Conv2d(fused_dim, 256, 3, padding=1, bias=False),
            nn.BatchNorm2d(256), nn.ReLU(),
            nn.Conv2d(256, num_health, 1)
        )

    def forward(self, x):
        batch_size, _, H, W = x.shape

        outputs = self.backbone(x, output_hidden_states=True)

        c1, c2, c3, c4 = outputs.hidden_states

        n4 = self.linear_c4(c4)
        n3 = self.linear_c3(c3)
        n2 = self.linear_c2(c2)
        n1 = self.linear_c1(c1)

        target_size = n1.shape[2:]

        n4 = F.interpolate(n4, size=target_size, mode='bilinear',
align_corners=False)
        n3 = F.interpolate(n3, size=target_size, mode='bilinear',
align_corners=False)
        n2 = F.interpolate(n2, size=target_size, mode='bilinear',
align_corners=False)

        fused = torch.cat([n4, n3, n2, n1], dim=1) # 1024 channels

        out_type = self.head_type(fused)
        out_level = self.head_level(fused)
        out_health = self.head_health(fused)

        out_type = F.interpolate(out_type, size=(H, W), mode='bilinear',
align_corners=False)
        out_level = F.interpolate(out_level, size=(H, W), mode='bilinear',
align_corners=False)
        out_health = F.interpolate(out_health, size=(H, W), mode='bilinear',
align_corners=False)

        return out_type, out_level, out_health
```

Hybrid CNN-ViT Model Architecture

```
class ResBlock(nn.Module):
    def __init__(self, in_channels, out_channels):
        super().__init__()
        self.conv1 = nn.Conv2d(in_channels, out_channels, 3, padding=1,
bias=False)
        self.bn1 = nn.BatchNorm2d(out_channels)
        self.relu = nn.ReLU(inplace=True)
        self.conv2 = nn.Conv2d(out_channels, out_channels, 3, padding=1,
bias=False)
        self.bn2 = nn.BatchNorm2d(out_channels)

        self.shortcut = nn.Sequential()
        if in_channels != out_channels:
            self.shortcut = nn.Sequential(
                nn.Conv2d(in_channels, out_channels, kernel_size=1, bias=False),
                nn.BatchNorm2d(out_channels)
            )

    def forward(self, x):
        residual = self.shortcut(x)
        out = self.conv1(x)
        out = self.bn1(out)
        out = self.relu(out)
        out = self.conv2(out)
        out = self.bn2(out)
        out += residual
        out = self.relu(out)
        return out

class ViTBottleneck(nn.Module):
    def __init__(self, in_channels, feature_map_size, num_layers=4,
num_heads=8):
        super().__init__()
        self.hidden_dim = in_channels
        self.seq_len = feature_map_size * feature_map_size

        self.pos_embedding = nn.Parameter(torch.randn(1, self.seq_len,
self.hidden_dim))

        encoder_layer = nn.TransformerEncoderLayer(
            d_model=self.hidden_dim,
            nhead=num_heads,
            dim_feedforward=self.hidden_dim*4,
            activation='gelu',
            batch_first=True,
            norm_first=True
        )
```

```
        self.transformer = nn.TransformerEncoder(encoder_layer,
num_layers=num_layers)

    def forward(self, x):
        B, C, H, W = x.shape
        x = x.flatten(2).transpose(1, 2) # (B, Seq, C)
        x = x + self.pos_embedding
        x = self.transformer(x)
        x = x.transpose(1, 2).reshape(B, C, H, W)
        return x

class HybridTransUNet(nn.Module):
    def __init__(self, in_channels=6, patch_size=64, num_type=3,
num_level=4, num_health=4):
        super().__init__()

        self.enc1 = nn.Sequential(nn.Conv2d(in_channels, 64, 3, padding=1,
bias=False), nn.BatchNorm2d(64), nn.ReLU(), ResBlock(64, 64))
        self.pool1 = nn.MaxPool2d(2)

        self.enc2 = ResBlock(64, 128)
        self.pool2 = nn.MaxPool2d(2)

        self.enc3 = ResBlock(128, 256)
        self.pool3 = nn.MaxPool2d(2)

        self.enc4 = ResBlock(256, 512)

        # Bottleneck (Transformer)
        # 64x64 -> /8 = 8x8 grid. 128x128 -> /8 = 16x16 grid.
        bottleneck_size = patch_size // 8
        self.vit = ViTBottleneck(in_channels=512,
feature_map_size=bottleneck_size)

        self.up1 = nn.ConvTranspose2d(512, 256, 2, stride=2)
        self.dec1 = ResBlock(512, 256)

        self.up2 = nn.ConvTranspose2d(256, 128, 2, stride=2)
        self.dec2 = ResBlock(256, 128)

        self.up3 = nn.ConvTranspose2d(128, 64, 2, stride=2)
        self.dec3 = ResBlock(128, 64)

        # Multi-Task Heads
        self.head_type = nn.Conv2d(64, num_type, 1)
        self.head_level = nn.Conv2d(64, num_level, 1)
        self.head_health = nn.Conv2d(64, num_health, 1)

    def forward(self, x):
        x1 = self.enc1(x)
```

```
        x2 = self.enc2(self.pool1(x1)) # 1/2
        x3 = self.enc3(self.pool2(x2)) # 1/4
        x4 = self.enc4(self.pool3(x3)) # 1/8

        x_vit = self.vit(x4)

        d1 = self.dec1(torch.cat([self.up1(x_vit), x3], dim=1))
        d2 = self.dec2(torch.cat([self.up2(d1), x2], dim=1))
        d3 = self.dec3(torch.cat([self.up3(d2), x1], dim=1))

        return self.head_type(d3), self.head_level(d3), self.head_health(d3)
```

Tree Recommendation Rule-Based Module

```
SPECIES_DB = {

    1: "Dipterocarpus grandiflorus (Apitong), Shorea contorta (White Lauan), Shorea negrosensis (Red Lauan), Pterocarpus indicus (Narra), Vitex parviflora (Molave), Artocarpus blancoi (Antipolo), Canarium ovatum (Pili)",

    2: "Lithocarpus spp. (Oak), Syzygium spp. (Stone Oak), Elaeocarpus cumingii (Elaeocarpus tree), Podocarpus rumphii (Rumph's Yellowwood), Cinnamomum mercadoi (Kalingag), Ficus balete (Balete tree), Weinmannia luzoniensis (Luzon Weinmannia)",

    3: "Dacrycarpus cumingii (Cuming's Yellowwood), Ilex philippinensis (Philippine holly), Vaccinium spp. (Blueberries), Rapanea spp.(Kalingag-kalingag), Clethra tomentella, Weinmannia subsessilis ",

    "General": "No Proper Tree Recommendation Provided"

}
```

*where 1: Lowland, 2: Lower Montane, and 3: Upper Montane.**

APPENDIX G

Evaluation Tool

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

Cristino L. Tiburan Jr., PhD
Professor
Institute of Renewable Natural Resources
College of Forestry and Natural Resources
University of the Philippines – Los Baños

Dear Dr. Tiburan,

This form is for the expert review of the technical and ecological decisions made in our study. We sincerely request your evaluation of the methodologies used to process satellite imagery and generate the ground-truth map labels (Forest Type, Level, and Health).

Please use the column on the right to provide your comments and suggestions based on your expertise in Remote Sensing and GIS. Your feedback is crucial in confirming the scientific validity of our data pipeline.

Topic of Validation	Topic Details	Comments and Suggestions
I. Satellite Data Processing (Composites)		
Generation of Yearly Mosaics	Utilization of Google Earth Engine (GEE) to generate Yearly Sentinel-2 Composites (2019 was for NDVI difference computation, and 2020-2021 was for experiment proper).	This is okay and no further comments.
Mosaic Function	The study employed the mosaic() function to aggregate pixels from an entire year. This process reduces cloud contamination and creates a consistent spectral baseline for the subsequent analysis.	This is an appropriate function to produce the composite image creating less cloud cover over the study area.
II. Forest Type Label Validation		

Source & Reclassification	Labels were derived from the ESA WorldCover's Land Cover Map. Where, Forested: Class 10 [Tree-Cover] and Non-Forest: All other classes (Built-up, Shrubland, etc.).	This is more appropriate for the land cover categories.
III. Forest Level Label Validation		
Data Source & Resolution	SRTM DEM was resampled from 30-meter to 10-meter resolution to match the Sentinel-2 pixel size.	This is commonly applied to ensure that pixel sizes are of similar resolutions across different raster datasets that are applied in the overlay process.
Elevation Zoning	The zoning was based on studies by Villanueva & Buot (2018) and Gevaña & Pampolina (2017), where: Lowland (< 300m), Lower Montane (301m – 700m), and Upper Montane (> 700m).	The classes for the elevation are okay.
IV. Forest Health Label Validation		
NDVI Difference Thresholds	Forest Health classes were defined by the change in NDVI values between the generated Yearly Composites, where: Degraded: Difference of < -0.10, Neutral: Difference between -0.10 to 0.10, and Regrowth: Difference of > 0.10 (Stroppiana et al., 2020).	These categories provide significant information on vegetation condition and are helpful for decision-making.
V. Overall Methodology Assessment		
Data Alignment	Assessment of the co-registration and spatial alignment between the generated Sentinel-2	These are standard protocols that must be conducted to ensure proper co-registration of

	Mosaics (RGB/NDVI) and the Forest Type, Level, and Health label maps.	the different layers used in the study.
Map Quality	Assessment of the overall clarity, spatial accuracy, and minimal presence of artifacts in the final three label maps.	The study provided an assessment on the overall accuracy of the process and this has been included in the study.

References:

- Gevaña, Dixon T. and Pampolina, Nelson M., "Broadleaf diversity and carbon estimates along the northeastern slope of Mt. Makiling, Philippines" (2017). *Journal Article*. 5184. https://www.researchgate.net/profile/Dixon-Gevana/publication/290020631_Broadleaf_diversity_and_carbon_estimates_along_the_northeastern_slope_of_Mt_Makiling_Philippines/links/5cd37d77458515712e9b5bb4/Broadleaf-diversity-and-carbon-estimates-along-the-northeastern-slope-of-Mt-Makiling-Philippines.pdf
- Stroppiana, D., Brivio, P. A., Meroni, M., & Boschetti, M. (2020). A threshold- and trend-based vegetation change monitoring algorithm (TVCMA) for detecting abrupt and gradual changes in vegetation. *Remote Sensing of Environment, 240*, 111675. https://doi.org/10.1016/j.rse.2020.111675
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- WorldCover. (n.d.). WORLDCOVER. https://esa-worldcover.org/en

Requested by:

JOHN ADRIAN LAPITAN

SARAH CLAUDINE V. NOOR

ANDRHEY U. DORADO
The Researchers

Reviewed by:

Cristino L. Tiburan Jr.
Professor

APPENDIX H

Grammarly Results

Forest-Classify-and-Recommend-v32-p1

by John Adrian Lapitan

General metrics

72,946	10,005	512	40 min 1 sec	1 hr 16 min
characters	words	sentences	reading time	speaking time

Score

99

This text scores better than 99% of all texts checked by Grammarly

Writing Issues

42	1	41
Issues left	Critical	Advanced

Plagiarism

This text hasn't been checked for plagiarism

Writing Issues

40	**Clarity**	
13	Intricate text	
1	Passive voice misuse	
26	Paragraph can be improved	
1	**Correctness**	
1	Incorrect noun number	
1	**Engagement**	
1	Word choice	

Unique Words

Measures vocabulary diversity by calculating the percentage of words used only once in your document

20%

unique words

Rare Words

Measures depth of vocabulary by identifying words that are not among the 5,000 most common English words.

48%

rare words

Word Length

Measures average word length

5.9

characters per word

Forest-Classify-and-Recommend-v32-p2

by John Adrian Lapitan

General metrics

66,226	9,375	405	37 min 30 sec	1 hr 12 min
characters	words	sentences	reading time	speaking time

Score

97

This text scores better than 97% of all texts checked by Grammarly

Writing Issues

97	5	92
Issues left	Critical	Advanced

Plagiarism

This text hasn't been checked for plagiarism

Writing Issues

76	**Clarity**	
7	Intricate text	
64	Paragraph can be improved	
5	Hard-to-read text	
7	**Correctness**	
2	Text inconsistencies	
1	Misuse of modifiers	
3	Incorrect punctuation	
1	Determiner use (a/an/the/this, etc.)	
10	**Engagement**	
10	Word choice	
4	**Delivery**	
4	Inappropriate colloquialisms	

Unique Words

15%

unique words

Measures vocabulary diversity by calculating the percentage of words used only once in your document

Rare Words

49%

rare words

Measures depth of vocabulary by identifying words that are not among the 5,000 most common English words.

Forest-Classify-and-Recommend-v32-p3

by John Adrian Lapitan

General metrics

43,613	6,128	262	24 min 30 sec	47 min 8 sec
characters	words	sentences	reading time	speaking time

Score

98

This text scores better than 98% of all texts checked by Grammarly

Writing Issues

52	7	45
Issues left	Critical	Advanced

Plagiarism

This text hasn't been checked for plagiarism

Writing Issues

40	**Clarity**	
8	Intricate text	
30	Paragraph can be improved	
1	Passive voice misuse	
1	Hard-to-read text	
7	**Correctness**	
1	Determiner use (a/an/the/this, etc.)	
1	Incorrect punctuation	
4	Ungrammatical sentence	
1	Wrong or missing prepositions	
4	**Engagement**	
4	Word choice	
1	**Delivery**	
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APPENDIX I

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Forest-Classify-and-Recommend-v33-Turnitin (1)

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APPENDIX J

Publishable Paper (IEEE Format)

Semantic Segmentation of Forest Condition in Mt. Makiling Using a CNN-ViT Hybrid with Native Tree Species Recommendation

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Article Info

Article History:

Received by
Revised by
Accepted by

Keywords:
Deep Learning
Forest Monitoring
Hybrid TransU-Net
NDVI
Semantic Segmentation
Sentinel-2

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ABSTRACT

This study developed and evaluated a deep learning framework for forest monitoring and restoration planning within the Mt. Makiling Forest Reserve (MMFR), utilizing Sentinel-2 satellite imagery and SRTM elevation data. The study implemented a multi-task semantic segmentation pipeline to classify Forest Type, Forest Level, and Forest Health using a 6-channel composite input (RGB + NDVI + EVI + NDWI). Three architectures were compared—U-Net (CNN), SegFormer (ViT), and TransU-Net (Hybrid)—across two patch resolutions (64×64 and 128×128). The results identified the Hybrid TransU-Net with 128×128 inputs as the superior architecture, achieving a near-perfect Mean Intersection-over-Union (mIoU) of 99.21% for Forest Health on the 2020 test set, successfully resolving critical minority classes such as "Regrowth" and "Degraded." However, temporal evaluation on an independent 2021 dataset revealed a lack of temporal robustness, with Forest Level accuracy dropping to approximately 55% and Forest Health mIoU declining to 33.62%. This indicates that while the model excels at characterizing the specific spatial and spectral conditions of the training year, it suffers from overfitting to static topographical features. The study further demonstrated a rule-based native tree species recommendation module, which functioned logically but was constrained by the upstream classification errors in the multi-temporal test. The research concludes that while the hybrid approach provides a powerful tool for precision forest monitoring, operational deployment requires multi-year training data to ensure temporal generalization.

1. INTRODUCTION

The Philippines is one of the world's 17 megadiverse countries, yet its forest ecosystems remain vulnerable to anthropogenic pressures and climate-induced degradation. The Mt. Makiling Forest Reserve (MMFR), an ASEAN Heritage Park, serves as a critical biological corridor. However, landscape changes driven by land conversion and natural disturbances necessitate innovative strategies for monitoring and restoration [1], [2].

Traditional forest monitoring relies on field surveys, which are labor-intensive. Remote sensing offers a scalable alternative, with indices like the Normalized Difference Vegetation Index (NDVI) used to assess canopy vigor [3], [4]. While effective, standard indices often lack the site-specific resolution to guide replanting. Furthermore, integration between condition analysis and restoration planning remains limited [3].

Deep learning (DL) has emerged as a transformative tool in remote sensing. Convolutional Neural Networks (CNNs) excel at spatial feature extraction but struggle with global context in heterogeneous landscapes [5], [17]. Conversely, Vision Transformers (ViTs) capture long-range dependencies but require extensive datasets [12]. Hybrid architectures, which fuse CNNs and ViTs, offer a solution for fine-grained mapping [14], [15].

This study aims to develop a proof-of-concept pipeline for precision monitoring in MMFR. Specifically, it seeks to: (1) Process Sentinel-2 imagery to generate multi-task labels; (2) Compare CNN, ViT, and Hybrid architectures; and (3) Implement a rule-based module to recommend native tree species. This supports UN Sustainable Development Goals 13 and 15 [27], [28].

2. RELATED WORKS

A. Vegetation Indices and Thresholds

NDVI is the standard for assessing vegetation health but can saturate in high-biomass forests. The Enhanced Vegetation Index (EVI) addresses this by incorporating the blue band [6]. The Normalized Difference Water Index (NDWI) is also critical for monitoring moisture stress [7]. Thresholding techniques allow for the categorization of forest health into classes such as degraded, stable, or regrowth [5], [8].

B. Deep Learning in Remote Sensing

CNNs like U-Net are dominant in segmentation. However, their limited receptive field hinders performance in complex scenes [11], [16]. Transformers like SegFormer [13] utilize attention mechanisms to model global context. Recent studies show that Hybrid models (e.g., TransU-Net) combine local texture extraction with global context, achieving state-of-the-art results [14], [15].

C. Ecological Zonation in Mt. Makiling

Restoration relies on site-specificity. MMFR studies document distinct vegetation stratification along elevational gradients [19]. Villanueva and Buot [18] identified plant communities for Lowland, Montane, and Mossy zones. Integrating these rules with geospatial data is essential for ensuring reforestation uses climatologically appropriate native species [22], [24].

3. METHODOLOGY

The study integrated multi-temporal remote sensing analysis for ground-truth label generation, a comparative analysis of three Multi-Task Learning (MTL) architectures: a baseline CNN (U-Net), a Transformer (SegFormer), and a Hybrid CNN–ViT (TransU-Net), and a rule-based module for native tree species recommendation. The methodology emphasized computational optimization through binary data serialization and systematic evaluation using pixel-wise diagnosis.

A. Study Area

The Mt. Makiling Forest Reserve (MMFR), located in Southern Luzon, Philippines, spans approximately 4,244 hectares. The area is officially delineated using boundaries from the World Database on Protected Areas (WDPA), adopted as the Area of Interest (AOI) in all data processing.

B. Data Acquisition and Preprocessing

The study utilized Sentinel-2 Level-2A Surface Reflectance imagery processed via Google Earth Engine (GEE) [26]. To ensure high-quality spectral inputs, the full archive intersecting the MMFR AOI was queried. A total of 245 satellite scenes were processed.

A rigorous masking algorithm using the Scene Classification Layer (SCL) removed clouds (classes 8, 9, 10) and shadows. Two compositing strategies were applied:

1. 2019 Baseline**:** Median composite (NDVI only) for pre-monitoring state.
2. 2020 (Training) & 2021 (Testing): Quality Mosaic sorted by cloud cover. This prioritized pixels from the clearest scenes to preserve texture. The final input was a 6-channel tensor: RGB + NDVI + EVI + NDWI.

C. Label Generation

Ground truth labels were derived from authoritative sources:

1. Forest Type: From ESA WorldCover [25], remapped to "Forested" and "Non-Forest."
2. Forest Level: From SRTM DEM (30m) resampled to 10m. Zones: Lowland (<300m), Lower Montane (300-700m), Upper Montane (>700m) [21].
3. Forest Health: Computed via NDVI differencing [9], [10]. Thresholds: < −0.10 (Degraded), -0.10 – 0.10 (Neutral), > 0.10 (Regrowth) [23].

D. Model Architectures

Three Multi-Task Learning (MTL) architectures were evaluated:

1. U-Net (CNN): Standard encoder-decoder with skip connections [17].
2. SegFormer (ViT): Hierarchical Transformer with Mix-FFN layers [13].
3. TransU-Net (Hybrid): ResNet backbone for local features + Transformer bottleneck for global context [14].

Models were trained on 64×64 and 128×128 patches using stratified random sampling to address class imbalance.

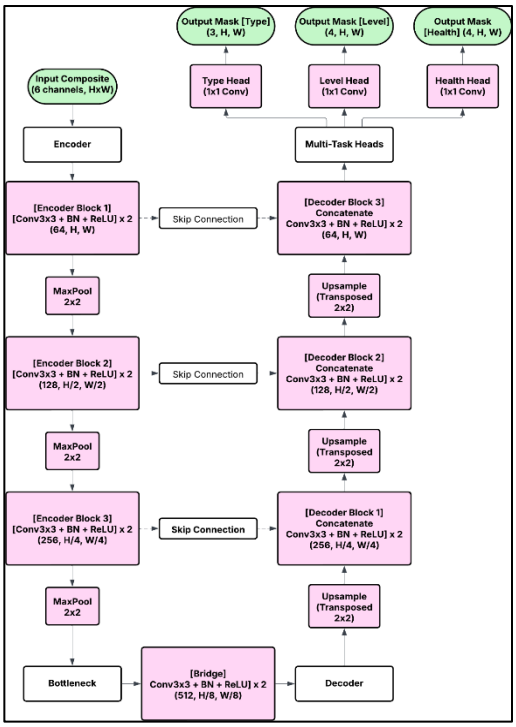


Fig. 1. CNN (Multi-Head U-Net) Model Architecture

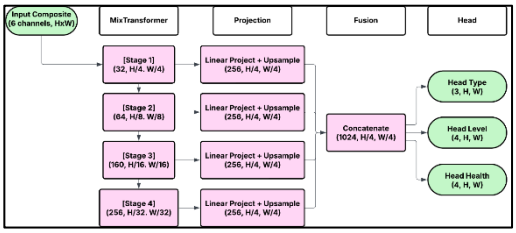


Fig. 2. ViT (SegFormer) Model Architecture

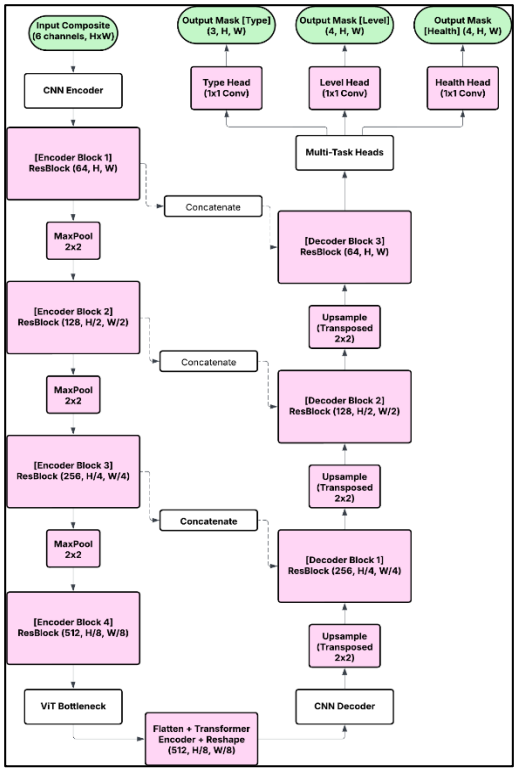


Fig. 3. Hybrid (Trans U-Net) Model Architecture

E. Model Evaluation

The models were evaluated on the held-out test sets (internal: 2020; external: 2021) using Accuracy (1), F1-score (2), and Intersection-over-Union (IoU) (3).

$$A = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$F_1 = 2 \cdot \frac{P \cdot R}{P + R} \quad (2)$$

$$IoU = \frac{TP}{TP + FP + FN} \quad (3)$$

Macro-averaged metrics were reported across classes, complemented by confusion matrices to identify misclassification patterns.

F. Tree Species Recommendation

A rule-based module was developed using ecological literature [2], [24]. It suggests species based on the intersection of "Degraded" status and "Forest Level."

Table I. Tree Recommendation Lookup Table

Forest Type	Forest Level	Recommended Tree Species	Source
Forested	Lowland (<300 m)	• Dipterocarpus grandiflorus (Apitong) • Shorea contorta (White Lauan) • Shorea negrosensis (Red Lauan) • Pterocarpus indicus (Narra) • Vitex parviflora (Molave) • Artocarpus blancoi (Antipolo) • Canarium ovatum (Pili)	Malabrigo et al. (2016) [2]; Haribon Foundat-ion (2010) [24]; Fernando et al. (2008) [22]
	Lower Montane (301–700 m)	● Lithocarpus spp. (Oak) ● Syzygium spp. (Stone Oak) ● Elaeocarpus cumingii (Elaeocarpus tree) ● Podocarpus rumphii (Rumph's Yellowwood) ● Cinnamomum mercadoi (Kalingag) ● Ficus balete (Balete tree) ● Weinmannia luzoniensis (Luzon Weinmannia)	Fernando et al. (2008) [22]; Villanueva & Buot (2018) [18]; Alcala (2021) [20]
	Upper Montane (700–1000 m)	● Dacrycarpus cumingii (Cuming's Yellowwood) ● Ilex philippinensis (Philippine holly) ● Vaccinium spp. (Blueberries) ● Rapanea spp.(Kalingag-kalingag) ● Clethra tomentella ● Weinmannia subsessilis	Buot & Okitsu (1998); Villanueva & Buot (2018) [18]; Alcala (2021) [20]

G. Ethical Considerations

Only open-access datasets (Sentinel-2, SRTM, ESA WorldCover) and open-source software (GEE, QGIS, Python) were used. No personal or field data were collected, ensuring compliance with research ethics and data privacy standards.

4. RESULTS AND DISCUSSION

This chapter presents the results of implementing the CNN–ViT hybrid semantic segmentation model for classifying forest type, forest level, and forest health in the Mt. Makiling Forest Reserve (MMFR). It also discusses the outputs of the rule-based recommendation module that suggests native tree species for degraded areas. Results were analyzed using both quantitative evaluation metrics (accuracy, precision, recall, F1-score, Intersection-over-Union, and confusion matrices) and qualitative visualizations of the classification maps and recommendations. Results presented here included both training/validation metrics used to monitor learning, and final evaluation metrics on the held-out test set. The discussion situates these results within the broader objectives of forest monitoring and restoration planning in MMFR.

A. Baseline Performance

The Hybrid TransU-Net with 128×128 inputs emerged as the superior architecture. As shown in Table I, it achieved near-perfect scores across all tasks.

Crucially, the Hybrid model resolved the "Regrowth" class (mIoU 0.9888), which standalone ViT models failed to detect (mIoU 0.5294). ANOVA confirmed that increasing patch size to 128×128 yielded statistically significant improvements ($p<0.001$).

Table II. Model Performance (2020 Baseline)

Model (Input)	Forest Level (Acc)	Forest Health (mIoU)	Regrowth (IoU)
CNN (64x64)	99.11%	0.7569	0.6773
ViT (64x64)	98.38%	0.6072	0.4757
Hybrid (128x128)	99.97%	0.9921	0.9888

B. Temporal Stability (2021 Test Set)

When applied to the 2021 dataset without retraining, the model (Hybrid 128x128) exhibited severe overfitting. Forest Level accuracy dropped to 55.54%, indicating the model memorized the 2020 topography rather than learning generalized terrain features. Consequently, Forest Health mIoU declined to 33.62% (Table III).

Table III. Temporal Stability (2020 vs 2021)

Metric	2020 Score	2021 Score	Drop (in %)
Forest Level Accuracy	99.97%	55.54%	-44.43%
Forest Health mIoU	99.21%	33.62%	-65.59%

C. Rule-Based Recommendation Module and Output Visualization

The recommendation module generated native species suggestions by cross-referencing the model's "Forest Level" and "Forest Health" predictions against the ecological rules defined in Table III.
For the 2020 baseline data, the system correctly identified degraded patches within the Lower Montane zone and recommended appropriate mid-elevation species such as Oak (*Lithocarpus spp.*) and Stone Oak (*Syzygium spp.*). Conversely, for the 2021 temporal test, the upstream classification errors (misclassifying Lowland as Upper Montane) caused the system to erroneously recommend high-altitude species like Yellowwood (*Dacrycarpus cumingii*) for lower-elevation zones. This confirms that while the rule-based logic is sound, its reliability is strictly dependent on the temporal robustness of the underlying segmentation model.

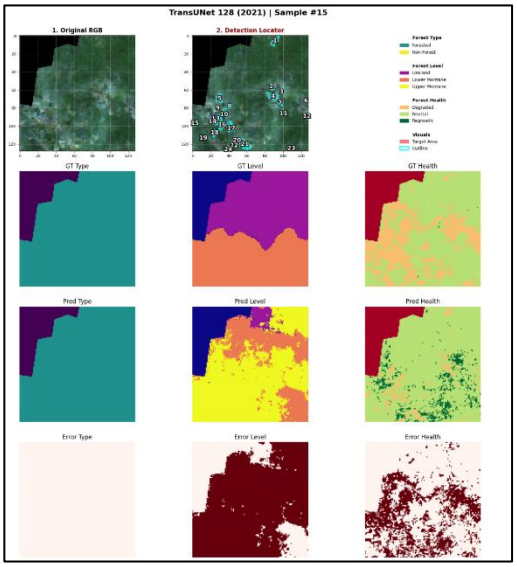


Fig. 4. Visualization of Hybrid Model's Classification

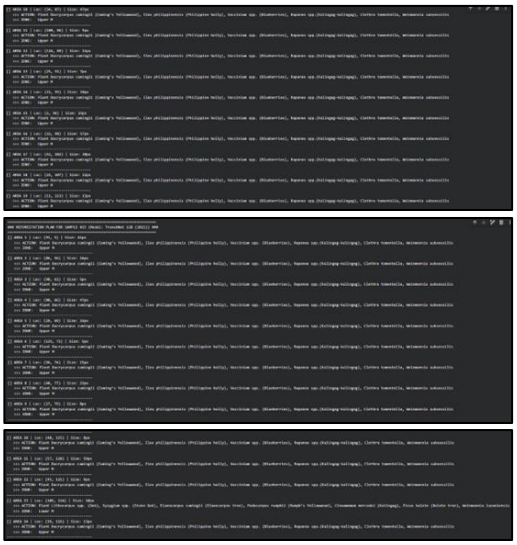


Fig. 5. Sample of Corresponding Tree Species Recommendations

Visual analysis of the segmentation masks confirms the quantitative findings. On the 2020 test set, the Hybrid model produced highly coherent maps, accurately delineating the boundary between Lowland and Montane forests and precisely isolating fragmented Degraded patches (Fig. 4).

However, visualization of the 2021 outputs reveals a distinct "spatial collapse" (Fig. 4). Large sections of the terrain were incorrectly shifted to higher elevation classes, and the precise detection of "Regrowth" areas vanished. This visual evidence supports the conclusion that the model learned to memorize the specific pixel coordinates of the 2020 training imagery rather than learning generalized, invariant topographical features.\

D. Summary of Findings

The study successfully identified the Hybrid TransU-Net (128x128) as the superior architecture for forest monitoring. It achieved near-perfect segmentation on the baseline year, significantly outperforming standalone CNNs and ViTs in detecting minority classes. However, the temporal evaluation revealed a critical vulnerability: the model lacks robustness to year-to-year spectral shifts without retraining. The results

diagnose spatial memorization as the primary cause of the performance drop on the 2021 dataset.

5. SUMMARY, CONCLUSION, AND RECOMMENDATIONS

A. Summary

This study developed a deep learning pipeline for forest classification and restoration planning in the Mt. Makiling Forest Reserve (MMFR). The system integrated a Hybrid CNN–ViT (TransU-Net) model with a rule-based ecological module.

Experimental results on the 2020 dataset were exceptional, with the Hybrid 128x128 model achieving a Forest Health mIoU of 99.21% and successfully resolving the difficult "Regrowth" class (0.9888 IoU). However, applying the model to the 2021 dataset without retraining resulted in a significant performance decline: Forest Level accuracy dropped to 55.54% and Forest Health mIoU to 33.62%. This confirmed that the current training strategy leads to overfitting on the specific spatial arrangement of the training year.

B. Conclusions

The study concludes that Hybrid Deep Learning is a scientifically valid and powerful approach for precision forest monitoring, capable of resolving fine-grained ecological features that traditional indices miss. The framework works effectively as a proof-of-concept for single-year analysis.

However, the study also concludes that these models are not "plug-and-play" for multi-temporal monitoring. The large discrepancy between the 2020 and 2021 results highlights a fundamental limitation in temporal generalization. The system essentially memorized the "where" (topography) rather than fully learning the "what" (spectral features). Therefore, operational deployment requires specific data strategies to prevent this spatial overfitting.

C. Recommendations

To address the identified constraints and prepare the system for operational use, the following refinements are recommended:

1. Transfer Learning: Investigate transfer learning from models pre-trained on large-scale, general remote sensing datasets (e.g., Sentinel-2-based land cover) to initialize the weights. This could help the model learn more general features before specializing on the small MMFR dataset, potentially mitigating overfitting
2. Expanded Spectral Inputs: The input tensor should be expanded beyond RGB+NDVI to include Near-Infrared (NIR) and Short-Wave Infrared (SWIR) bands from Sentinel-2. These bands provide direct physical measurements of moisture and structure, reducing the model's over-reliance on elevation data.
3. Geospatial Reconstruction: For practical deployment, a post-processing module must be developed to stitch the predicted 128x128 image chips back into a seamless, georeferenced raster. This will allow the system to export specific GPS coordinates of degraded zones for ground-level intervention.
4. Cross-Location Validation: To rigorously test generalizability, the model should be evaluated on spatially distinct ecosystems (e.g., Mt. Banahaw or Sierra Madre). This will determine if the learned features are transferable across different tropical mountain terrains or if site-specific retraining is mandatory for every new location.

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APPENDIX K

Curriculum Vitae

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